Relationships between stock market equities and their dynamics

ZHIYIN (GRACE) YAN
SID: 510437747

Supervisor: Dr. Joseph T. Lizier
Associate Supervisor: Dr. Chetan Gohil

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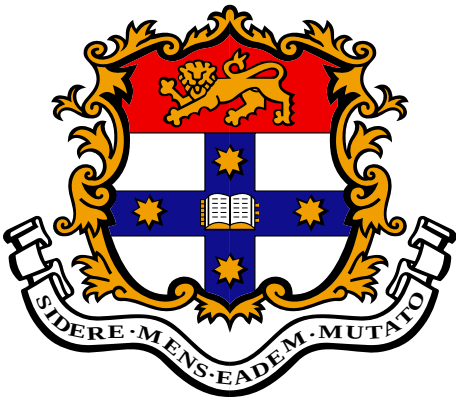

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Abstract

Practitioners, researchers and policymakers continually seek to decode market behaviours and economics by analysing dynamic trends of stock markets. Stock markets fluctuate in response to evolving shocks and can be modelled as dynamic complex networks. With existing approaches focusing on individual stocks, there is an inadequate understanding of the collective market structure and dynamics. Computational neuroscience (CN) utilises more mature methodologies, such as Functional Connectivity (FC), to analyse complex systems. This study proposes a novel dynamic FC network approach to assess relationships between stocks in the New York Stock Exchange (NYSE) and their dynamics. We compute Pearson correlation coefficients as the FC measure between stocks to construct an FC network, with stocks as nodes and correlations as edges. This network illustrates the time-averaged market structure, with market sectors identified via clustering. To capture the time-evolving market structure, we apply Sliding-Window analysis and the Hidden Markov Model. Both methods uncover market states, which are distinct market structures of market conditions defined in prior research. A time lag correlation analysis assesses the market states and transitions with a directed measure of FC. Our results reveal significant implications for identifying potential market-driving stocks and sectors, as well as analysing the impact of crises on relationships between stocks and sectors. This study lays the groundwork for future research on the CN network approach to financial data analysis to understand market behaviour, evaluate systemic risk, and refine investment strategies.

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Glossary

Automated Teller Machine (ATM): An electronic telecommunications device that enables customers to perform financial transactions.

CBOE Volatility Index (VIX): A real time market index representing the market's expectation of 30-day forward-looking volatility for the S&P 500.

Cluster: A group of objects (stocks in FC and windows in DFC) that are more similar to others within the same group than objects in other groups.

Computational Neuroscience (CN): An interdisciplinary field that applies statistical models, theoretical analysis and computer simulations to understand how the nervous system processes information.

Coronavirus Disease 2019 (COVID-19): An infectious disease caused by the coronavirus in 2019, leading to a global pandemic in 2020 and ending in 2023.

Covariance: A measure of how two random variables change together (co-movement).

Dot-Com Bubble: A period of speculative investment in internet-based companies around 1995-2000. The market crashed due to the boundless excitement over the internet's possibilities.

Dwelling Time: The length of time in days of a particular market state being active consecutively before transitioning into another market state.

Dynamic Functional Connectivity (DFC): Statistical relationships between objects in the environment evolve in time.

Federal Reserve Bank (FED): The central banking system of the United States (US).

Fractional Occupancy: The proportion of total time that a particular market state was being active.

Functional Connectivity (FC): Statistical relationships between objects in the environment.

Global Financial Crisis (GFC): A severe worldwide economic crisis around 2007-2009 and was the most serious financial depression since 1930s.

Hidden Markov Model (HMM): A generative model that can capture the underlying dynamics without sliding-window.

Idiosyncrasy: Idiosyncratic risk is a type of investment risk that is specific to a particular asset or group of assets, meaning it's not related to broad market movements.

Independent Component Analysis (ICA): A computational technique to conduct a linear transformation that produces source signals which are as independent from one another as possible.

Leader-Follower Relationship: A dynamic in which movements of one object (leader) systematically precede and help predict the subsequent movements of another object (follower) in the system.

Market Sentiment: The overall attitude of investors towards a particular stock or sector, revealling whether participants feel bullish (optimistic) or bearish (pessimistic). It often drives short-term price movements beyond what fundamentals alone would suggest.

Market State: Also called "State", models a unique market structure, reflecting a particular market condiction.

Market Structure: The statistical relationship among stocks and market sectors.

Module: A cluster identified on the static FC matrix that represents a market sector.

National Association of Securities Dealers Automated Quotations (NASDAQ): A global automated stock exchange that has most of the world's technology giants listed.

New York Stock Exchange (NYSE): One of the world's largest equities exchange by market capitalisation, located on Wall Street.

Pearson Correlation: Or "Correlation", a statistical measure that describes the degree and direction of a relationship between two random variables.

Power Spectural Density (PSD): A spectural analysis to examine the nature of the time series and ensure assumptions for statistical properties such as stationarity (constant mean and variance) have been satisfied.

Principal Component Analysis (PCA): A statistical analysis to identify the most significant components in explaining variances of the dataset.

Probability Distribution Functions (PDF): Describes how probability density is assigened over possible outcomes of a random variable.

Spillover: The transmission of shocks to returns from one sector or stock to another.

Stand and Poor 500 (S&P 500): A stock market index that tracks the performance of the top 500 market capitalised companies in the US.

Stock: Also called "Share" or "Equity", is a unit of ownership in a company where the buyer entitles certain rights and economic benefits.

Switching Rate: The frequency of the market state transitions across the entire time frame.

Stock Return: Also called "Logarithmic/Log Return" or "Return", measures how much an investor earns or loses from holding a share of stock over time. In this study, we refer to the daily return.

CHAPTER 1

Introduction

1.1 Background

The stock market is a dynamic complex system where buyers and sellers trade equities (or "shares", "stocks") of publicly listed companies, continuously reflecting the collective information of all market participants [2]. Interactions and relationships among individual stock prices give rise to emergent phenomena, such as herding behaviour, in which groups of investors follow prevailing trends rather than independent analysis [3]. Therefore, it is essential to understand these relationships between stocks to discern the dynamic trends in the market.

1.1.1 Trend analysis of financial markets

Market trends are the dominant directions of market movement as time progresses, integrating broader economic and contextual perspectives to offer valuable insights into markets beyond price fluctuations [4]. Analysing time-evolving changes in relationships between stocks is a common approach to detecting market trends [5–9]. Trend analysis of stock markets has been traditionally regarded as a fundamental and essential aspect of financial research and practice [4, 6, 10].

Investors would apply trend analysis to gauge market sentiment [10] - whether other investors think optimistically (bullish) or pessimistically (bearish) about the market, to strategically determine the optimal timing of entry and exit and manage overall investment risk. Economists and researchers would use them to explore critical matters such as market efficiency and investor behaviour. Hence, studying the change in relationships between stocks as a trend analysis of stock markets is vital for a broad range of stakeholders [4].

1.1.2 Current approaches

Current approaches utilise various measures in market trend analysis, including the moving average of price [6], log consumer price index [5], and coupling degree [9]. In this study, we aim to reveal the trend of stock markets by establishing statistical relationships between stocks. The relationship between stocks changes over time for both idiosyncratic (internal) and systemic (external) reasons, as seen in Figure 1.1. In Atasoy and Özkan's study, contagion is a measure of shock or disturbance in the stock market. It is well-established that high contagions are associated with market crises, such as the Global Financial Crisis and COVID-19. Nevertheless, they did not capture the change in relationships between stocks across the market during the transition of market crises.

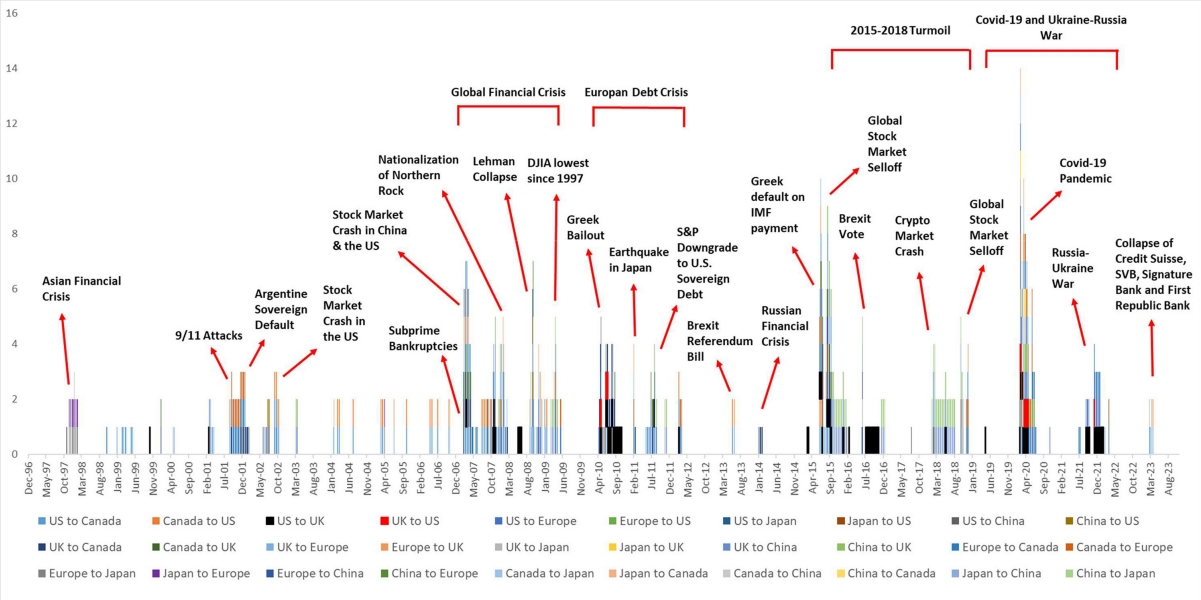


FIGURE 1.1. Aggregated contagion graph.
Reprinted from Figure 5 of Atasoy and Özkan's study [11].

Since relationships between stocks are changing over time due to disturbances, it would be beneficial to understand the underlying dynamics. These dynamics are underexplored in prior studies as they hold notable limitations, including simplified assumptions, overreliance on volume historical data, and inadequate remarks on overall market structure and dynamics [4–9, 12, 13]. More specifically, existing models cannot capture the sudden shifts in the temporal market structure. The current approaches fail to reveal whether the distinct market structure has reoccurred over time or when it reoccurred. Consequently, there is a constant demand for analysing market trends more robustly and sophisticatedly.

1.1.3 Computational neuroscience techniques for studying the dynamics of relationships

In this study, we aim to systematically analyse the relationships between stocks in the market and how they change over time, which overcomes the gap in current approaches. This investigation is akin to studying the relationships between the activity of different brain regions and how they change over time, techniques for which have been well-developed in the domain of computational neuroscience (CN). Researchers have increasingly adopted techniques from CN to model the structure of stock relationships and their dynamics. However, more efforts are required to discern distinct market structures from the change in relationships between stocks [14, 15].

Functional Connectivity (FC) is a CN technique that measures how brain regions interact by measuring their statistical relationships. FC applies the functional network inference that aims to model complex networks of pairwise statistical relationships between nodes as edges. This functional network can quantify either the undirected dependence using the Pearson correlation coefficient and mutual information or the directed dependence using time-lagged correlation and mutual information, transfer entropy and Granger causality. This study will focus on an approach using the Pearson correlation coefficient.

By applying a clustering algorithm to the FC network, brain regions with the most similar statistical relationships are grouped and referred to as modules. These modules reveal functional groups in the brain, such as visual, somatomotor and frontoparietal [16]. Modelling FC over the brain provides valuable insights into the network architecture of the brain and how it might change across tasks and pathological conditions.

As brain networks undergo subtle and time-varying dynamics, the FC between brain regions gradually changes. The **Dynamic Functional Connectivity (DFC)** methodology is instrumental to identifying the recurrence of similar FC network patterns throughout the given timeline. The most typical DFC analysis involves a sliding window analysis over the time series and clustering the windowed FC networks to identify distinct brain states.

This methodology is evidenced as practical and facilitates understanding of how information is transmitted, processed, and integrated within a dynamic network, which is beneficial for studying cognition, perception, and behaviour [16, 17].

1.2 Contributions of this research

This study introduces a novel methodology applying the concept of the DFC network, developed in neuroscience, to analyse relationships between stocks in the market and their dynamics. Techniques in CN were designed to identify subtle and dynamic relationships in complex systems, and they are remarkably beneficial for stock data analysis in uncovering the evolving relationships among stocks, sectors, or indices. We model the market structure by the statistical relationships between stocks. This new approach aims to capture the high-dimensional and time-varying nature of the stock market more effectively and offer a better understanding of how market crises relate to distinct market structures.

We conduct FC analysis to describe the typical market-wide relationships among 50 stocks observed over 32 years. We identify modules of stocks quantitatively, where there are strong within-module correlations and interesting cross-module relationships. These modules are related to established market sectors, including Financial Services, Healthcare, and Technology. Our analysis reveals the collective behaviour of stocks in these sectors and provides insights into their interrelationships at a macro level.

A novel DFC analysis is performed to analyse how the structure of relationships in the market changes over time. We capture the temporal market structure for each unique time window from the Sliding-Window analysis. Some market structures reoccur in multiple windows, and we identify the unique market structures from them, referred to as market states. Our market states correspond to known events, such as the Global Financial Crisis and COVID-19, and we can identify these states with known periods of stable, crisis and transitioning market behaviours. Furthermore, we compare our DFC results with those of the Hidden Markov Model (HMM), a generative model that predicts market states. This comparison offers insights into market states and their transitions with more resolution.

The contribution of this study is to bridge the gap in stock market analysis via techniques from CN to understand the relationships between stocks and their dynamics. The results of this study have demonstrated the value of using the HMM in inferring states that represent unique market structures and analysing their transitions. With a further time lag analysis, this study also uncovers the impact of global market events on the dynamics of directed relationships between stocks in the market.

1.3 Research outline

In this thesis, *Chapter 2 Literature Review* will explore and elucidate FC and DFC analysis. *Section 2.1* will introduce undirected and directed measures of FC and demonstrate network inference with clustering modules to model the underlying system. The Sliding-Window analysis will be explained to infer the DFC network, followed by a presentation of the procedures for computing states and transitions. *Section 2.2* will showcase the current adoption of CN techniques in Finance. Despite some empirical studies in Finance demonstrating similar motivations, the inadequate adoption of CN techniques cannot reveal the temporal market structure and its transitions over time, leaving many gaps for further investigation.

Chapter 3 will introduce the dataset of this study, consisting of daily closing prices from the New York Stock Exchange. *Section 3.1* will describe the data source and dimensions, outlining the procedures for preprocessing data. Dynamics of the preprocessed data will then be explored in *Section 3.2* using a Power Spectral Density analysis. We will also evidence the modular structure in our dataset using a Principal Component Analysis in *Section 3.3*.

The methodology and results will be presented in *Chapters 4–6*. *Chapter 4 Static Networks in Financial Markets* will introduce the Pearson correlation coefficient as our FC measure and conduct FC network inference and clustering for market sectors. Additionally, we will investigate whether some stocks or sectors act as a "hub" (driver) in the market.

Chapter 5 Network Dynamics in Financial Markets will explain the DFC analysis, including Sliding-Window analysis and clustering for market states. Predictions of market states and their transitions will be correlated to the Volatility Index. These correlations are compared across market states between the DFC and HMM methods.

Chapter 6 Time-Lagged Connectivity in Financial Markets will repeat the entire pipeline of FC and DFC analysis on the dataset with a one-day time lag. This chapter will apply the lagged correlation as a directed measure of FC to investigate the directed relationships between stocks, offering a different perspective on modelling the stock market structure.

Chapter 7 Conclusions and Discussion will summarise key findings and contibutions in *Section 7.1* and examine limitations in *Section 7.2* to demonstrate the study design has minimised them. Finally, *Section 7.3* will inform future research based on the study design and results presented in this work.

CHAPTER 2

Literature Review

In this chapter, we aim to establish the state of prior studies on the network of relationships in financial market data and the dynamics of these relationships, highlighting new opportunities to extend our understanding of stock markets. Existing studies in finance have attempted to understand the market by inferring a network structure based on the statistical relationships between stocks. However, most of the study designs did not capture the temporal and sudden shifts in the market network and, hence, were unable to relate the impact of market events on the market network over time.

We will first trace the development of the DFC methodology in CN and its applications in finance. An FC network of a complex system is constructed based on the FC between entities, where entities are nodes and FCs are edges. FCs are the pairwise statistical relationships between entities and can be undirected or directed measures. Next, we will examine how the DFC methodology captures the dynamics of this FC network to analyse state transitions in both CN and finance domains.

At the end of this chapter, we will synthesise existing DFC analysis in finance to assess and identify shortcomings in current applications of the DFC network in modelling financial markets. Simultaneously, we will critically evaluate the strengths and limitations of the DFC network to inform our study design. These gaps will motivate our methodology of this study to analyse the relationships between stocks in the market and relate market events to their dynamics.

2.1 Functional Connectivity

Detecting and quantifying relationships in multivariate data is crucial for analysing high-dimensional datasets, with increasing attention from both academic and industrial fields [18]. Analysis of statistical relationships is a prominent tool across multiple domains for data interpretation and harvesting significant insights [19].

FC was initially defined by Karl Friston as "temporal correlations between remote neurophysiological events" [20], a measure of the pairwise statistical dependence between brain regions that provides insights into the brain network dynamics during tasks or resting states [21].

We can select a measure of the FC to either quantify undirected statistical dependence using correlation or Mutal Information [22], or directed dependence using Transfer Entropy or Granger Causality [23] in Figure 2.1. These FC measures will be explored in *Section 2.1.1.1* and *Section 2.1.1.2*.

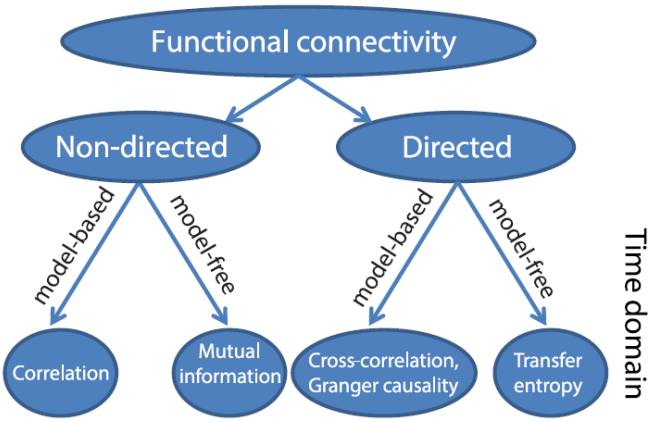


FIGURE 2.1. A taxonomy of popular methods for quantifying functional connectivity. *Reprinted from Figure 1 of Bastos and Schoffelen's study* [24].

2.1.1 Networks and modules

This section will explore both directed and undirected measures of FC and demonstrate how they can be used to construct the FC network. As part of the FC analysis, modules on the FC network will be identified and utilised to explain the relationships between functional groups within the system.

2.1.1.1 Undirected measures

An undirected measure is a metric that treats the relationship between two variables as symmetric, meaning that it does not distinguish between the direction of the relationship. It aims to capture the strength of the relationship between two variables without considering which one influences the other. For an undirected measure, M, always satisfies:

$$M(X,Y) = M(Y,X), \text{ for random variables } X \text{ and } Y. \tag{2.1}$$

correlation is a statistical measure that quantifies the relationship between variables. In a bivariate context, correlation determines the linear interdependencies between two variables, where a change in one variable is associated with a change in the other. The most well-known measure of the bivariate correlation is **Pearson's correlation coefficient**. Karl Pearson proposed a linear correlation based on covariance and the standard deviations of two serially independent variables, X and Y [25, 26]:

$$r = \frac{\sum(X_i - \overline{X})(Y_i - \overline{Y})}{\sqrt{\sum(X_i - \overline{X})^2 \sum(Y_i - \overline{Y})^2}}, \quad -1 \leq r \leq 1 \tag{2.2}$$

where X_i and Y_i are individual data points and $\bar{X}$ and $\bar{Y}$ are the means of X and Y respectively.

The statistical significance of the correlations was evaluated using the Bonferroni Correction. The **Bonferroni Correction** addresses the problem of multiple comparisons, i.e., the chance of finding at least one significant result just by random chance increases when performing several statistical tests [27]. For our analysis, the features are the stocks.

$$\text{Adjusted significance level for each test} = \frac{\alpha}{m} \tag{2.3}$$

where $\alpha = 0.05$ (rule of thumb) and $m = \dfrac{(\text{number of features})(\text{number of features} - 1)}{2}$.

Any correlations identified as non-statistically significant were replaced with zero using a 0.05 p-value threshold, as we aim to clean the network while maintaining statistical rigour.

Mutual Information (MI) is a widespread measure that quantifies the reduction in the uncertainty of one variable given another variable, defined as [22]:

$$I(X;Y) = \sum\sum p(X_i, Y_i) \log_2 \left(\frac{p(X_i, Y_i)}{p(X_i)\, p(Y_i)} \right) \tag{2.4}$$

where $I(X;Y) \geq 0$ and zero MI indicates that X and Y are statistically independent.

2.1.1.2 Directed measures

In contrast to the undirected measure, a directed measure attempts to infer directional influence beyond the interaction between two variables. For a directed measure, D, it always satisfies:

$$D(X \to Y) \neq D(Y \to X), \text{ for random variables } X \text{ and } Y. \tag{2.5}$$

The simplest directed measure is the cross-correlation with a **time lag** variable (k). Given the two time series X_i and Y_i, having constant means $\overline{X}$ and $\overline{Y}$ and constant variances σ_X^2 and σ_Y^2 respectively. The cross-correlation coefficient with a lag k at current time T is [28]:

$$r(k) = \frac{\sum_{i=1}^{T-k}(X_{i+k} - \overline{X})(Y_i - \overline{Y})}{\sqrt{\sum_{i=1}^{T-k}(X_{i+k} - \overline{X})^2 \sum (Y_i - \overline{Y})^2}}, \quad k = 0, 1, 2, \ldots \text{ and } -1 \leq r \leq 1 \tag{2.6}$$

where X_i and Y_i are individual data points and $\bar{X}$ and $\bar{Y}$ are the means of X and Y respectively.

Transfer Entropy (TE), is used to quantify the direct exchange of information between two variables evolving in time [23]. Formally, the TE from a source process X to a target process Y is defined as [29]:

$$T_{X \to Y}(k) = I(X_i; Y_{i+1} \mid Y_i^{(k)}) = \mathbb{E}\left[\log_2 \frac{p(y_{i+1} \mid y_i^{(k)}, x_i)}{p(y_{i+1} \mid y_i^{(k)})}\right] \tag{2.7}$$

where $I(X_i; Y_{i+1} \mid Y_i^{(k)})$ is the conditional MI between the previous sample X_i of the source and the following sample Y_{i+1} of the target, given the target's past k values $Y_i^{(k)}$. This measure excludes information from common history and effectively distinguishes driving and responding variables [23].

TE is equivalent to **Granger Causality (GC)** for Gaussian [30], exponential Weinman, log-normal, and certain parametrisations of Generalised Gaussian distributed variables [31]. GC was developed by Clive Granger, originally for measuring causal lag and strength in econometrics with the assumption of linear relationship. For random variables X and Y, he defined X is "Granger causing" Y ($X_i \Rightarrow Y_i$) if [32]:

$$\sigma^2(Y \mid U) < \sigma^2\left(Y \mid \overline{U - X}\right), \text{ where } U \text{ is all information in the universe.} \tag{2.8}$$

X_i is said to Granger cause Y_i if Y is better predicted using the universe of information than if X was excluded. Must note that GC does not imply true causality in the philosophical sense, but rather a predictive relationship based on past data [32]. This measure identifies directed functional interactions from time-series data, providing insights beyond correlations [33].

2.1.1.3 Network inference

The **functional network inference** aims to model complex networks of pairwise statistical relationships between entities in complex systems. This scope encompasses multivariate systems in which each entity generates its time series of activity, collectively forming a multivariate time series that captures the dynamics of the entire system. In these networks, nodes represent entities (e.g. stocks in a financial market, regions in neural fMRI brain imaging time series) with an FC measure of statistical dependence used to compute the weight of each edge [34]. The statistical dependence can be undirected or directed depending on the measure [35–40], as explored in the previous section.

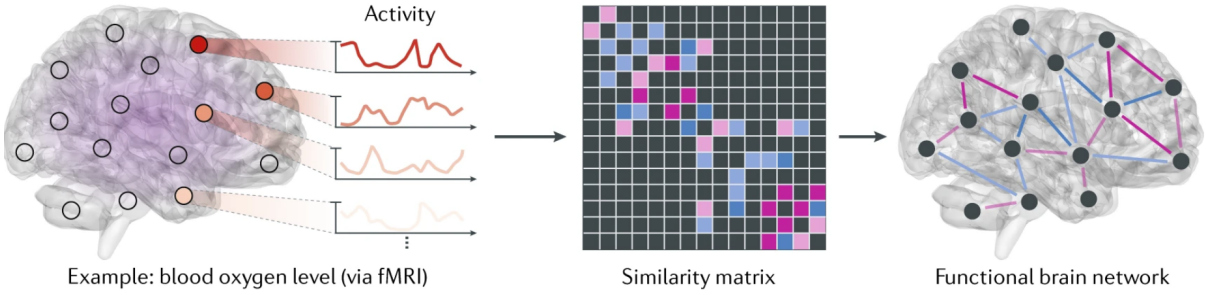


FIGURE 2.2. Measuring and modelling brain network function.
Reprinted from Figure 2a of Lynn and Bassett's study [41].

Conventionally, the FC network is visualised in a symmetrical correlation adjacency matrix, as seen in Figure 2.2. This example illustrates that the measurement commences with experimental data that specify the activity of neurons or brain regions, commonly measured using functional magnetic resonance imaging (fMRI). The similarities measured by MI, TE or GC between pairs of activity time series are encoded in a similarity matrix (coloured by similarity). This matrix defines an FC network constructed from the original measurements of neural activity.

In contrast, Novelli and Lizier found that multivariate TE captures essential features of all network structures for longer time series, and bivariate methods achieve better sensitivity for shorter time series [36]. Multivariate methods prompt the direction into effective networks. Effective connectivity refers to the influence that one neural system exerts over another [42]. Since we will concentrate on pairwise FC first, multivariate methods will be outside the scope of this study.

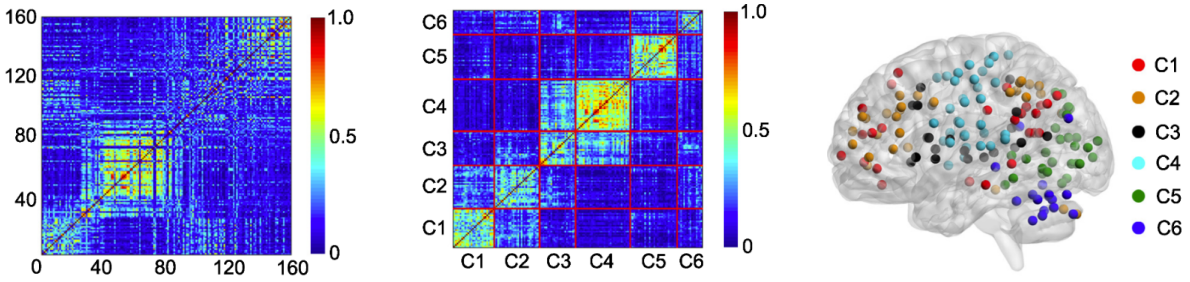
Given the FC network, various types of graph-theoretic analysis are possible to address questions about how the brain's statistics are structured. One obvious step is to examine whether we have clear modules where the regions are acting similarly.

2.1.1.4 Clustering

Many real networks have modular structures; thus, identifying modules on the FC network helps understand the relationships between entities and the connectivity patterns [43, 44]. A modular structure of the FC network can support the twin pillars of brain function, functional segregation and integration [37]. To uncover modules of the FC network, we will apply a clustering algorithm.

A clustering algorithm partitions data into groups (clusters) so that observations in the same cluster are more similar to each other than observations in different clusters [45]. In neuroscience, we aim to cluster rows of the FC network so that brain regions are grouped if they exhibit similar relationships (functionality) to those in the same cluster [46]. The clustering coefficient can also be considered a simple measure of the redundancy of a network, representing the repetition of identical elements within a system, which helps understand the functional robustness and stability of networks [38].

For example, by clustering FC measurements, the network in Figure 2.3 can be decomposed into modules that are expected to have high intrinsic connectivity and weak extrinsic connectivity [16, 47], presented as the squares with lighter shades in Figure 2.3(b). A strong intrinsic connectivity results in a high clustering coefficient, suggesting that nodes share standard functions, whereas a weak extrinsic connectivity indicates that these functions are segregated from each other [47]. FC modules can also be spatially distributed across the brain (Figure 2.3(c)). The choice of clustering algorithms will result in considerable differences in the module decomposition.



(a) Before clustering (b) After clustering (c) Brain functional groups

FIGURE 2.3. Brain FC network

Reprinted from Figure 3 of Jia and Gu's study [48].

Hierarchical clustering is a popular agglomerative clustering method that merges every data point initially as a separate cluster in a greedy fashion and eventually forms one single cluster based on the similarity criterion [49]. This approach, therefore, requires defining a similarity criterion for each row of the FC matrix. All pairwise similarities are then computed to generate an $N \times N$ similarity matrix and a dendrogram [47]. Hierarchical clustering does not require a predefined number of clusters, supporting a variety of distance measures, such as Euclidean, Manhattan, and cosine, making it adaptable to different types of data and applications [47]. On the other hand, it heavily depends on the choice of distance measure, which may result in different dendrogram structures, and it is sensitive to outliers that can misclassify clusters and affect the entire hierarchy [50].

Another mainstream clustering method of quantifying modularity is **K-means analysis**, where it partitions the N-dimensional population into K sets, and it is required to define a constant for the number of clusters that will be extracted from the dataset [51]. K-means clustering is efficient and straightforward to implement, with scalability and interpretability, making it widely applicable in various domains [52]. Despite this, it requires a predefined number of clusters, which can be challenging to determine optimally beforehand [53]. The algorithm is sensitive to the initialisation, and it assumes spherical clusters that may not be true in real-world datasets [54].

2.1.2 Dynamic Functional Connectivity

Given the dynamic nature of brain activities, it is anticipated that the FC network will exhibit variation over time [17, 35, 36, 40, 46, 49, 55–62]. Hence, this section will showcase how a DFC network characterises the relationship between entities by the change in temporal FC networks over time.

The most frequently performed approach to infer a DFC network is the **Sliding-Window analysis**, which can detect temporal fluctuations in brain connectivity based on FC from different time windows [17, 55–57, 61, 63]. Figure 2.4(A) illustrates a static FC matrix computed based on correlations of the fMRI time series, as explored in the previous section. We then apply the Sliding-Window approach to the time series by shifting a fixed-duration window that can be overlapped or non-overlapped between successive windows. An FC network is obtained per window to capture the temporal structure of the relationships between brain regions, as shown in Figure 2.4(B).

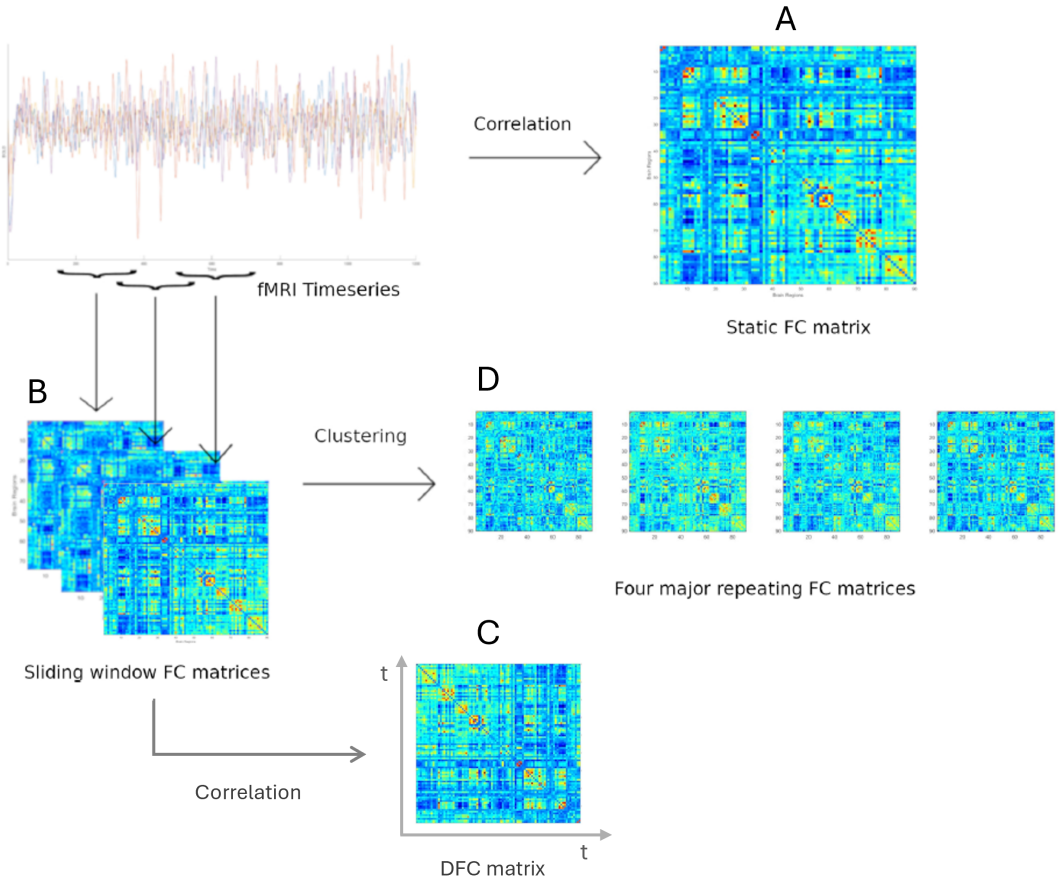


FIGURE 2.4. Static and dynamic FC matrices on fMRI time series.
Adapted from Figure 1 of Menon and Krishnamurthy's study [64].

By computing the correlation between these windowed FC networks, a DFC network is constructed in Figure 2.4(C). The DFC network is time-by-time, where each data point presents the correlation between a window pair. For example, the value with the coordinate (t_1, t_2) in the DFC network presents the correlation between the two FC networks for time windows that begin at t_1 and t_2. Based on the DFC network, we can assess the similarities between the temporal FC networks.

Similar windowed FC network could repeat over time, representing patterns of brain regions exhibiting similar dynamic behaviours [17, 56, 57, 62]. The windowed FC networks are clustered to obtain major repeating FC networks, referred to as states, as seen in Figure 2.4(D). We can apply the clustering algorithms explored in *Section 2.1.1.4*. Notably, most empirical studies conduct K-means clustering to identify states. States are the average patterns of clusters identified, and they usually represent a particular condition of the brain, such as the resting state and epileptic state [57].

With FC states identified, we are interested in determining the active time of their occurrences, as it would indicate state transitions and facilitate the interpretation of results. Beyond observing brain states and recurrences individually, many empirical studies of DFC pursue to model transitions of brain states [17, 35, 36, 46, 57, 59–61, 65]. Each state will have a binary active time vector indicating when it is active (1) and inactive (0). The active time vectors of all states will be plotted against time, and thus, we can observe the change in states, as seen in Figure 2.5.

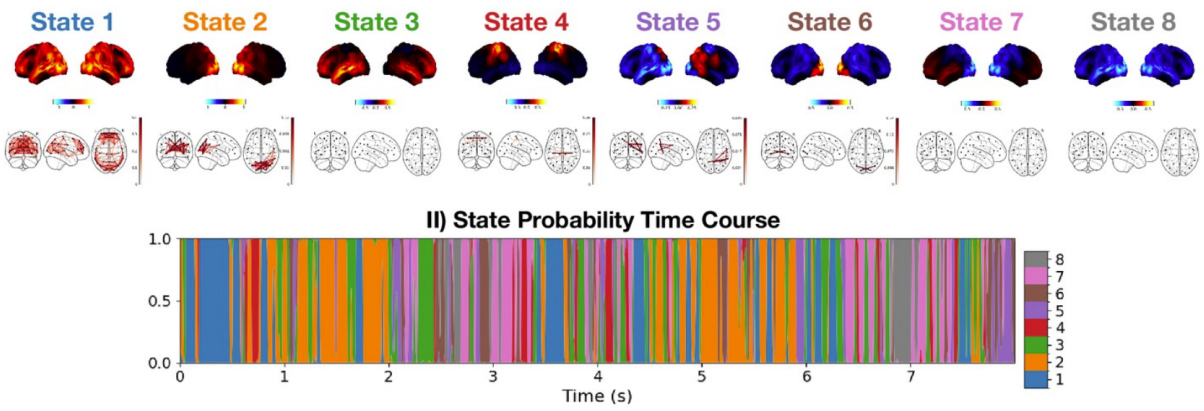


FIGURE 2.5. Dynamic network detection.

Reprinted from Figure 2A of Gohil et al.'s study [60].

Nevertheless, the limitation of Sliding-Window analysis is the use of fixed window length, a trade-off between specificity and sensitivity [61]. An ideal size would be sufficiently large to accommodate the relatively slow frequencies of the data while sufficiently small to be sensitive to transient changes in FC [17, 65]. Many empirical studies found that it often produces negative correlation coefficients from "out of phase" observations in the window [56, 65, 66], which may confound the identification of consistent patterns [67].

Based on the Sliding-Window analysis, we can only observe transitions between windows where it is possible that an event occurred in the middle of a window and caused the shift in brain states. On the other hand, the **Hidden Markov Model (HMM)** is a technique that can identify states at each time step without assuming a fixed window length. The HMM is a probabilistic model wherein observations are considered to be generated from a process that moves among unobserved states [59, 68, 69].

Thus, an HMM is a doubly stochastic process [68]:

(1) an underlying Markov chain, $Q = (q_1, q_2, ..., q_T)$ of hidden states, with known transition probabilities, $a_{ij} = P(q_{t+1} = j \mid q_t = i)$, and

(2) a sequence of observations, $O = (o_1, o_2, ..., o_T)$, each generated from the current hidden state based on the observation probability distribution, $b_j(o_t) = P(o_t \mid q_t = j)$.

An HMM involves estimating the state properties, state transition probabilities, and the active time point for each state [57, 59], as seen in Figure 2.5. When modelling for state transitions, the HMM labels a sequence of observations with hidden states by finding the state that is most likely to generate the observed data. Based on the mathematical definition of the HMM, if we are in state j, then the chance of seeing a particular observation o_t is given by a zero-mean Gaussian, whose spread and orientation are determined by the covariance matrix Σ_j.:

$$P(o_t \mid q_t = j) = \mathcal{N}(0, \Sigma_j) \tag{2.9}$$

An HMM can effectively model transient changes in the time series and segment the data into states in a completely unsupervised fashion [69]. In situations when there is limited prior knowledge about the underlying physiological state, unsupervised methods are instrumental. HMMs are well-established tools with widespread applications in biomedical fields, and they have been utilised in sleep staging, nonstationary biomedical signal segmentation, brain-computer interfacing, and speech recognition [69]. Nevertheless, HMM possesses limitations, such as difficulties in capturing long-range temporal dependencies and the assumption that states are mutually exclusive [59].

2.2 Financial markets

Similar to CN, the financial market can be viewed as a complex dynamical system [70] and is innately influenced by a series of macro- and micro-economic factors, such as fiscal and monetary policies, government decrees, business operation policies, and investor psychology [71–73]. We aim to understand the relationships between stocks and how they change over time, which traditional financial approaches in *Appendix A Traditional approaches to studying markets* do not focus on [71, 74]. Raised from these gaps, CN methodologies are specifically designed for this and, thus, have been utilised to some extent in finance. This section will assess the nature of stock data and explore current applications of DFC methodology in finance to identify gaps that motivate our study.

2.2.1 Stock market data

Stock markets often indicate changes in economic growth, consumption, and investment patterns, and they inherently exhibit noisy data from high-frequency trading. Noises are short-term fluctuations caused by behaviour quirks and measurement errors that do not reflect any change in stock prices. In addition to noises, volatilities, which are changes in stock return over time caused by news events, political developments, or abrupt economic changes, can distort trend signals and lead to false positives.

The daily closing price is the most commonly used by academia and industry [75, 76]. The closing price more accurately reflects the stock's value for the day, as it consolidates all trading activities and information throughout the day. A typical stock dataset consists of the daily closing prices of S&P 500 stocks from major stock markets, such as the New York Stock Exchange (NYSE) and the National Association of Securities Dealers Automated Quotations (NASDAQ) [77–81]. The time series is usually preprocessed using the logarithmic transformation of price differences (log return) to mitigate nonstationarities [78, 81, 82]:

$$x_n = \ln(S_n) - \ln(S_{n-1}), \text{ where } S_n \text{ is the closing price of } n^{th} \text{ trading day.} \tag{2.10}$$

Building robust models was particularly challenging due to the nonlinear and nonstationary nature of the financial data, commonly stocks, market indices, commodity futures, and foreign exchange [72, 73, 77]. This step helps standardise the properties of the data, ensuring that statistics are comparable across different time windows [83]. This step is vital if the price has fluctuated to other levels in various periods, and it also accounts for the general gradual rise in prices over time.

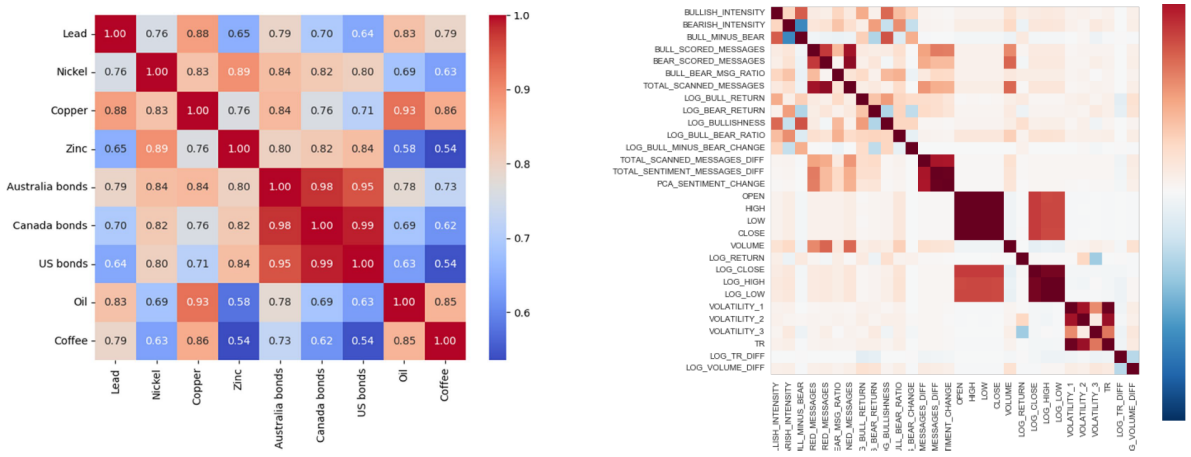
2.2.2 Market dynamics

To identify gaps in current approaches, we will showcase some examplar of applications of DFC methodology in financial markets. This section will uncover the DFC methodology by examining how prior studies constructed static FC and DFC networks using directed and undirected measures.

2.2.2.1 Static FC networks

Existing literature has adapted CN techniques for use in finance, where FC is predominantly assessed using measures such as MI, TE or GC [11, 77, 78, 84–90]. The statistical significance of measures is tested using surrogate datasets with similar statistical properties but without the interrelationships [84].

In Figure 2.6(a), Magzumov and Kumral's study investigated and computed a correlation matrix of commodity markets [91]. Stocks were manually ordered, revealing a block of high correlations for Australian, Canadian, and US bonds, indicating that they exhibit similar price movements. However, this study did not discuss the relationship between groups of commodities, for example, the correlation between metal and bond commodities. Moreover, a niche focus on commodity markets cannot explain the network structure of the entire stock market. Alternatively, Manfield et al.'s study in Figure 2.6(b) demonstrated a correlation matrix of market features from both social media and finance perspectives rather than individual stocks. This approach addresses the overall market structure but does not offer a direct view of the shifts in stock relationships.



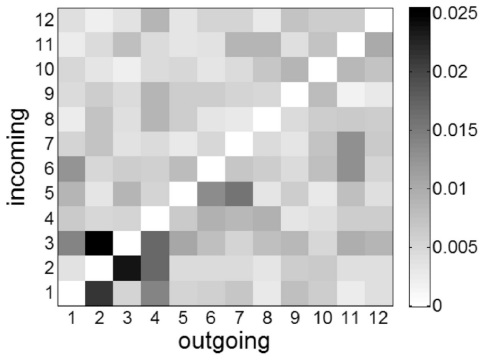
(a) Correlation matrix of commodity markets. *Reprinted from Figure 4 of Magzumov and Kumral's study* [91].

(b) NASDAQ correlation matrix of social & financial features. *Reprinted from Figure 4 of Manfield et al.'s study* [92].

FIGURE 2.6. Examples of undirected FC matrices in finance.

Similar to undirected measures, many studies have applied directed measures, such as TE, focus solely on FC without further exploring modules [78, 84, 86]. Nevertheless, they primarily aimed to investigate the information transfer between individual stocks rather than their classification by sector.

Figure 2.7 [78] used the daily closing price of 12 stock indices in Figure 2.7(b) to infer an effective (directed) matrix based on bivariate TE in Figure 2.7(a), quantifying the information flow between different stock indices. According to the colour scale, darker shades indicate stronger information transfers. The directed matrix is asymmetrical because the effects of incoming and outgoing flows for the same stock are different. For example, there is a strong information transfer from stock 1 (NDX) to stock 2 (NYA) but not in the other direction. It is also possible to identify a strong information transfer in both directions, for example, between stock 2 (NYA) and stock 3 (GSPC).



(a) Transfer entropy for the closing price data.

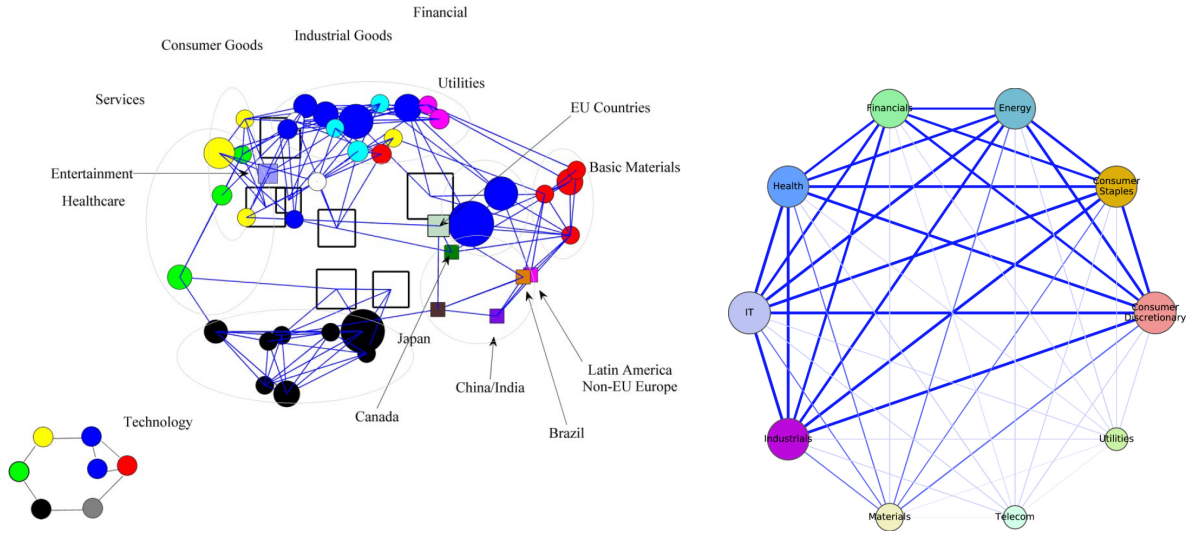
America	1	NDX	US
	2	NYA	US
	3	GSPC	US
	4	GSPTSE	Canada
Asia	5	HSI	China
	6	KS11	Korea
	7	N225	Japan
	8	BSESN	India
Europe	9	ATX	Austria
	10	FCHI	France
	11	FTSE	UK
	12	SSMI	Switzerland

(b) List of 12 stock indices.

FIGURE 2.7. An example of a directed FC matrix.
Reprinted from Figure 5 of Mao and Shang's study [78].

All studies above were not designed to classify stocks into market sectors or analyse the relationships between market sectors. Analogous to applications in CN, **clustering algorithms** are utilised in financial FC to group stocks that demonstrate high correlations, with popular clustering algorithms being hierarchical and K-means clustering [79, 80, 93], as explored in *Section 2.1.1.4 Clustering*. "Modules" in a financial FC matrix reveal clusters of stocks that exhibit strong correlations based on price movements. They often align with specific **industry sectors** as stocks within the same sector tend to correlate due to similar market forces and business environments [7, 8, 94, 95]. For example, companies in the oil industry, such as Chevron, Texaco, and Exxon, form a closely related cluster because they are affected by similar market forces, including global oil prices [94]. However, clustering can alternatively group stocks according to other factors [96], such as market capitalisation and geographical location.

With hierarchical clustering, the **hierarchical importance** can be identified and reveal the potential relationship between sectors, where major leading sectors act as "central hubs" with a significant amount of connections to the rest of the market. These central hubs are typically market-driving sectors, such as Financial Services and Energy, that play a central role in the market [95].



(a) Correlation network structure of the stock market. *Reprinted from Figure 2 of Leibon et al.'s study* [80].

(b) MI network of the stock market for year 2013. *Reprinted from Figure 7 of Liu et al.'s study* [97].

FIGURE 2.8. Examples of undirected FC matrices in finance.

Figure 2.8 showcases two studies that investigated modules of the market network based on undirected measures. Figure 2.8(a) illustrates the connectivity between market sectors (circles) and geographical regions (squares). Edges (blue lines) connect cluster centroids that nearby nodes represent highly correlated groups, for example, Utilities and Basic Materials. The cyclic arrangement ("ring" topology) of the macro-clusters network supports the phenomenon of capital flowing sequentially through sectors over the economic and investment cycles, referred to as **sector rotations** [80].

Sector rotation might be a factor that causes the relationships between sectors to change. It is an investment strategy that involves periodically shifting portfolio allocations between different market sectors to capitalise on their volatility over the economic cycle and momentum signals. Figure 2.9 illustrates the cumulative return index of Portfoilo-US and SPY ETF between 1999 and 2019 while highlighting the US Federal Reserve (FED) policy regimes of easing and tightening. Portfoilo-US is the cumulative value of a monthly rebalanced momentum portfolio. SPY ETF is the cumulative total return of the S&P 500 index.

Figure 2.9 evidences that the sector rotation strategy in the US provided both higher cumulative returns and smoother drawdowns (fall in investment value) than a passive S&P 500 investment during the given period. Therefore, there are significant incentives for market participants to engage in sector rotations, which may lead to the reoccurrence of particular market structures.

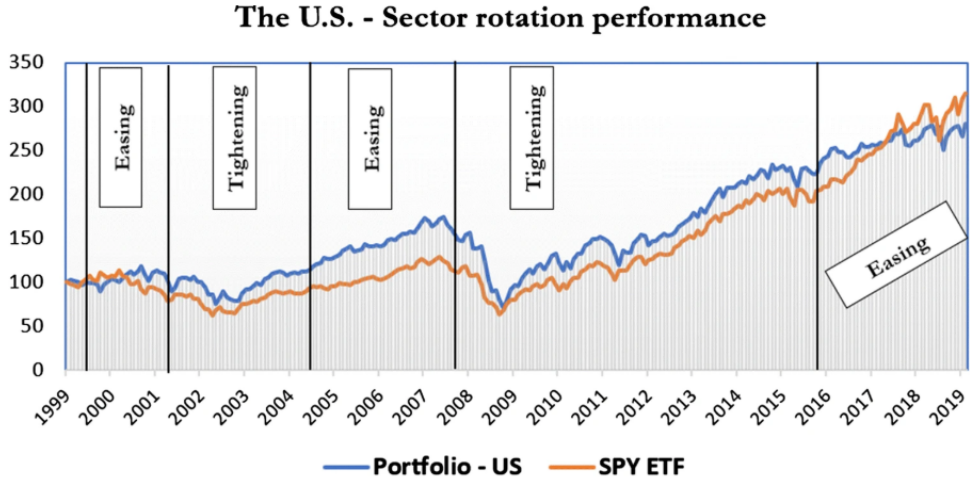


FIGURE 2.9. US momentum-based sector rotation strategy performance.
Reprinted from Figure 2 of Alexiou and Tyagi's study [98].

Nonetheless, the multi-dimensional view of the market network is challenging to decipher. Figure 2.8(b) provides a more intuitive way of visualising the market network, focusing on the relationships within market sectors. In this study, thicker edges represent higher MI between sector pairs, indicating that intra-day return dynamics in these sectors tend to co-move strongly. This network identifies seven "hub" sectors: Industrials, IT, Health, Financials, Energy, Consumer Staples, and Consumer Discretionary, as every pair shares a relatively high MI.

The key limitation of undirected measures is that the market network cannot indicate the spillover, which is the flow of influence from one sector to another. Spillovers help us understand how information or shocks propagate from one sector to another, which is essential for portfolio construction and diversification, market integration and regulation, as well as behavioural and structural insights [99].

Sandoval's study was conducted on the daily log returns of the top 197 most extensive financial-sector stocks at the NYSE by market capitalisation from January 2003 to December 2012 [87]. He applied the Effective TE (ETE) with a one-day lag to infer a directed network, as shown in Figure 2.10. ETE is a cleaner measure than TE as it subtracts the average TE from a false causal link by computing TE on randomly shuffled surrogates of each time series. Figure 2.10 focuses on geographical region clustering of the network: US (1-79), Canada (80-89), Europe (91-152), Japan (153-165), Hong Kong (166-174), Singapore (177-179), and Australia (180-197) [87].

The off-diagonal, brighter blocks reveal directional flows between regions, for example, from North America (the US and Canada) to Europe. Although we expect to compute an asymmetrical network from directed measures, Figure 2.10 seems highly symmetrical. This observation could be due to mutual spillovers and strong within-cluster coupling that the influence on regions from each other is almost equal.

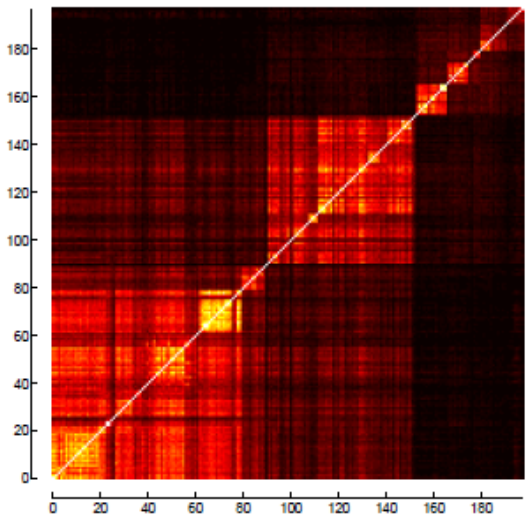


FIGURE 2.10. Effective Transfer Entropy from lagged to original stocks.
Reprinted from Figure 9b of Sandoval's study [87].

Correlation matrices in Figure 2.6 and Figure 2.7 appear to be easier to gauge the statistical relationships between stocks and sectors both qualitatively, through cluster position and colour scale, and quantitatively, through the correlation scale. As the matrix representation presents more insights than maps with nodes and edges in Figure 2.8, this study will adopt this approach in visualising our market networks.

2.2.2.2 DFC analysis

In the previous section, we demonstrated how prior studies inferred the static market network using both undirected and directed measures with an analysis of the identified modules. As we discovered that sector rotation may result in reoccurring market structures, in this section, we are interested in uncovering the dynamics of the market network via the DFC analysis.

Based on existing studies in finance, only a few have attempted to investigate market states, and barely any studies analyse state transitions [90, 100]. Across all these studies, the general procedures were Sliding-Window analysis to capture the temporal market structure, computing similarity measures on these window networks to construct a DFC network, and clustering the DFC network to identify states. Lastly, studies obtained the active times of each state by applying statistical methods such as Markov processes. We will showcase the application of each step in financial studies.

Sliding-Window analysis

Similar to CN, the Sliding-Window analysis is extensively utilised to capture the dynamics of the financial complex network [9, 40, 80, 87–90, 93, 100, 101]. Given the quantitative nature of finance, studies often involve statistical methods and centrality analysis to quantify uncertainties and similarities. Continuing with Sandoval's study from Figure 2.10, he derived pairwise ETE over 6-month windows [87]. As the window shifts in time, the windowed ETE reveals the changing network of stocks, as seen in Figure 2.11. Brighter colours represent higher values of ETE and vice versa. Per observation, there were highly similar windowed networks in Figure 2.11. However, Sandoval did not assess any similarity measures between different windows to compute a DFC network nor identify market states.

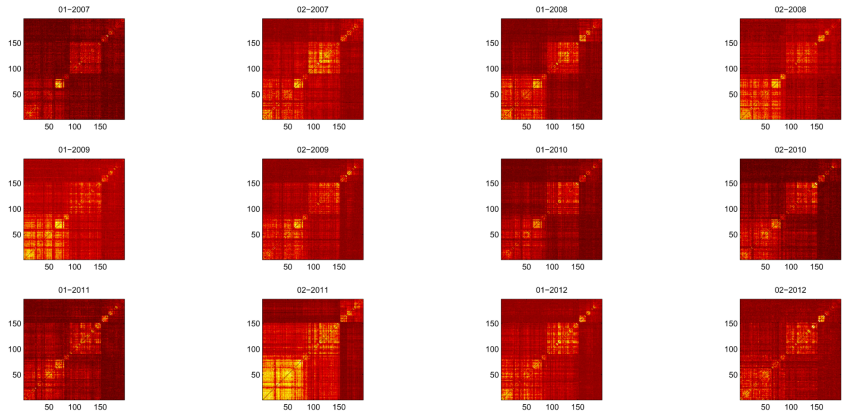


FIGURE 2.11. Effective Transfer Entropy from Sliding-Window (2007-2012).
Reprinted from Figure 18 of Sandoval's study [87].

DFC network

Zhao et al.'s study was conducted on daily log returns of 322 S&P 500 stocks from 1st January 1994 to 1st January 2014 [100]. They constructed a DFC network based on the pairwise distance, $D_{t,t'}$, among stock return correlations and a Sliding-Window analysis over 1-month windows. A smaller value of $D_{t,t'}$ indicates a higher similarity in the FC networks.

Their DFC network in Figure 2.12 exhibits sharp ridges ($D_{t,t'} > 0.5$) during crises, such as the Dot-Com Bubble in 2000 and the Subprime Mortgage Crisis in 2007, and remains stable ($D_{t,t'} < 0.1$) in normal periods. This observation suggests that the network structure reconfigures dramatically in just one month during crises.

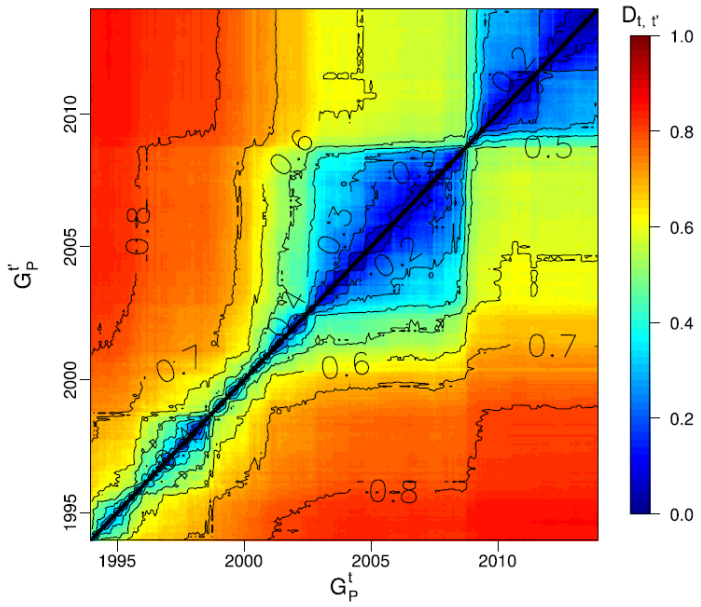


FIGURE 2.12. The distances between all window networks.
Reprinted from Figure 14 of Zhao et al.'s study [100].

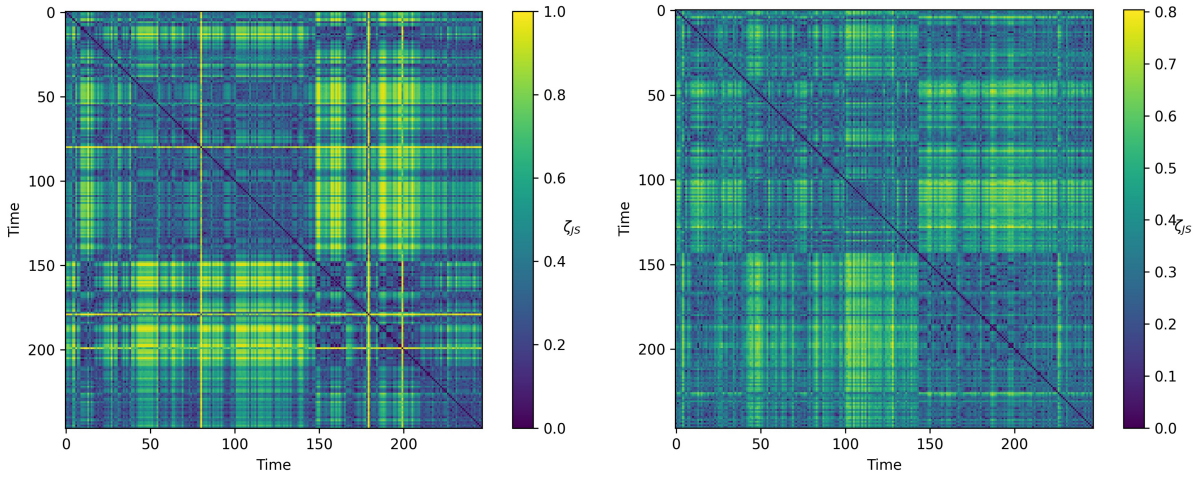
This study has attempted to model the market states of the two crises. However, the map representation can only present the market structure of an individual event rather than distinguishing unique "crisis" and "non-crisis" states to identify the transitions between them.

Market states and transitions

Pérez and Corona's study was conducted on the daily log returns of the 24 most extensive US stocks by market capitalisation from 7th March 2018 to 7th March 2019 [90]. They used an ordinal-pattern-based measure called transcript-synchronisation, which quantifies the strength of synchrony between every pair of stocks to construct the static FC network.

For each trading day, they computed the histogram of degree and eigenvector centrality (EVC) to measure how well-connected each stock is within the FC network. They then calculated the Jensen-Shannon (JS) distance between every pair of daily histograms to construct a network with days as nodes and edges representing the JS distance between nodes, as seen in Figure 2.13.

The high JS distances (brighter off-diagonal) represent days in the decentralised season, where potential hub companies dominate the market. The coherent blocks with low JS distances along the diagonal correspond to days within the centralised season, where the market moves more diffusely. This time-by-time matrix is similar to a DFC network, but it focuses on capturing similarity in the node's properties instead of relationships in the network.



(a) JS distance matrix in degree.

(b) JS distance matrix in EVC.

FIGURE 2.13. An example of DFC in finance.
Reprinted from Figures 7 and 8 of Pérez and Corona's study [90].

Hierarchical and K-means clustering were applied to identify market states from the JS distance network. A Markov model was used to assign market states and identify outliers. Each state thus corresponds to a characteristic degree of market-wide synchronisation. There is a state assigned with the highest synchronisation, which is active during the decentralised season. Similarly, there is another state that represents the centralised state with the lowest synchronisation. All others are transitioning states with different levels of synchronisation.

The transition of states is visualised in a time vector, as seen in Figure 2.14, where it illustrates the market moves from one state (y-axis) to another on a timeline (x-axis), with outliers highlighted in red. It is possible to align the occurrence of each state with known events, but they only focus on centralised and decentralised seasons.

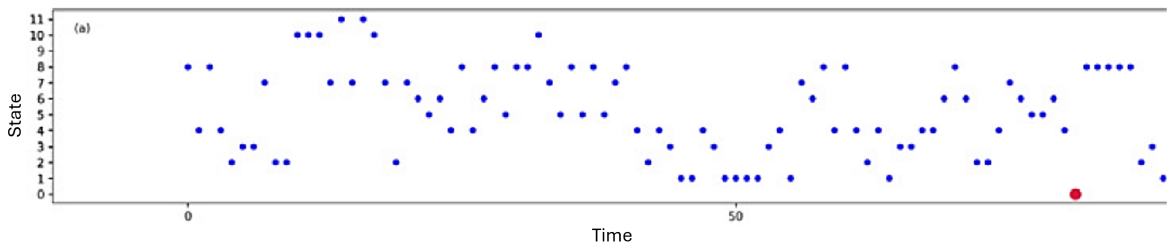


FIGURE 2.14. example of a time vector of states.

Excerpted from Figure 12a of Pérez and Corona's study [90].

2.2.3 Gaps of current work

Based on this literature review, we found that there exist many gaps in analysing the relationships between stocks and relating their dynamics with well-established market events. Traditional approaches in *Appendix A* focus on price predictions. Current adoptions of the DFC network approach in *Section 2.2.2* either focus on the static market structure or changes in the market structure before and after a few selected crises. Most of these studies focus on investigating cross-sector relationships to identify hub sectors rather than changes in the overall market structure over time.

For all studies, the Sliding-Window analysis remains a significant limitation. It is possible that a market event occurred between two windows, making it challenging to capture the complete information about this event. Although having a smaller window duration would help with this issue, the trends may be distorted by the greater impact of noises within the window. Furthermore, in analysing real-life data, latent factors that cannot be measured or observed directly may hinder trends.

Only Pérez and Corona's study was designed to uncover the shifts between market states. The use of degree and EVC measures to identify days in centralised and decentralised seasons does not uncover market sectors, as they did not conduct a clustering on the static FC network. Using the JS distance to measure differences between networks was not indirect, whereas applying a more direct similarity measure, such as correlation or MI, may provide a more intuitive and practical approach. The analysis of this study was based on histograms of the degree and EVC vectors from networks, where the binning process of histograms is subject to uncertainties and can affect the results [**90**].

Henceforth, there are still plenty of opportunities for novelty in modelling the structure of relationships between stocks and analysing the change in these relationships over time. Notably, further efforts are required to enhance the validity of the study design, including measures to infer the FC network by identifying market sectors, approaches to capture the temporal market structure and identify state transitions. There is also a need to increase the interpretability of results obtained using network approaches in finance.

CHAPTER 3

Financial Market Dataset

In this chapter, we aim to explore the nature of our dataset and assess the statistical assumptions and potential modular structure for conducting further analysis. This study is designed to analyse relationships between stocks in a single market across a wide range of industries and over a sufficiently long period to allow DFC analysis to identify recurrent patterns in different periods. The NYSE is a suitable candidate for meeting this criterion, and it provides accessible data.

3.1 NYSE data

The data was retrieved using the `yfinance` Python package, which allows the download of historical market data from Yahoo! Finance. We select 50 key stocks listed at the NYSE from 11 market sectors, considering their market capitalisations and data availabilities for consecutive trading days throughout the period. Yahoo! Finance is a well-established source of financial datasets [78, 80].

All data are downloaded and imported into a data frame, which is ready for further analysis. Our data frame consists of 8069 trading day daily closing prices from 23rd February 1993 to 12th March 2025 of the 50 stocks, where the index is the date and column names are stock names (8069 rows $\times$ 50 columns). These 50 stocks will be referred to as the system ("market") of our analysis.

Additionally, for comparative analysis of different approaches in identifying market state transitions, we will need to use the Chicago Board Options Exchange (CBOE) Volatility Index (VIX) data of the same period (8069 rows $\times$ 1 column). VIX, often referred to as the market's "fear gauge", is a real-time market index that measures the market's expectation of 30-day forward-looking volatility for the S&P 500 Index [102]. Hence, if we obtain a high correlation with VIX, it is suggested that our unsupervised predictions of active states reflect real-world market sentiment. Otherwise, we can flag a potential misspecification.

3.1.1 Preprocessing

Preprocessing should begin by omitting stocks with missing data, as they often alter the results from the underlying relationships in the time series, leading to unreliable trends and inaccurate predictions of market movements. Hence, it is ensured that the data of all stocks extracted should be numeric with no missing values. Further, the filtered time series will be transformed by the logarithmic differences of daily prices (*Equation 2.10*) to mitigate nonstationarities, as discussed in *Section 2.2.1*.

3.2 Stock price dynamics

Before model construction and analysis, it is essential to conduct a **Exploratory Data Analysis (EDA)** to critically examine the underlying dataset, including understanding the data structure and dynamics. Since FC analysis focuses on the statistical relationship in a highly volatile time series, it is a common approach to conduct a **Power Spectral Density (PSD)** analysis in neuroscience [103].

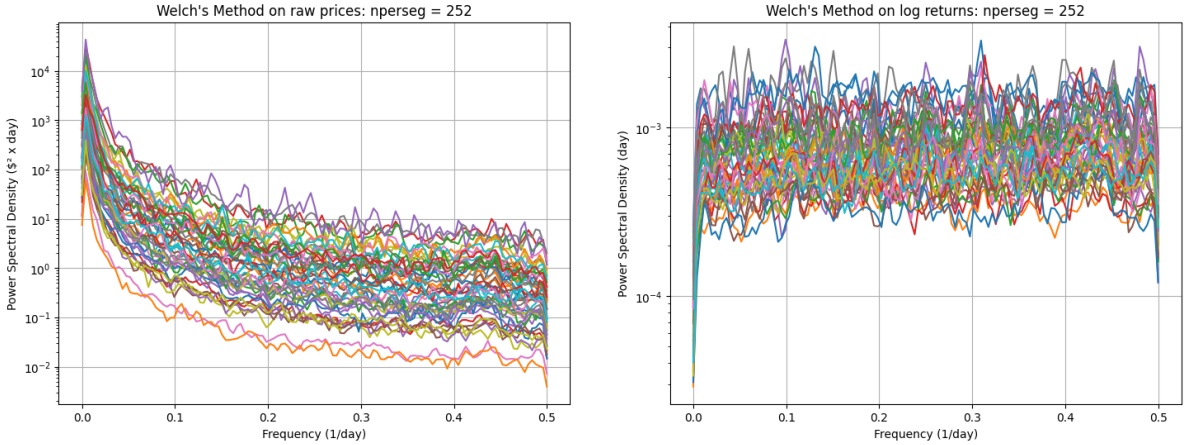
PSD is used to examine the nature of the time series and ensure that assumptions for statistical properties, such as stationarity (constant mean and variance), have been satisfied. Considering the stock data is also highly volatile and noisy, exhibiting temporal structure and complex dynamics, it is valuable to adopt PSD to this analysis. This study utilised Python SciPy's implementation of Welch's method (`scipy.signal.welch`).

Welch's method is an approach to estimating a signal's PSD by averaging periodograms from window segments to reduce its variance [104]. The full-time series is divided into window segments of length `nperseg`, which is the number of samples per segment. The windows are non-overlapping by default to keep the independence assumption between segment samples for computing statistical tests on connectivity measures in later sections.

The sampling frequency, `fs`, is set to 1, as we want the function to compute the PSD on a daily basis, aligning with the daily stock data. The number of samples per segment, `nperseg`, is set to 252 (the number of annual trading days) as it is the most common option in financial analysis [80, 93, 105], and we aim to maintain comparability with other studies.

We computed two different PSD graphs to give a complete view of the dataset. Figure 3.1(a) is a PSD graph plotted using the raw price data of all 50 stocks, showing a low-frequency dominance where every line spikes near zero frequency. This PSD graph suggests that slow fluctuations carry the most power, implying the presence of long-timescale trends.

As the raw price data is volatile and hard to assess trends, a PSD graph plotted using the log returns of all 50 stocks in Figure 3.1(b) shows an almost perfectly flat spectrum around a constant level (10^{-3} days), and hence, no dominant oscillations identified. The averaged PSD graphs were plotted based on the raw price data and log returns to glance at a clearer view of the market trend. See more details in *Appendix B*.



(a) Raw Price (b) Log Return

FIGURE 3.1. PSD of all stocks.

3.3 Modular structure

In this section, we aim to conduct a statistical analysis to determine whether there are systematic, low-dimensional drivers of market dynamics. A **Principal Component Analysis (PCA)** is undertaken to offer insights into the percentage of variance explained by each principal component (PC) and PCA loadings per PC. PCA is a classic statistical technique whose primary purpose is to reduce the dimensionality of a dataset while preserving the maximum variability ("information") as possible [106]. PCs are a new set of mutually uncorrelated variables transformed from the original variables.

The **percentage of variance explained by each PC** tells how much each PC captures the original dataset's total variability after applying PCA. We could potentially perform dimensionality reduction by maintaining only 80% of the underlying structure to improve generalisation since the stock market has 11 sectors. According to Yahoo!Finance [107], we set the number of PCs to be 11 using the Python package, `sklearn.decomposition.PCA`.

In Figure 3.2, we can see that the sum of all percentages of variances equals 79% of the original dataset's total variability, suggesting that the optimal clusters for this dataset are likely to be 11 as well. Furthermore, we can see that PC1 explains 29.11%, indicating its dominance over the other PCs and suggesting it may be the key driver of the market structure. After PC1, there is a steep drop-off with diminishing returns for each extra PC.

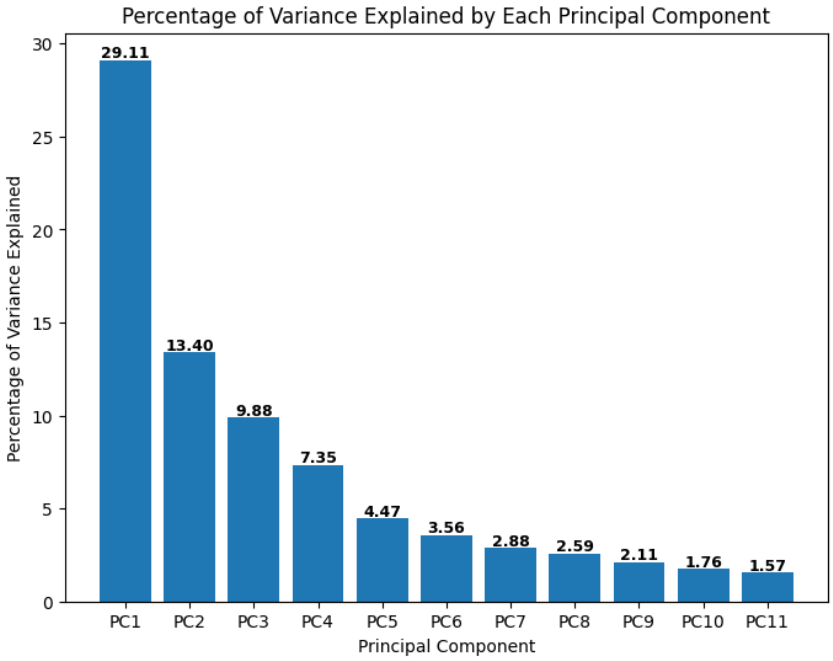


FIGURE 3.2. Percentage variance explained by each principal component.

To better understand the PCA results, this percentage of variance graph is used in conjunction with a PCA loading graph, as we are interested in determining the composition of each PC, expressed in terms of the loadings of the original variables. **PCA loadings** represent the correlation between stocks and the PCs. A high positive or negative loading indicates that the stock is making a significant contribution to this PC.

For example, in Figure 3.3, the stocks identified as dark red in PC1 are from the Financial Services sector, suggesting the Financial Services sector is the major contributor to PC1. We can also see that stocks on the top of PC1 are in blue, representing a negative contribution to PC1. Those stocks are from the Utilities, Healthcare, and Consumer Defensive sectors, with some stocks, such as AT&T Inc. (T) and Verizon Communications Inc. (VZ) from the Communication Services sector, represented in grey, indicating no contributions to PC1. Apart from those that are grey, all stocks are contributing to PC1 positively or negatively, with Bank of America Corporation (BAC) being the most significant contributor. This result is due to Bank of America's central and systemically important role in the US market [**108**].

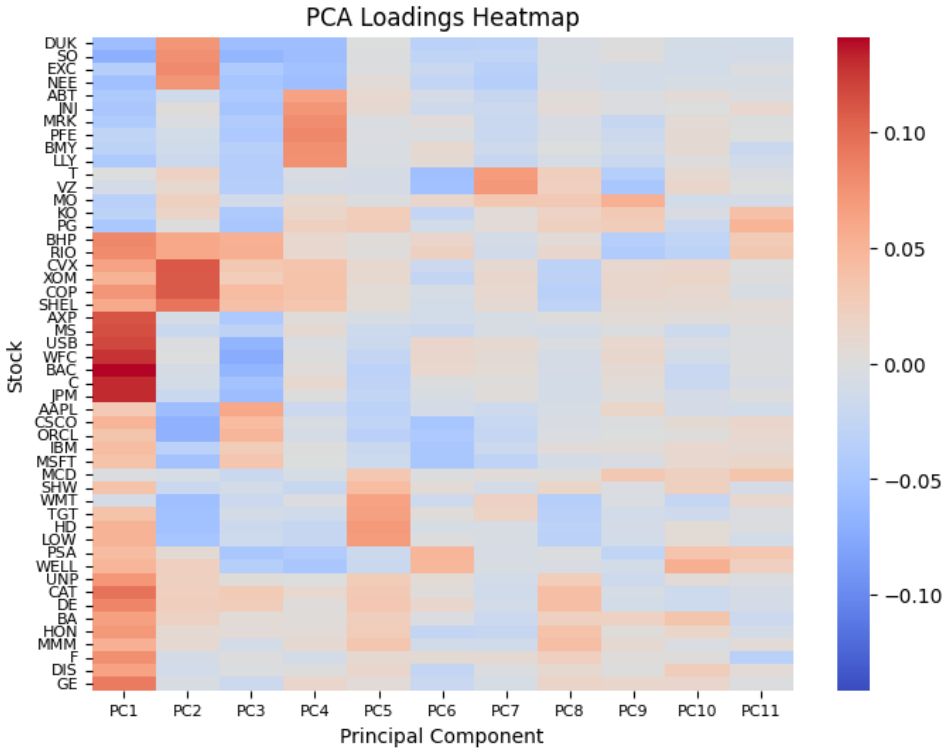


FIGURE 3.3. PCA Loadings per principal component.

Figure 3.3 also makes it clear to see the underlying relationships between market sectors, i.e., whether they will fluctuate in the same direction or exhibit no relationship, which can help us reaffirm our previous understanding of cross-sector relationships. Whilst the PCA suggests relationships, it does not specifically measure them nor identify any clustering within them. Therefore, we will construct an FC network to explore relationships between stocks and sectors.

CHAPTER 4

Static Networks in Financial Markets

In this chapter, we aim to use the preprocessed dataset from the last chapter to infer an FC network that characterises the structure of pairwise relationships between stocks and identify market sectors. Based on this FC network, we can highlight and analyse the market drivers and hub sectors.

4.1 Static FC network

After exploring and understanding the raw price dataset, we decided to proceed with their log returns as our analysis dataset for computing **FC**. We will start with the most basic statistical measure, **Pearson Correlation Coefficient**, to represent the FC of the stock price changes (log returns). A symmetric matrix visualisation of the **FC network** will also be presented as the outcome of this section.

Methodology

By calculating Pearson correlation coefficients between each pair of the 50 stocks as per *Equation 2.2* in *Section 2.1.1.1 Undirected measures*, FCs were identified as the bivariate linear relationships between stock returns. We then obtained a 50×50 symmetrical correlation matrix that can be visualised in a heatmap, as explored in Lynn and Bassett's study [41] from *Section 2.1.1.3 Network inference*.

To facilitate the presentation and interpretation of FC, we applied a hierarchical clustering algorithm (`scipy.cluster.hierarchy`) to reorder the stocks into their clusters on the FC network visualisation. **Hierarchical clustering**, as explored in *Section 2.1.1.4 Clustering*, is a popular agglomerative clustering method that greedily merges data points to form a single cluster based on the linkage criterion [49]. The linkage criterion was chosen to be Ward's method, that instead of merging two clusters with the smallest pairwise distance, it combines the pair that minimally increases the total within-cluster variance. This criterion relies on Euclidean distance to define variance.

As hierarchical clustering heavily depends on the distance measure, this may result in different dendrogram structures, and it is sensitive to outliers, which can misclassify clusters and affect the entire hierarchy [50]. Since the hierarchical clustering provided results for any number of clusters, we want to determine the optimal number of modules (clusters) that represent the market sectors on the FC network.

We applied three different methods to identify the most optimal number of clusters in our FC matrix:

- The **Silhouette Score** measures how similar an object is to its cluster compared to other clusters [109]. The score ranges from -1 to 1, where 1 indicates perfect clustering, 0 indicates overlapping clusters, and -1 indicates incorrect clustering.
- The **Calinski-Harabasz Index** is a ratio comparing cross-cluster separation (how far apart clusters are from each other) and within-cluster cohesion (how close points are within each cluster) [110].
- The **Davies-Bouldin Index** measures the average "similarity" between clusters, where similarity is the ratio of within-cluster distances to between-cluster distances [111]. A lower Davies-Bouldin Index value indicates better clustering.

The mode of these three methods was used to obtain cluster labels. 11 was identified as the optimal number of sectors. Taking the Silhouette Score in Figure 4.1 as an example, when the number of clusters is 11, it gives the first highest score. Lastly, we evidenced that every correlation was statistically significant using a Bonferroni correction, as explored in *Equation 2.3* in *Section 2.1.1.1 Undirect measures*.

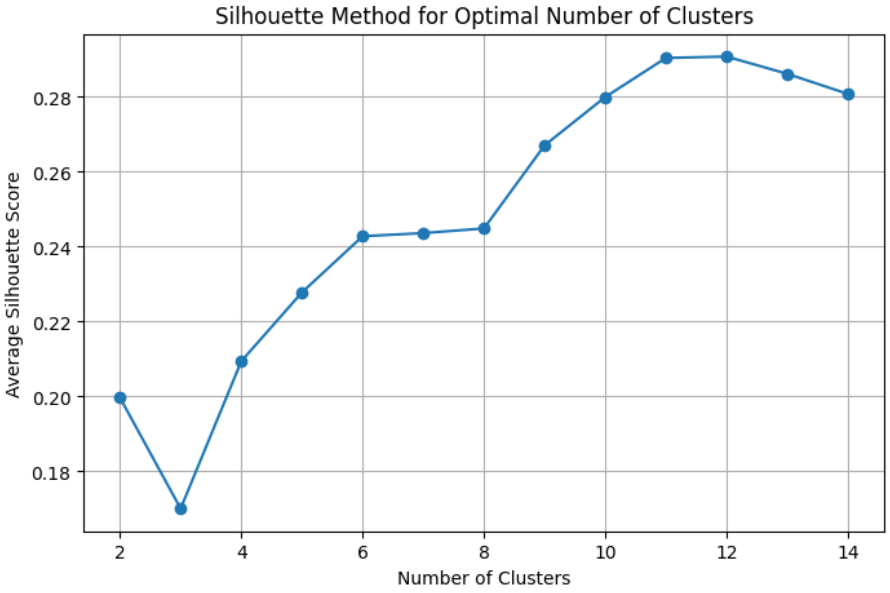


FIGURE 4.1. Silhouette Score of the optimal number of market sectors on FC.

Results

The dendrogram, output from the hierarchical clustering with 11 clusters, is visualised in Figure 4.2 to inspect the separation of stocks into clusters. The modules were identified on the FC visualisation bounded by red boxes. We have assigned market sector labels through manual identification from existing market sectors on Yahoo!Finance, see company names and market sectors of each stock in *Table C.1*, matching what we expected. Figure 4.3 is the static FC network over the 32 years. The rows and columns are reordered according to the 11 clusters identified, following Jia and Gu's study [48] (see Figure 2.3).

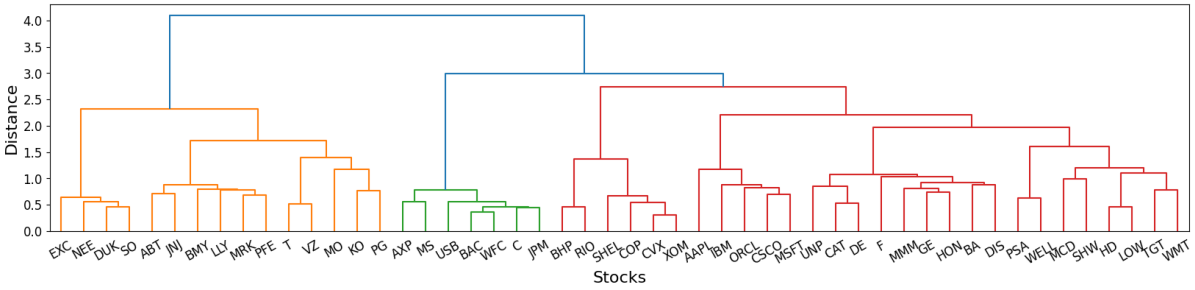


FIGURE 4.2. Hierarchical clustering dendrogram.

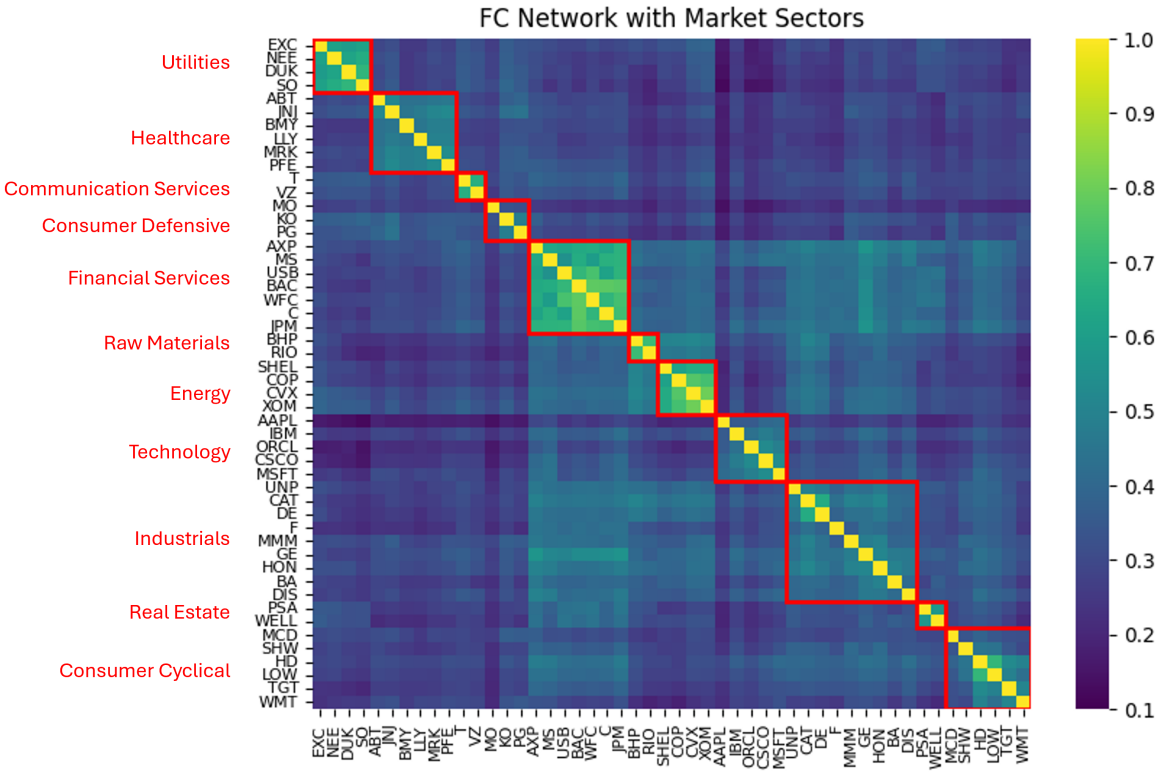


FIGURE 4.3. Static FC network on log returns.

There were 11 market sectors in Figure 4.3 static FC matrix, identified by the hierarchical clustering algorithm. This result also aligns with the 11 PCs identified in *Section 3.3 Modular structure*, reaffirming that 11 clusters were optimal in representing the original dataset while still maintaining meaningful interpretation. The market sectors were all aligned with Yahoo!Finance mostly corresponds to the sectors identified in prior studies, as shown in Figure 2.8, and explored in *Section 2.2.2.1 Static FC networks*, due to the different number of sectors.

There were only a few stocks misclassified into the wrong market sector. For example, Disney (DIS) should be classified under Communication Services, but it was misclassified under Industrials. This observation is meaningful, as Disney's daily returns are more correlated with the Industrials than its sector. A reason may be the complex nature of classifying sectors so that daily returns only capture the part of its business that is closer to being in Industrials.

From an individual stock's perspective, we observed that Apple (AAPL) has a relatively lower correlation with other stocks, as indicated by the darker bands. Apple is most uncorrelated with the Southern Company (SO) in Utilities and the Altria Group (MO) in Consumer Defensive due to their different sensitivities to macroeconomic factors, such as interest rates and inflation, as well as their exposure to economic cycles. They exhibit fundamentally distinct regulatory and competitive dynamics [**112**].

Regarding market sectors, Utilities, Financials, Raw Materials, and Energy sectors have revealed high within-sector correlations, suggesting potential stock comovements within each sector. The Financial Services sector is highly correlated with other clusters, visualised by the brighter horizontal and vertical bands around it. Particularly, Financial Services are more correlated with Industrials compared to other sectors. This observation aligns with prior economic studies that have found a connection between sectors and GDP growth, as well as interest-rate dynamics [**113**]. Additionally, Industrials often depend on bank lending and capital markets to finance their plants, equipment, and inventory. Intensive industrial activity typically means greater loan growth and fee income for financial institutions [**12**].

Furthermore, by examining market sectors collectively, Utilities, Healthcare, Communication Services, Consumer Defensive, Financial Services, Raw Materials, and Energy form a hierarchical structure, with Financial Services, Raw Materials, Energy, Technology, Industrials, Real Estate, and Consumer Cyclical in another structure (see in Figure 4.2). This result illustrates the two sector groups, highlighting the importance of Financial Services, Raw Materials, and Energy to the entire market as they are represented in both groups. Fluctuations in these three sectors could result in a market-wide shock [**114**].

4.2 Sector-level FC network

In the previous section, we analysed the static FC qualitatively by identifying the most obvious relationships between sectors. In this section, we aim to conduct a quantitative analysis of the entire static FC network by comparing it to its pairwise averaged FC network.

Methodology

We averaged the correlations between stock pairs within each sector (excluding the diagonal) and across sectors from the identified market sectors. The averaged correlations indicate how much information is lost or smoothed out by replacing every value with its sector average.

Results

Figure 4.4 presents a smoother depiction of the market, offering high-level insights into market sectors.

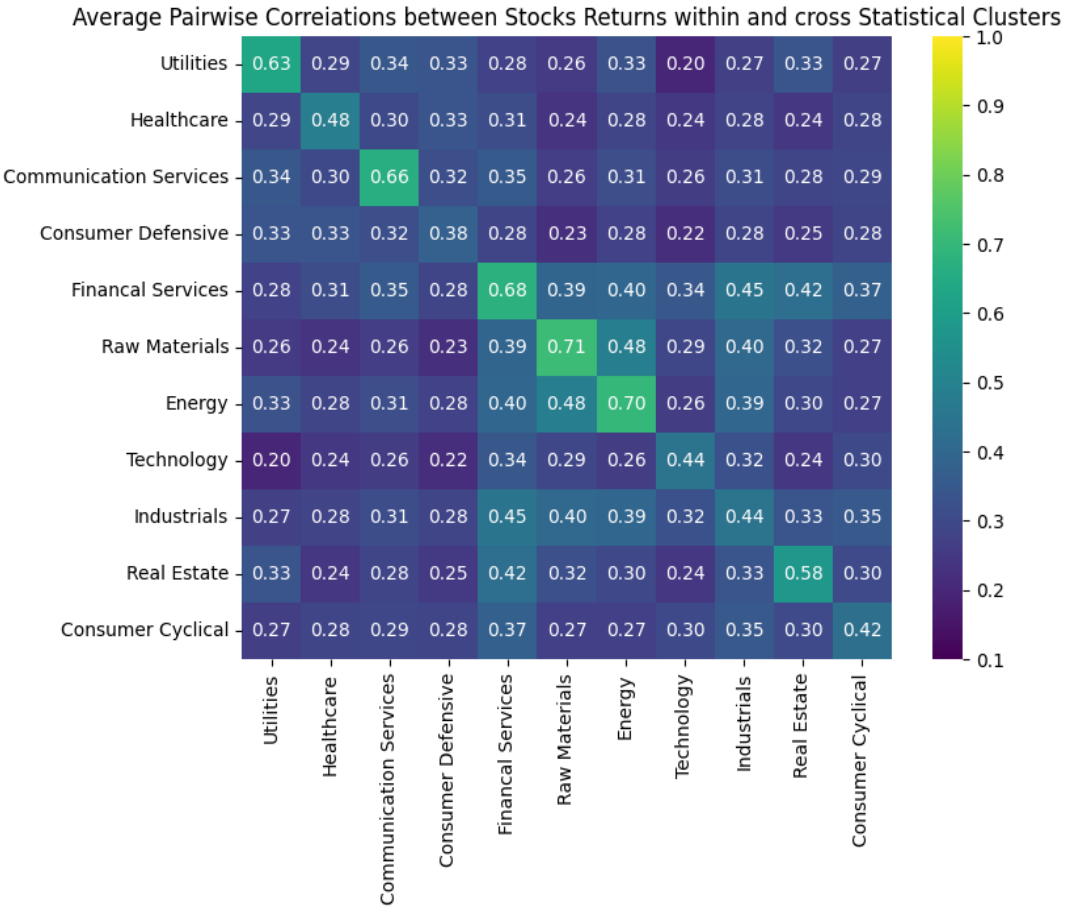


FIGURE 4.4. Average pairwise FC per market sector.

The Raw Materials, Energy, and Financial Services sectors exhibit stronger internal correlations in their dynamics, as evidenced by the highest within-sector correlations. This observation may be due to the uniform price movements in these sectors. The Consumer Defensive, Consumer Cyclical, Technology, and Industrials sectors have the lowest within-cluster correlations, suggesting that there may be market-driver stocks that move differently from other stocks within the sector. It is more challenging to make sector-wide generalisations across these sectors.

Turning our attention to cross-sector correlations, it is reaffirmed that the Financial Services sector is more correlated with the rest of the market, especially showing a higher correlation with the Real Estate, Energy, and Industrials sectors. Compared to the previous section, we can now conclude that the Financial Services sector correlates most closely with the Industrials sector, with a correlation coefficient of 0.45. It is also evidenced that the Technology and Utilities sectors are the most uncorrelated in the market, with a correlation of 0.2. Additionally, it is noted that the Raw Materials and Energy sectors are the most correlated in the market, with a correlation coefficient of 0.48. This observation was also evidenced in Magzumov and Kumral's study [91] in Figure 2.6(a) from *Section 2.2.2.1 Static FC networks*.

Since Figure 4.4 is based on average correlations, we want to investigate the amount of information omitted to ensure our conclusions remain valid. We then assess the **averaged FC matrix** in Figure 4.5.

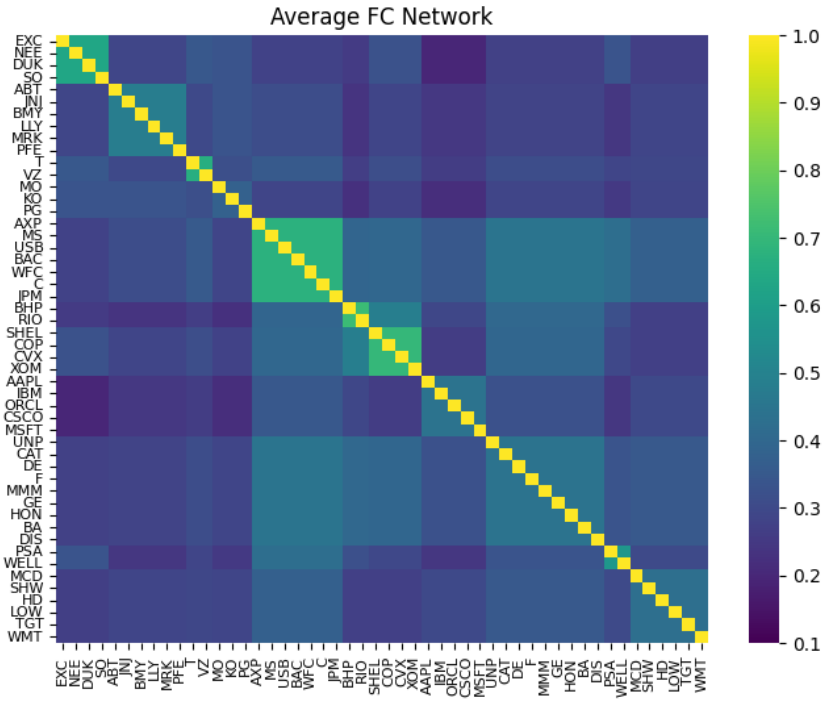


FIGURE 4.5. Average static FC network.

From here, the correlation differences between the original (Figure 4.3) and averaged (Figure 4.5) FC networks can be determined by taking the difference between the two matrices. The mean correlation difference of the entire matrix is 0.0312 (4 d.p.), suggesting that we are not omitting a significant amount of information by taking averages over sectors. This difference is roughly 10% of the typical correlation, indicating a high degree of low dimensionality in the dynamics.

Moreover, we can also examine the mean correlation difference (4 d.p.) within each sector, as visualised in Figure 4.6. It is noted that the Communication Services, Raw Materials, and Real Estate sectors have no difference in taking the averages. The Consumer Cyclical sector has the highest absolute difference from the original FC network, as it may contain sector-driving stocks that do not co-move with other stocks within this sector. Nevertheless, all differences are quite small, indicating that our previous remarks about cross-sector relationships are reasonable.

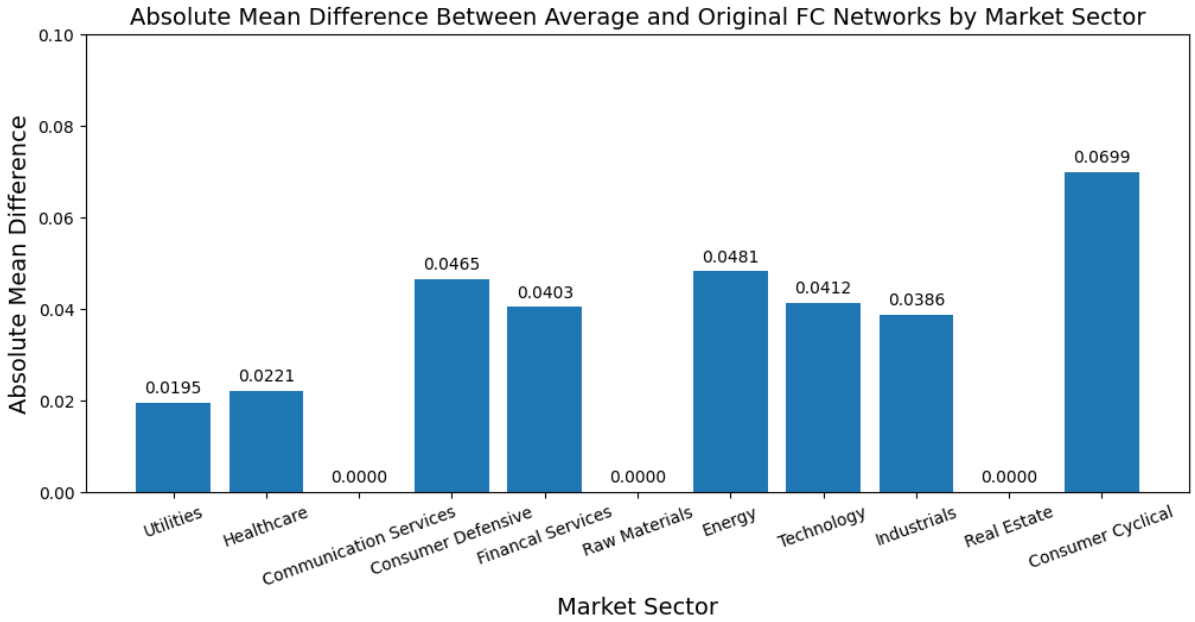


FIGURE 4.6. Mean differences in averaging FC networks per market sector.

Up to this point in our pipeline, we have a static FC network with identified modules, revealing the 32-year-averaged market structure. As we are interested in investigating and analysing the dynamics of the FC network, we will continue to unpack how the relationships between stocks change over time.

CHAPTER 5

Network Dynamics in Financial Markets

In *Chapter 4 FC Analysis*, we inferred and analysed the static FC network, specifically exploring the underlying relationships between stocks and sectors. However, the market is constantly evolving over time. This chapter aims to enhance our understanding of how these relationships vary over time, uncover common states for those patterns and identify when they recur.

5.1 DFC network

In this section, we will continue following the approach in Menon and Krishnamurthy's study [64] from *Section 2.1.2 Dynamic Functional Connectivity* to compute a DFC network based on Sliding-Window analysis. The **Sliding-Window analysis** is a general technique for examining how features evolve by dividing the time series into windows with a fixed length, where we identified that our study results were generally consistent across different window lengths.

Methodology

We selected the most common window lengths for financial analysis: 1 month (21 trading days) [7], 3 months (63 trading days) [113], 6 months (126 trading days) [77] and 1 year (252 trading days) [80, 93, 105]. As explored in *Section 2.1.2 Dynamic Functional Connectivity*, the algorithm loops over the entire time series and splits the data into designated window-sized batches. For each windowed data, we constructed an FC network the same way in *Section 4.1 Static FC network* and stored it into a list, as illustrated in Figure 2.4(B). Each **windowed FC network** is labelled by its **starting date**. Therefore, we can compute the pairwise correlation among all windowed networks to form the DFC network, as shown in Figure 2.4(C). The DFC is time-by-time, consisting of the starting date (yyyy-mm-dd) of each windowed FC network. For all FC network visualisations throughout the analysis, we set the same correlation limits to obtain a consistent colour bar, thereby ensuring comparability and interpretability.

Results

The result analysis in this section will focus on the **6-month window** as its output balances the data resolution and readability with all other window length outputs in *Appendix D DFC Networks with Other Time Windows*.

The DFC, based on the 6-month window length in Figure 5.1, illustrates the correlation between pairs of FC networks from different time windows. A dark horizontal line is observed between 2008-02-27 and 2008-08-26, and another noticeable dark line appears at 2020-03-03. Since the DFC is symmetrical, these two horizontal lines are replicated vertically at the exact dates, demonstrated by the cross shape on the DFC. The dark lines indicate that those windows have a poor relationship with almost all others, meaning that the FCs at those times are anomalous compared to other periods. They are shocks from well-established events, such as the Global Financial Crisis (GFC) in 2008 and the COVID-19 pandemic in 2020 [11]. Hence, our DFC network reflected shocks from well-established global market crises.

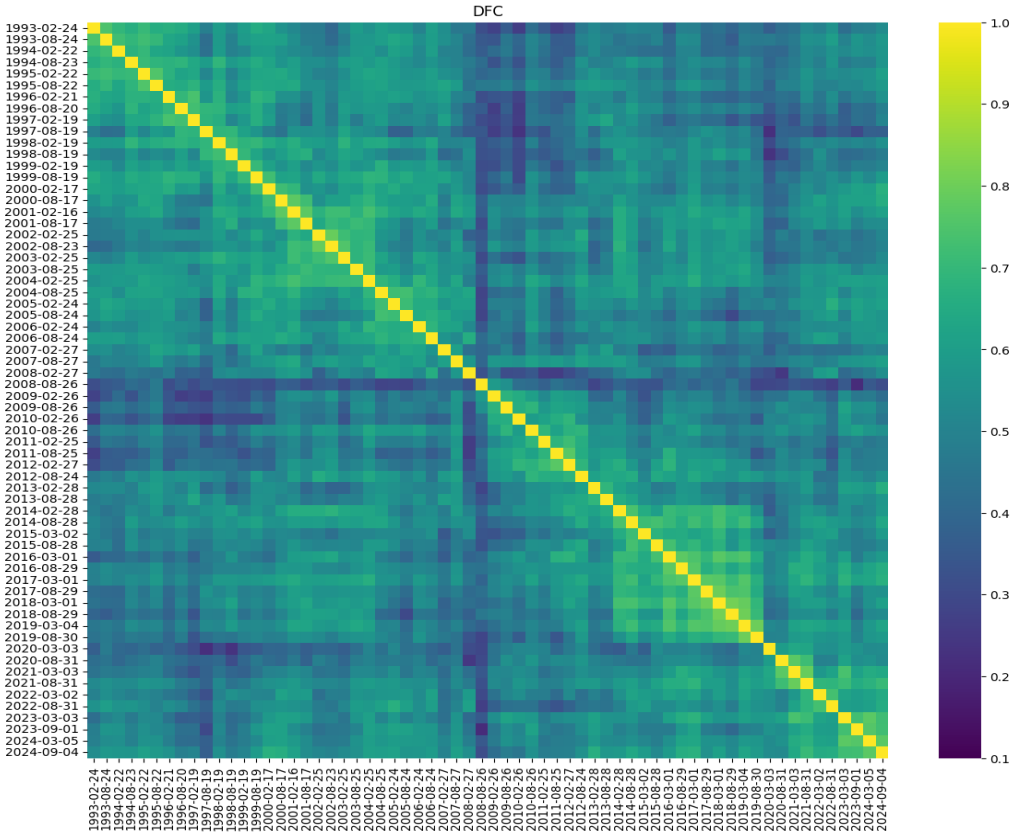


FIGURE 5.1. DFC using the 6 Months (126 trading days) window.

Additionally, the 2010-2012 Europan Debt Crisis and 2015-2018 Turmoil in Atasoy and Özkan's study [**11**] were also captured by our DFC as a brighter square area bounded by dark boundaries around the respective periods in Figure 5.1. The same insights were identified from the DFC based on other window lengths in *Appendix D DFC Networks with Other Time Windows*.

Nevertheless, the 6-month window may not provide a clear-cut distinction between each crisis period. For example, Figure 5.1 cannot identify components of the 2015-2018 Turmoil, including the Global Stock Market Selloffs in 2015 and late 2018, the Brexit Vote in 2016, and the Crypto Market Crash in early 2018. Therefore, it is valuable to examine smaller window lengths.

The 1-month Window DFC in Figure A.1 was, in principle, able to capture the short-term temporal dynamics of the market and provided the highest data resolution. There were more precise dark boundaries between crises that revealed the same market conditions, aligning with our hypotheses.

5.2 Market states and transition

In *Section 5.1 DFC Analysis*, it was observed that the market structure changes over time, and we explored how the Slinding-Window analysis can capture market dynamics. Nevertheless, the DFC network indicates which time windows have similar market structures, but it does not reveal what those market structures are. In particular, we would like to summarise the underlying common market structures (states) when they occur and understand the transitions between them. Henceforth, this section aims to investigate the **market states** and uncover their **transitions** as time progresses.

Methodology

The market states from a DFC network can be identified by a **K-means clustering** algorithm. K-means clustering is preferred over hierarchical clustering because market states are not expected to have nested relationships, unlike stocks and sectors. We used a K-means clustering explored in *Section 2.1.1.4 Clustering*, to cluster rows of the DFC network in Figure 5.1, in a similar fashion to clustering of the rows in *Section 4.1 Static FC network*, including the same methods to identify the optimal number of clusters. For example, Figure 5.2 suggests that the optimal number of market states is 3.

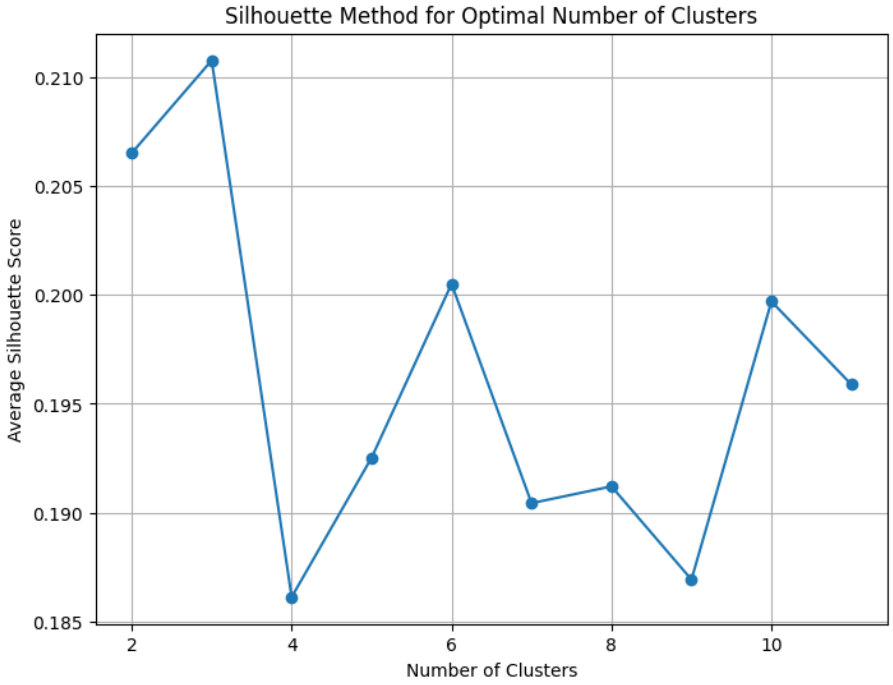


FIGURE 5.2. Silhouette Score of the optimal number of states - 6 Months window.

The K-means algorithm produced a new list of cluster labels for market states from the DFC network so that we could classify windows, as explored in Figure 2.4(D) from *Section 2.1.2 Dynamic Functional Connectivity*. For windows in the same market state, we took the average across their FC networks to generate the **representitive FC network** of this market state. Therefore, if we identified three market states, we will have three representative matrices, as shown in Figure 5.3. The order of stocks is the same as the static FC network in Figure 4.3 for greater interpretability and comparability.

Results

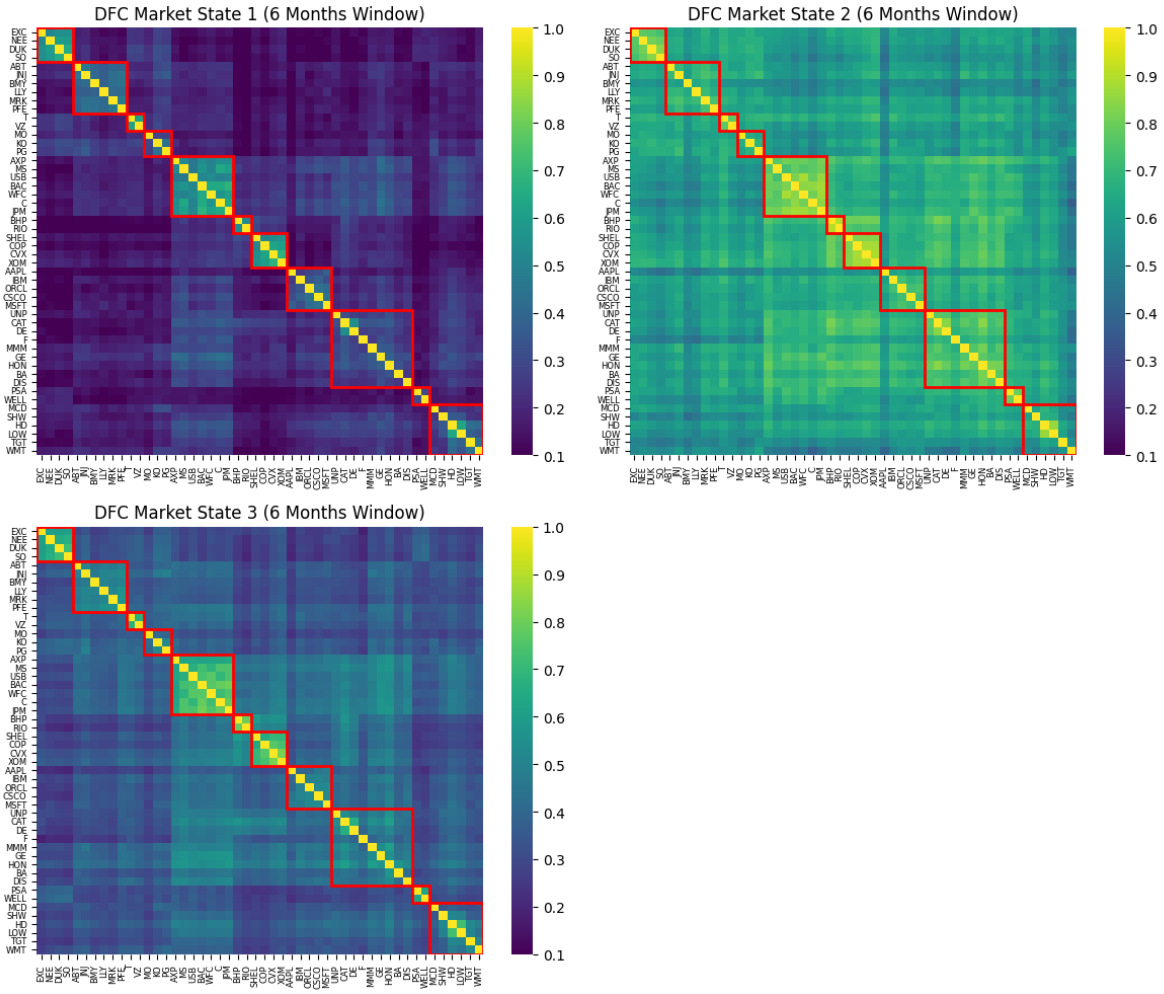


FIGURE 5.3. Market States using the 6-month (126 trading days) window.

The identified market states should present different market structures based on correlations, and these may differ from the static FC network as well. In Figure 5.3, we can see that State 1 has minimal overall correlations, indicated by the darker colour of the matrix. This observation suggests that when State 1 is active, stocks in the market, especially those from different sectors, are less dependent on one another and exhibit a higher degree of idiosyncrasy (company-specific characteristics). During State 1, less systematic risk dominates the market, and individual stock news plays a larger role in driving returns. State 1 also offers opportunities for diversification benefits, which can substantially reduce overall portfolio volatility by combining stocks.

Stocks tend to be strongly correlated during market crises, as evidenced in Zhao et al.'s study [100] from *Section 2.2.2.2 DFC analysis*. We observed that State 2 has the highest correlations, anticipating that it may be active during market crises. State 3 has broad market correlations, but less than State 2, whilst retaining a modular structure to some extent similar to State 1. Therefore, State 2 is likely to be the state that was active when market crises happened. We expect to see State 3 being active before and after State 2 as a transition state to capture market sentiment.

To substantiate our anticipated results, the bar plot, coloured by states in Figure 5.4, illustrates the transition of active states across the entire timeline of the dataset. Each bar represents the active window (in this case, 6 months) for that state. We can see that State 1 (blue) occupied most of the timeline except near market crises, depicting the "normal" or "ordinary" market structure. State 2 (orange) represents the "crisis" market structure, which was only active during market crises, such as the GFC and COVID-19, aligning with our hypothesis about strong correlations across the market in State 2. State 3 (green) was both active before and after State 2, as expected, representing the "crisis-transition" market structure of the build-up and recovery phases of crises. Thus, the time vector in Figure 5.4 supports our expectations.

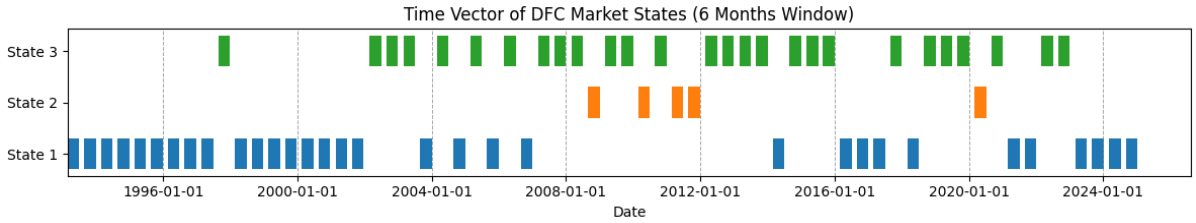


FIGURE 5.4. Time Vector of the 6 Months (126 trading days) window.

On the other hand, it was also observed that in all three states (see Figure 5.3), a few prominent sectors dominated the market. We can identify that the Utilities, Financial Services and Energy sectors had strong within-sector correlations regardless of time-varying market structure. Therefore, the overall market behaviour has less influence on these sectors.

The **Utilities** sector was evidenced to be the least sensitive to changes in interest rates and GDP growth, making it defensive in turbulent times [98]. Additionally, the surge in electricity demand from AI and data centres, combined with a favourable regulatory environment for clean-energy adoption, has significantly increased the importance of utilities despite market dynamics [98].

It was also noticed in Figure 5.3 that the **Financial Services** sector exhibited systematic importance as it acted as a "hub", which plays a central role in both amplifying and dampening shocks through the economy as they distribute stresses and risks across the market [13, 115]. The nature of obtaining highly liquid assets, diversified investments and hedging make Financial Services seem to be unaffected by market crises [116].

The **Energy** sector has higher within-sector correlations as the energy companies are exposed to standard oil-price shocks, and they tend to move together, regardless of the market movement [117]. Unexpected changes in oil prices cause similar profit and effects across the sector, driving up correlations among energy stocks. These features of the Energy sector boost the pairwise within-sector correlations when the overall market exhibits low idiosyncratic volatility.

As shown in *Appendix E*, DFCs using other window lengths may produce a different number of market states. For the 3-month window, its three states in Figure E.3 offered similar patterns and insights to those in the 6-month window. For the 1-month window, only two states were identified in Figure E.1, representing the most distinct market structures: the "normal" state and the "crisis" state. Nevertheless, the high within-sector correlations were also demonstrated in the Utilities, Financial Services, and Energy sectors, validating our remarks for the 6-month window.

In contrast, the 1-year window provided a 10-state prediction in Figure E.5, which was not comparable to the 6-month window result. Although the optimal number of states was 10, as shown in Figure 5.5, there was no significant difference between the Silhouette Scores of 3 and 10 states. Hence, it is worthwhile to try computing three states for the 1 Year window.

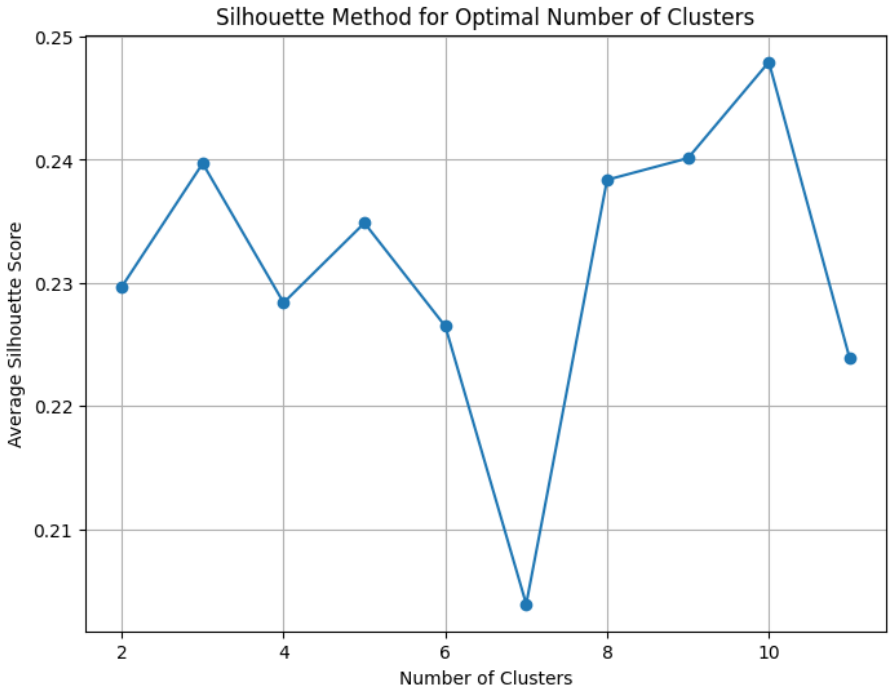


FIGURE 5.5. Silhouette Score of the optimal number of states - 1 Year window.

Results are presented in Figure 5.6 and Figure 5.7, showing the "normal" market structure in State 1, the "crisis" market structure in State 2, and the "transition" market structure in State 3, aligning with the output of 6 Months window. However, compared to the 6-month window, State 1 in Figure 5.6 had overall lower correlations in the market (darker colour). This observation could be due to the improved estimation reliability from noise reduction as the window length increased. With a larger window length, the Sliding-Window analysis can yield more distinct states and emphasise low-frequency components (slow dynamics).

The DFC network enabled us to identify recurring states that remained stable across the window length used. We have uncovered three distinct market states and traced their active times aligning with known market events. State 1 represents the typical market structure, which is highly modular. State 2 reveals the crisis market structure with global correlations. Lastly, State 3 was the transitioning market structure, which is moderately modular and exhibits higher correlations across modules.

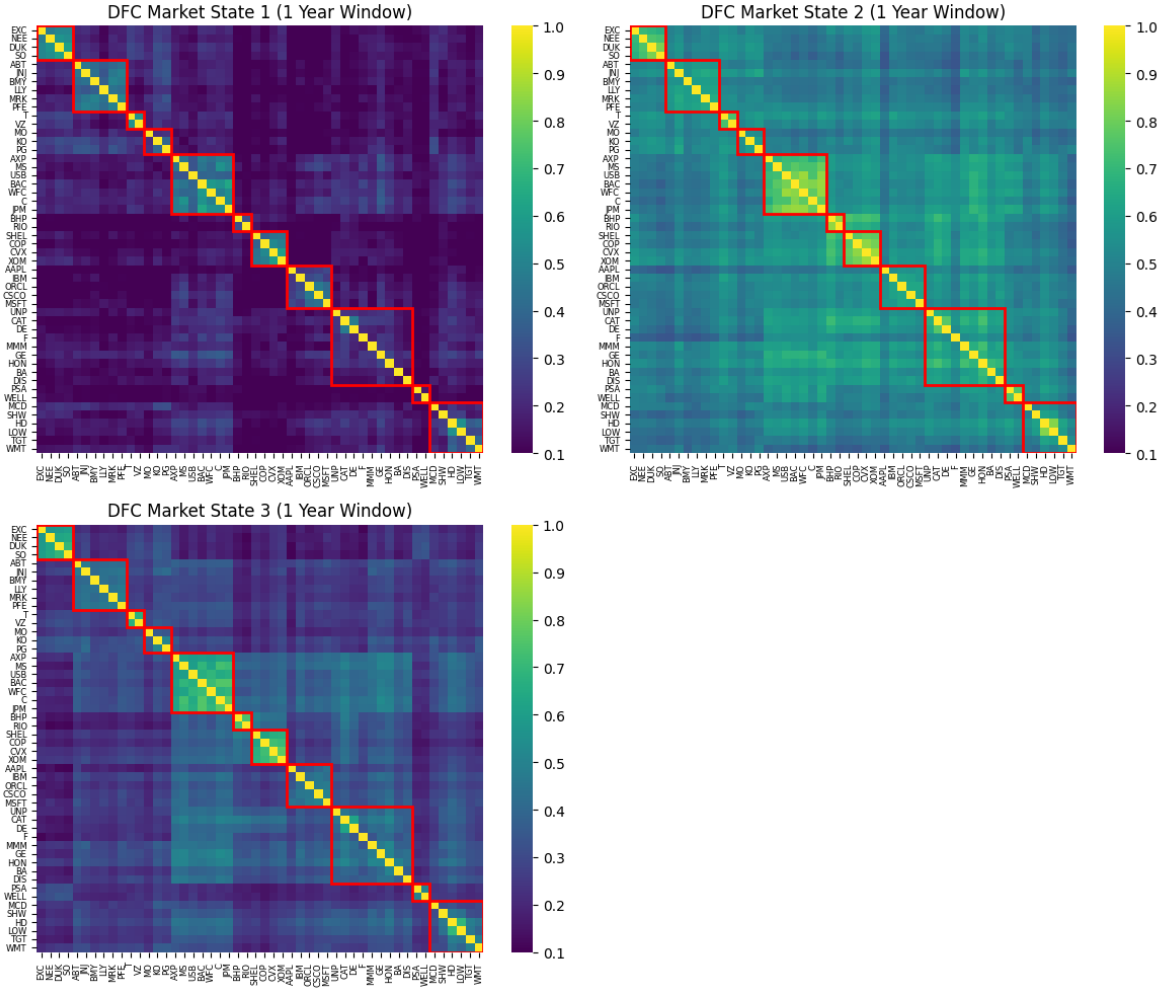


FIGURE 5.6. 3 Market States using the 1 Year (252 trading days) window.

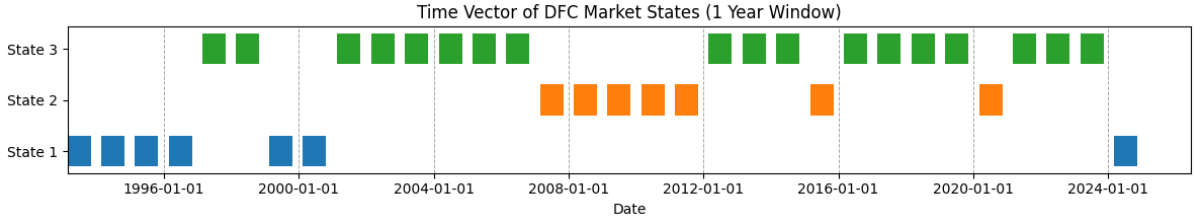


FIGURE 5.7. Time Vector of the 1 Year (252 trading days) window - 3 States.

5.3 Sliding-Window vs. Hidden Markov Model

In the last section, we have explored and analysed the market states and their transitions identified based on the DFC network. Nevertheless, the DFC involves a Sliding-Window analysis that carries a significant limitation of a fixed window length, causing the trade-off between specificity and sensitivity [61] as discussed in *Section 2.1.2 Dynamic Functional Connectivity*. Many empirical studies found that this approach often produces undesirable correlation coefficients [56, 65, 66]. Further, as the phase of time series data shifts, measurements taken within the window may reflect different parts of the cycle, which may confound the identification of consistent patterns [67]. Therefore, it is valuable to employ an alternative approach that does not require a fixed window length to capture the market's dynamics.

5.3.1 DFC networks using HMM

We will explore the results of the **Hidden Markov Model (HMM)** using the same pipeline as the DFC analysis. HMM is a generative probabilistic model as discussed in *Section 2.1.2 Dynamic Functional Connectivity* that moves among unobserved states and does not require a fixed window length [59, 68].

Methodology

In this study, the HMM was applied using the Python package, `osl_dynamics`, which takes a normalised time series and the number of market states to output the predicted FC structure of market states and their active time vectors. We set the number of states for the HMM algorithm to be 3, as we want to compare with the DFC states in Figure 5.3. Since the HMM is a machine-learning model that generates slightly different results in each run, we matched the HMM states by correlating them with DFC states that obtained the highest similarity. This approach can maintain the consistency and comparability of our outputs in facilitating analysis. Regarding the statistics of the time vectors, we computed the fractional occupancy (2 d.p.), mean dwelling time (2 d.p.), and switching rate (4 d.p.) of the predicted market states, as shown in Table 5.1.

- $\text{Fractional Occupancy}_i = \dfrac{\text{Total Time in State } i}{\text{Total time}}$
- $\text{Mean Dwelling Time}_i = \dfrac{\text{Total Consecutive Occurrences in State } i}{\text{Number of Consecutive Occurrences}}$
- $\text{Switching Rate} = \dfrac{\text{Number of State Transitions}}{\text{Total Time}}$

Results

The HMM states in Figure 5.8 reveals a similar and comparable market structure to the DFC states in Figure 5.3, with State 1 representing the "normal", State 2 representing the "crisis", and State 3 representing the "transition" market structures.

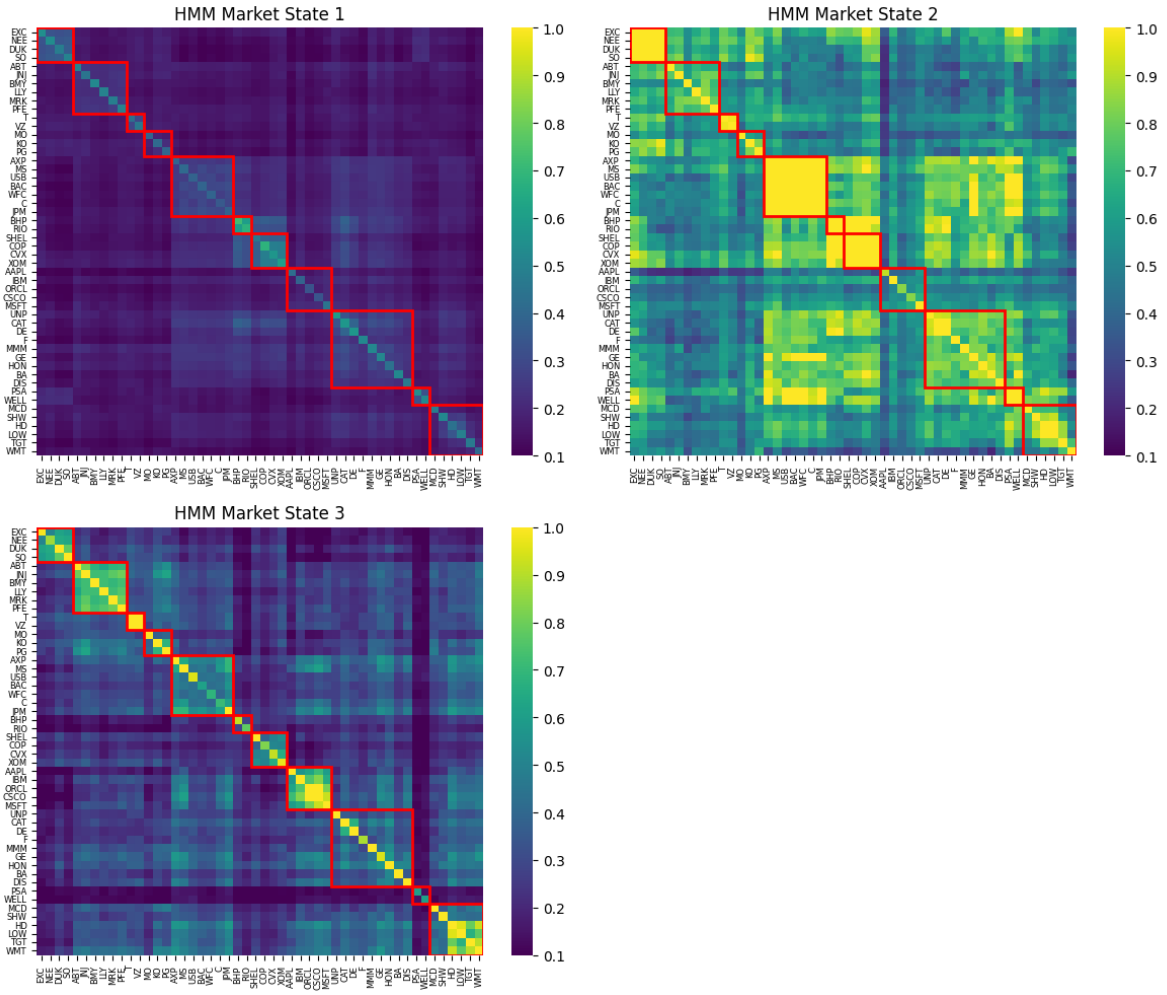


FIGURE 5.8. FC matrix for each market state detected by the HMM.

The alignment of DFC and HMM time vectors was expected as each unique market structure should have its distinct active times and, in combination, reflect the overall market conditions. However, the HMM algorithm provided a prediction of state labels per trading day, so it is possible that multiple states occurred in each DFC window (6 months). Thereby, there were less clear distinctions between State 1 and 3 active times, which makes the alignment of time vectors complex in Figure 5.4 and Figure 5.9.

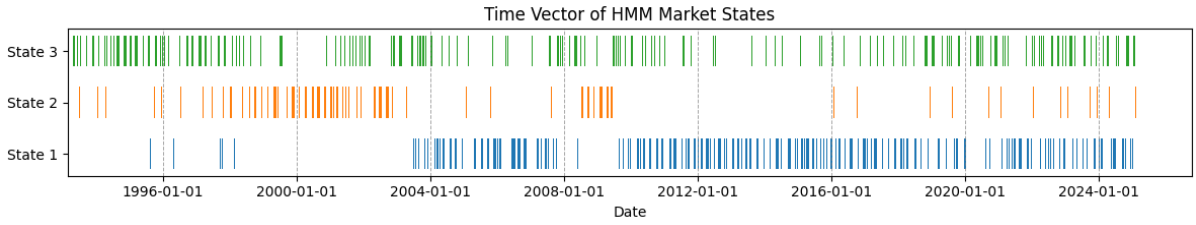


FIGURE 5.9. HMM time vector of active states.

Alternatively, we have attempted to match DFC and HMM states by correlating their time vectors. The results in *Appendix F* show an opposite alignment for State 1 and State 3. It was clearly in contrast to the clearer matches of their FC matrices. This ambiguity may be due to the more fine-grained timing of the HMM predictions, in which fluctuations have a substantial influence in measuring their temporal overlap.

Furthermore, it is interesting to notice that the HMM suggests the crisis mode occurs in earlier parts of the timeline than the DFC had suggested. This phenomenon may have been caused by the more transient dwelling of the market state at these times, which the HMM is sensitive to but the DFC is not. These earlier periods may reflect the late Asian Financial Crisis and Dot-Com Bubble, as identified in Atasoy and Özkan's study [11].

Statistics	DFC States			HMM States		
	1	2	3	1	2	3
Fractional Occupancy (%)	50.00	7.81	42.19	43.03	15.96	41.01
Mean Dwelling Time (Day)	366.55	157.50	243.00	5.01	3.65	3.40
Switching Rate	0.0035			0.2504		

TABLE 5.1. Statistics of DFC and HMM market states.

The dwelling times of DFC states will be longer than those of HMM states, as the Sliding-Window method only captures information per window. Thus, the minimum dwelling time for DFC states is one window (6 months), whereas for HMM states, it is one day. Similarly, the switching rates of DFC states were shorter than those of the HMM states, as it only captured the transitions between windows.

Nonetheless, we can still compare trends in these statistics. In both DFC and HMM states, the fractional occupancy and mean dwelling time of State 1 is the highest, with State 2 being the lowest. This finding suggests that the "normal" market structure exhibits the highest occurrences and stability over time.

5.3.2 Relating DFC states to market volatility

Another comparative analysis between DFC and HMM was the correlation with the **Volatility Index (VIX)** as introduced in *Section 3.1 NYSE data*. In this approach, we aim to investigate whether the identified market states are correlated with market volatilities. Figure 5.10 visualises the time series of VIX, illustrating periods of various market conditions [118]. We expected that our "normal" market state should reflect volatilities below the green line, the "transition" market state should reflect volatilities around the yellow line, and the "crisis" market state should reflect volatilities above the red line.

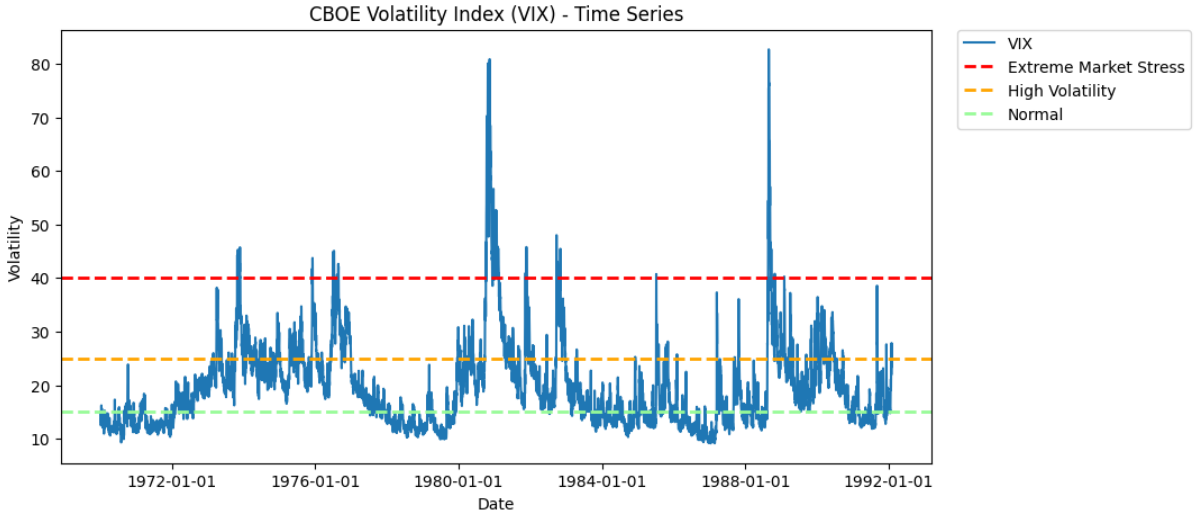
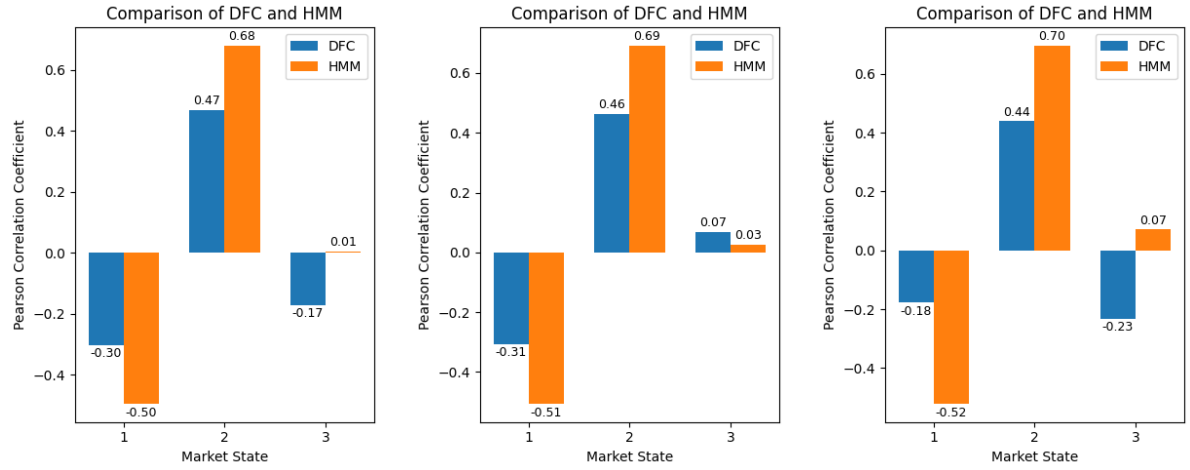


FIGURE 5.10. VIX Time Series.



(a) 3 Months Window Rolling Average (b) 6 Months Window Rolling Average (c) 1 Year Window Rolling Average

FIGURE 5.11. Correlation with VIX - DFC vs HMM.

An active time vector is a binary time series consisting only of 0s and 1s, where 0 denotes an inactive state and 1 denotes an active state. We computed an active time vector for each DFC and HMM state and used it to calculate the Pearson correlations with VIX using 3-month, 6-month, and 1-year rolling averages separately. The correlations (2 d.p.) for the matching states of DFC and HMM were visualised and compared in Figure 5.11.

State 1 is the "normal" market structure, which is typically associated with low volatility (below average). Since the active time vector is 1 for State 1 being active at low volatilities, the correlation is likely negative. This result aligns with what we see in Figure 5.11, which may reflect the more fine-grained temporal modelling of the HMM.

Contrary to State 1, both States 2 and 3 had a positive correlation with VIX as they reflected high volatilities (above average). State 2 is the "crisis" market state, characterised by extreme volatility. State 3 is the "transition" market state, which is associated with intermediate to high volatilities. The HMM offered more insights into temporal periods of volatility, enabled by the more subtle timing property than the DFC for State 2 in all window lengths, as it yielded significantly higher correlations. Nevertheless, for State 3, the DFC prediction yielded a more reasonable correlation using the 6-month window, as it potentially matches the natural timescale of the "transition" market structure.

For the DFC method based on a Sliding-Window, the 3-month and 1-year windows were less suitable for predicting market transitions, as they do not align with the natural time cycle of state transitions and may identify market transitions during an active period. The 3-month window was too short to obtain a reliably isolated pattern, as the connectivity estimates are noisy. The 1-year window was too long, as it oversmoothed the state transitions.

We also observed that as the window length increases, the HMM performance improves for all states. This trend may be due to the nature of HMM, which typically infers states that persist over multi-month horizons. Thus, a longer VIX window exposes more slow-moving state transitions that align better with the HMM time scale.

CHAPTER 6

Time-Lagged Connectivity in Financial Markets

In *Chapters 4–5*, we have completed the pipeline in investigating the market dynamics, including identifying distinct market states and their transitions. Nonetheless, this pipeline only focuses on the symmetrical relationships between pairs of stocks and pairs of sectors. In this chapter, we aim to explore further the time-lagged relationships within the static and dynamic FCs using the same dataset.

6.1 Static FC network: lag-1

In the **Time Lag-1 Analysis**, the FC matrix became directed and asymmetrical, where Y-axis represents Day 1 returns and X-axis represents Day 2 returns.

Methodology

For matrix visualisations, the colour bar is set to be symmetrical, with 0 being the centre, as we want to focus on the impact of Day 1 returns on Day 2 returns. The lagged FC matrix retains the same market sectors (clustering) and stock orders on the axes as the original FC matrix in Figure 4.3 for improved interpretability.

To determine whether the negative skew in Figure 6.1 was due to the nature of stock returns rather than algorithm bias related to log differencing, we applied the same algorithm using a reconstructed dataset based on Gaussian noise in Figure H.1.

Nevertheless, we should also confirm the statistical significance of these values to ensure that we are drawing conclusions based on valid data rather than random noise. The Bonferroni correction applied in this section is different from *Section 4.1 Static FC network* as $m = (\text{number of features})^2$ with features being the stocks using *Equation 2.3*, where self-correlation is meaningful for directed measures.

Results

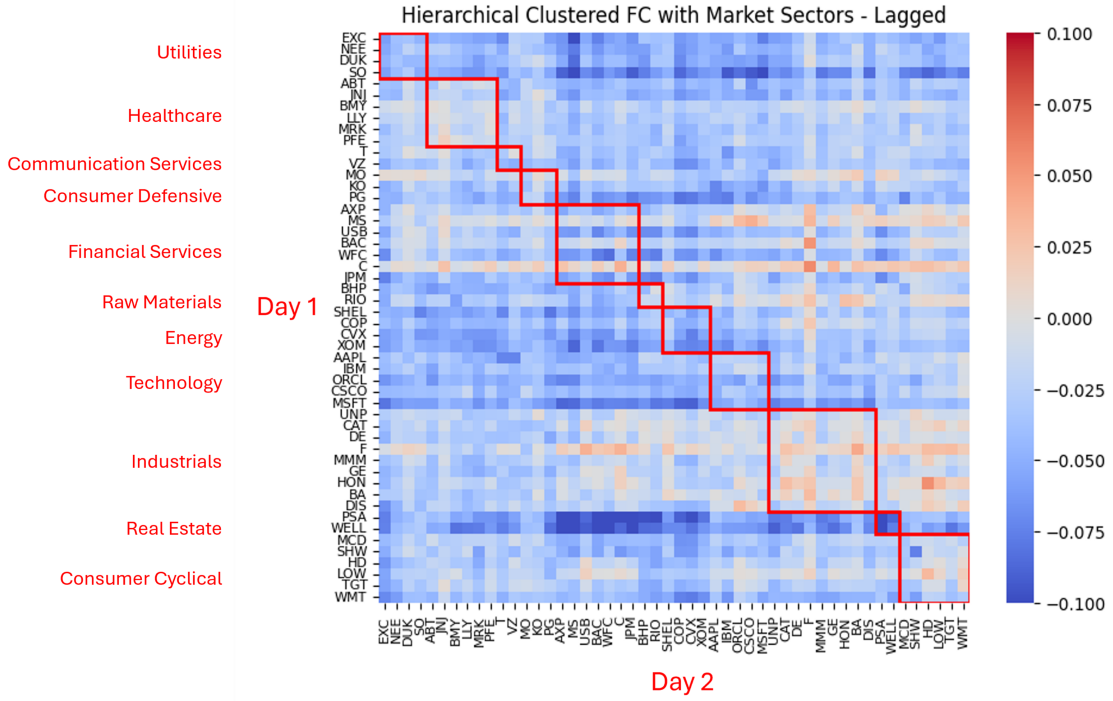


FIGURE 6.1. Static FC of lag-1 log return.

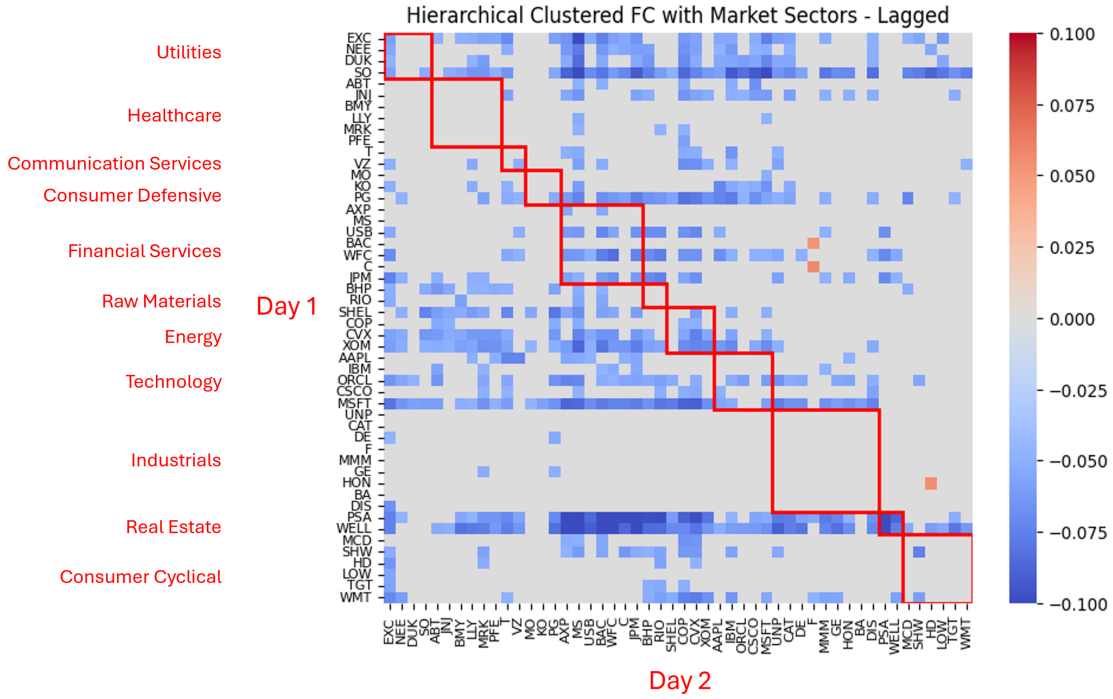


FIGURE 6.2. Static FC of lag-1 data with Bonferroni correction.

In Figure 6.1, all correlations between Day 1 and Day 2 were very small, much smaller than the non-lagged correlation in Figure 4.3, and exhibited more and greater negative values. Figure H.1 shows that Gaussian noise produced an evenly spread of positive (red) and negative (blue) correlations, indicating that the negative skew is meaningful rather than an algorithmic bias. Figure 6.2 with non-significant correlations set to 0 shows that there were indeed mostly negative correlations. There were only 3 positive correlations: (BAC, F), (C, F), and (HON, HD) in ($Day1$, $Day2$) directed pairs.

Bank of America (BAC) and Citi Group (C) tend to react immediately to FED announcements, earnings or credit-market shocks on Day 1 [119]. Roughly 37% of all US loans are originated by commercial banks [120]. Bank of America and Citi Group were the major banks in terms of trading volumes and analyst coverage, making their price changes cleaner with fewer noisy signals of credit-cycle shifts. It remains one of the largest providers of both automotive loans and corporate credit in the US.

Automotive manufacturers like Ford Motor (F), whose revenues depend more on consumer sentiment and credit availability, may absorb shocks overnight and adjust on Day 2. Ford Motor is the most financially sensitive automotive company, so its investors may interpret the "credit-good" news as stronger expected automotive sales and an increase in Ford Motor's price on Day 2 [121].

The Home Depot (HD) receives macroeconomic signals from the Industrials sector as a Consumer Cyclical [122]. A strong return from Honeywell International (HON) on Day 1 signals increases in industrial activity, leading to an expectation of increased consumer spending on Day 2. This phenomenon could also be attributed to sector rotations resulting from institutional rebalancing, which aims to encourage downstream beneficiaries to catch up by transitioning from Industrials to Consumer Cyclical [123].

There were no within-cluster correlations in the Healthcare and industrial sectors, suggesting that their price changes on Day 1 had no impact on Day 2. This observation is likely due to all companies within sectors having homogeneous risks and a similar direction for price changes [124].

In contrast, positive returns on Day 1 of Real Estate had a negative impact on the Day 2 returns of the bulk of stocks in the market. More specifically, the Day 2 returns of Financial Services and Real Estate were most significantly impacted. When Real Estate posts a gain on Day 1, this often reflects looser credit conditions (lower interest rate). As Financial Services profit from the spread of loan yields and funding costs, a pivot towards lower interest rates tends to dampen their returns on Day 2 [125, 126]. For the Real Estate sector, this effect may be due to the overreaction on Day 1, producing negative autocorrelation in subsequent returns [127].

6.2 DFC network: lag-1

Analogous to the symmetrical relationship analysis, a DFC should be constructed to uncover the dynamics that have evolved. In this section, we are interested in discovering whether the influence of Day 1 returns on Day 2 returns has reoccurred over time and when the transition in the effect occurred.

Methodology

By repeating the previous procedures in *Section 5.1 DFC analysis* but with lagged correlations as the FC measure, we obtain a DFC in Figure 6.3 based on the 6 months window. We applied the Bonferroni correction in *Section 4.1 Static FC network* with features being the windows. High correlations were represented in red, while low correlations were represented in blue, indicating changes in the directed relationships.

Results

Except for self-similarities (diagonal), Figure 6.3 shows clusters of high correlations around 1997-2000 and 2006-2009 along the diagonal. These red clusters reveal epochs of a persistent structure of lag-1 correlations between stocks over successive windows. A market crisis likely contributed to this stability in the directional influence, as the leader-follower relationships among stocks remained essentially unchanged.

The Dot-Com Bubble drove the 1997-2000 cluster, whereas the GFC was the major contributor to the 2006-2009 cluster. During market crises, the idiosyncratic fluctuations are less dominant than the systemic market shocks, resulting in the emergence of a synchronous market structure and, hence, strengthened uniform spillovers [**100**]. Since these crises had a multi-year influence on the market structure, the directional relationship among stocks was stable during these crisis periods.

COVID-19, as a global crisis event, did not result in a cluster of high correlations in Figure 6.3, as it did not directly impact a particular sector or sectors. Many studies have found that while overall market co-movement and volatility spiked, the directional spillovers among stock markets changed surprisingly little. COVID-19 was an external health crisis rather than a financial system failure, as it affected all sectors simultaneously, driving a common global fear rather than a new leader-follower relationship [**128**, **129**].

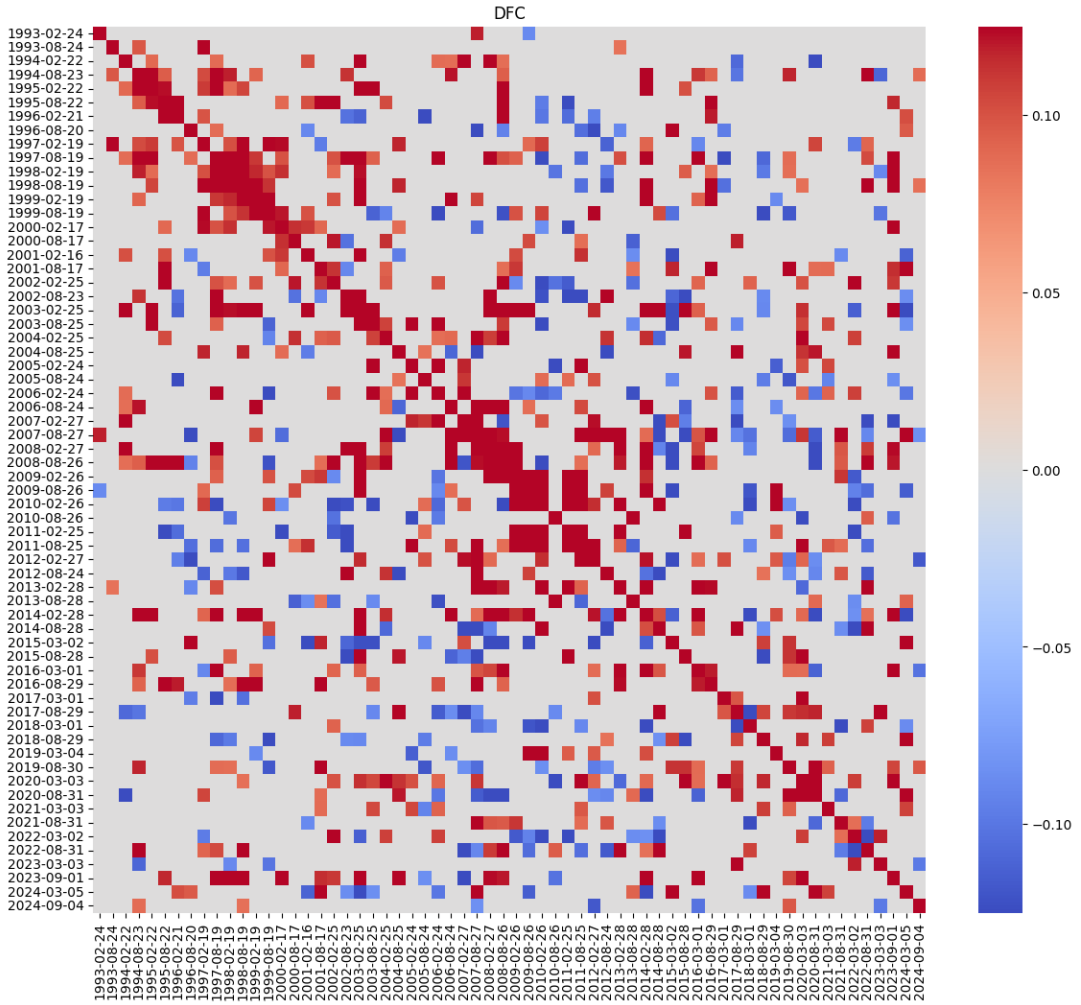


FIGURE 6.3. DFC of lag-1 data - 6 Months Window.

Moreover, unlike the collapse of GFC, the most acute market shock from COVID-19 unfolded over a few weeks in early 2020. Hence, the directional patterns did not have time to restructure stably, especially within the 6-month window. The DFC network based on other window lengths can be seen in *Appendix H DFC Networks with Lag-1 Data*. Notably, the 3-month window DFC in Figure H.3 unveils a small cluster of high correlations around 2020. Nonetheless, central banks and governments rolled out corresponding fiscal and monetary support within weeks of the COVID-19 outbreak. This synchronised intervention aligned market movements and prevented the change in leader-follower relationships due to COVID-19 shocks [**130**].

6.3 Market states and transitions: lag-1

For off-diagonal values, Figure 6.3 suggests an alternating pattern of high (red) and low (blue) correlations across the entire timeline from 1993-02-24 to 2024-09-04. This observation provides insight into the potential recurrence of leader-follower relationships within the market. Hence, in this section, we will explore the unique directed FC structures and their temporal transitions.

Methodology

For further analysis, the same procedures in *Section 5.2 Market states and transitions* were applied in this section. The optimal number of states may not be 2 using the method in *Section 4.1 Static FC network* as various categories of market events would result in different impacts on the directed FC. However, since we want to focus on identifying transitions between the "non-crisis" and "crisis" states in this section, the number of states is set to be 2 for this particular analysis. We first computed the most distinct states in Figure 6.4, representing the unique leader-follower relationships. We then confirmed the market cycles by plotting the state transitions in Figure 6.5.

Results

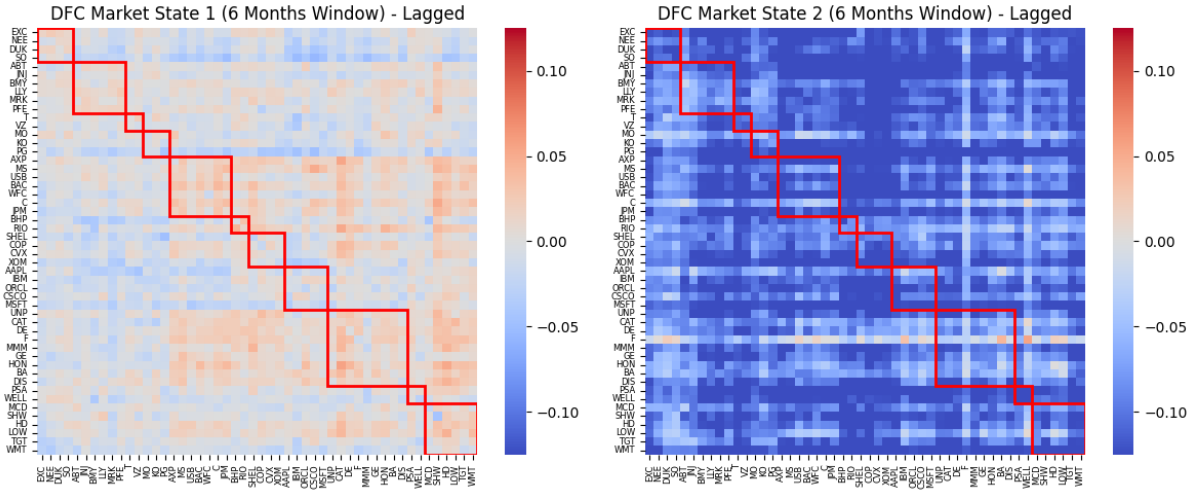


FIGURE 6.4. States of lag-1 data - 6 Months Window.

In Figure 6.4, we can see that State 2 exhibits relatively higher lag-1 correlations for the entire market, representing the "crisis" spillovers. State 1 reveals that the lag-1 relationships between stocks do not show a market-wide comovement, as evidenced by the overall lower lag-1 correlations, which represent the "non-crisis" leader-follower relationships within the market.

Several reasons, such as short-term reversion and liquidity-driven bounce-back, can potentially cause the negative lag-1 correlations in Figure 6.4. Short-term reversion occurs after a significant price move, during which traders often adjust their positions, causing prices to return to a "fair value" the next day [131], resulting in a negative lag-1 correlation. Liquidity-driven bounce-back typically occurs during crises, as the intensification of crisis periods increases liquidity demand. When funding liquidity is tight during recessions, sellers tend to depress prices sharply on Day 1, with selling pressure relieved on Day 2, allowing prices to recover and produce reversals (negative lag-1 correlation) [132].

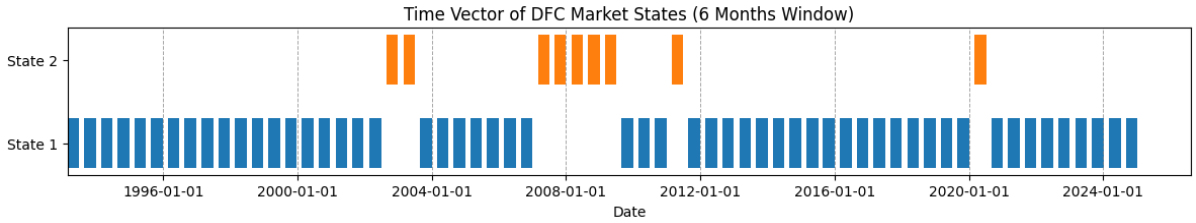


FIGURE 6.5. Time Vector of lag-1 data - 6 Months Window.

The transitions between the two states are visualised in Figure 6.5, showing that all occurrences of State 2 were during crisis periods. This finding suggests that market crises will also lead to changes in directional influence among stocks. Similar remarks were also demonstrated based on the 1-year window, as seen in *Appendix I Market States and Transitions with Lag-1 Data*.

In this chapter, we examine the directed relationships between stocks via lag-1 correlations and analyse how the leader-follower relationships in the market change over time using the same methodology as the non-lag analysis from *Chapters 4–5*. We found that most of the statistically significant lag-1 correlations are negative in the static FC network. We observed that leader-follower relationships persisted for a more extended period during the Dot-Com and GFC. We also identified the "crisis" and "non-crisis" market states and investigated the potential reasons for the negative lag-1 correlations during crises.

CHAPTER 7

Conclusions and Discussion

As we have completed our analysis of relationships in stock market structures and explored their dynamics, this chapter will conclude our study findings. We will discuss the potential limitations of our study design and the choices made to minimise them. Key limitations will serve as the basis for future work in refining the DFC methodology in finance.

7.1 Conclusions

In this study, we have analysed statistical relationships in the market structure and how it changes over time. Our study design is based on the daily closing price data on Yahoo!Finance of 50 stocks listed at NYSE over the period from the 23rd February 1993 to the 12th March 2025.

We measure FC using the Pearson correlation coefficient between stock returns and infer a network representation of the static market structure with nodes being the stocks and their correlations as edges. By applying hierarchical clustering, 11 modules are identified, indicating market sectors that mostly correspond with classifications on Yahoo!Finance. This FC network is visualised as a matrix, offering insights on similarities (high correlations) in stock returns and sector behaviours. We observe that Financial Services act as a hub sector, with Bank of America being the leading stock (*Section 4.1 Static FC network*).

The dynamics of the stock market are revealed through sliding-window analysis and the HMM. In Sliding-Window analysis, we compute an FC network for each window, illustrating temporal market structures. We analyse results based on 1-month, 3-month, 6-month and 1-year window lengths and observe consistent patterns across all window lengths in *Section 5.1 DFC network*. Some windowed FC networks exhibit high similarities, suggesting that there exist some "representative" market structures reoccurring over time. We define these unique and repeating market structures as market states.

From clustering windowed FC networks, we compute three market states representing the "normal", "transition", and "crisis" market structures, with each occurring at a specific level of volatility in *Section 5.2 Market states and transition*. The active times of each market state are traced on the same timeline and compared against well-established market events in Figure 1.1 from Atasoy and Özkan's study [**11**].

In contrast to Sliding-Window analysis, the HMM overcomes the limitation of using a fixed window length, which may not capture sudden and irregular shifts in market states. We use HMMs to assign market states to our time series on a daily time scale. The results of the two methods are compared using correlations against VIX to evaluate how much the results align with volatilities in *Section 5.3 Sliding-Window vs. Hidden Markov Model*. We found that the HMM presents a more accurate view of market state classifications, as it provides predictions with higher resolution. For example, it is common for states to alternate during a period that Sliding-Window has to assign the entire period to the state that first occurred.

Lastly, the time lag of one cross-correlation analysis provides insight into the directed leader-follower relationship between stocks. Applying the same pipeline as the undirected analysis, we compute two distinct market states that indicate the "non-crisis" and "crisis" market structures of spillovers between stocks in *Section 6.3 Market states and transitions: lag-1*. We also explore the transitions between these two states and analyse the impact of three major market events (Dot-Com, GFC and COVID-19) on the time-lagged market structures. We observe that these three events exert a similar influence on the leader-follower relationship between stocks. During a crisis, the Day-1 stock returns are negatively correlated with the Day-2 stock returns market-wide, indicating high market volatility that may result from short-term reversion and liquidity-driven bounce-back, where day-to-day trading fluctuates in opposite directions.

This study showcases the utility of a novel network approach from CN in analysing relationships in the stock market structure and revealing its dynamics. Compared to prior studies in finance, this network approach can capture a distinct network representation of known market events, thereby providing a better understanding of how these events triggered shifts in temporal market structures.

7.2 Limitations of this research

This section aims to demonstrate rigour, acknowledge potential weaknesses and critically evaluate our data, design and analysis. We will contextualise our findings by setting the boundaries within which our results hold and explicitly state what our design does not address to mitigate potential misinterpretation of our conclusions. Nonetheless, our study design can be generalised to analyse any daily price datasets and execute on any software systems that have the Python interpreter.

Data

This study design relies on historical daily price data of 50 stocks retrieved from a single exchange, the NYSE, which can introduce threats to both internal and external validity. As this study focuses on the dynamics of the market structure, missing values in the data will alter the fundamental market dynamics, affecting the validity of our findings. Thus, our design performs a check on the imported data to ensure it has a price value for all consecutive days.

The daily price data is known to be highly volatile, and the noise may hinder the underlying relationships in the market structure. Our model computes the log returns of the raw price data to minimise this concern and make statistical relationships more stable. However, this approach ultimately altered the investigation from analysis based on stock price to stock returns (price changes). The 50 stocks were a mix of the largest and medium companies in terms of market capitalisation from each market sector, aiming to balance sectors and represent the underlying structure of the stock market. However, the inferred network may overrepresent well-covered and highly liquid companies and underrepresent smaller, less liquid companies due to selection bias.

Although the NYSE is one of the world's largest stock exchanges and should be representative of the global stock market, data from different exchanges can yield diverse insights, as they have different listing compositions and inherent sector biases. For example, the NASDAQ is more technology-heavy, whereas the NYSE has a greater focus on financial services and industrials. Even with the same set of stocks, sector contributions towards the overall market will be different. Hence, the coverage of stocks and the choice of exchange will inevitably impact the generalisability of our findings, and the model's performance could be inflated relative to a broader universe.

FC measure

On the other hand, the validity of our inferred FC network is highly dependent on the FC measure. The main drawback of using the Pearson correlation coefficient is that it only captures pairwise linear relationships between stocks, whereas stock data are nonlinear and multivariate in nature [133]. The leverage effect occurs when a stock reacts more sharply to another stock, for example, an exponential increase in price. This phenomenon causes the nonlinear relationships between stocks. Moreover, the change in stock price could be due to the impact of multiple factors rather than just one stock, resulting in a multivariate relationship.

Thus, the measure of FC needs to suit the nonlinear and multivariate nature of stock data, ensuring the validity of our analysis. Information-theoretic measures, such as MI, TE, and GC, introduced in *Section 2.1 Functional Connectivity*, are more suitable for capturing nonlinear and multivariate dependencies between stocks. Find more details in *Section 7.3 Future work*. Notably, correlations do not imply causation. The conclusions of this study did not suggest any causal relationships between stocks or sectors.

Furthermore, there exist confounding events, such as sector-specific regulations and financial crises, having external impacts on market dynamics. These events may distort the market structure and dynamics beyond price fluctuation patterns. This threat was minimised by corroborating our results against existing studies, such as Figure 1.1 from Atasoy and Özkan's study [11], which had identified major market events during the period.

Sliding-Window analysis

The limitation of Sliding-Window analysis remains that the use of a fixed window length poses a threat to construct validity, as the optimal window length to maintain feature sensitivity and uncover the intended market structure is unknown. Additionally, the Sliding-Window analysis cannot account for latent variables, as they cannot be directly measured.

Henceforth, the application of the HMM in our study design is significant in ensuring the construct validity. As explored in *Section 2.1.2 Dynamic Functional Connectivity*, the HMM does not require windows and can infer latent variables from observed data, allowing us to model complex systems with fewer parameters and uncover hidden structures and relationships. Consequently, the HMM improves capturing market sentiment that truly maps onto what we intend to study.

Time lag analysis

Nevertheless, no Python package of the HMM is available to conduct the time lag analysis. The comparison between the DFC and HMM results in time lag analysis is omitted. It is a necessary procedure to perform in future work as the HMM may provide more insights into the lagged relationships between stocks. Even the time lag of one analysis provided a directed perspective on the changes in stock market structures, as it only measured the correlation between Day-1 and Day-2 returns rather than conveying that Day-1 returns caused Day-2 returns.

Statistical power

Statistical power is a concern regarding conclusion validity, as we want to ensure our findings remain meaningful beyond noise. The use of a Bonferroni correction is a conservative approach that helps bolster the validity of conclusions by controlling the overall risk of obtaining a false positive when running multiple statistical tests. This approach controls the family-wise error rate, reduces spurious findings and makes significant results more robust. Nonetheless, this approach may be overly conservative, resulting in a loss of power and increasing the probability of failing to detect real effects (Type II errors).

Nevertheless, generalising this study design requires further efforts on a broader selection of stock data.

7.3 Future work

Based on the limitations identified in the previous section, this section will suggest potential improvements and directions for future research.

Regarding the data limitation, using stocks from the S&P 500 can better represent the market as it includes the top 500 US companies with the largest market capitalisation, representing a broad section of the US economy [134].

Undoubtedly, the main improvement for this study design will be the measure of FC. Future research is encouraged on nonlinear and multivariate measures of FC, such as information-theoretic measures. For undirected measures, MI quantifies the total statistical dependency between two or more random variables by comparing the joint distribution to the product of marginals. A conditional MI can also be used when a known condition is present. For example, we have random variables X, Y, and Z, and we want to investigate how much X explains Y given Z. This approach can help measure the FC between stocks that are affected by external shocks, such as rising interest rates.

For directed measures, TE or GC quantifies the information flow in the time series. For random variables, X and Y, TE detects any influence of past values of X on future values of Y, with conditional probabilities of entire history vectors. TE can be extended to a network by conditioning other variables to be multivariate. GC with nonlinear predictors, such as kernel methods and nonlinear autoregressive models, can capture nonlinear, directed influences. With a vector-autoregressive (VAR) form, GC can account for multivariate time series by testing whether off-diagonal blocks of the VAR are jointly zero.

Furthermore, Independent Component Analysis (ICA) can be applied to assess the "normal" market structure without crises. ICA decomposes stock returns into statistically independent components to better understand latent drivers of the market. We can remove the independent components corresponding to the three crises — Dot-Com, GFC, and COVID-19 — to assess the market structure without their influence on stock returns. Hence, we can potentially uncover subtler drivers of the market, such as sector rotation, interest rate movements and liquidity cycles. See more information about ICA in *Appendix J*.

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APPENDIX A

Traditional Approaches for Studying Markets

For decades, there have been high pursuits of financial forecasting, especially in market pricing and average movement of financial assets, in which researchers have aimed to develop better models based on computer science, statistics and economics [71]. These models were designed to help investors gain efficient information and identify potential investment opportunities [72]. To date, financial analysis has primarily relied on two major streams of techniques, without incorporating the principles of DFC: statistical methods and machine learning methods [72].

Statistical Models

Most prevalent statistical methods to construct dynamic econometric models are the Autoregressive Moving Average (ARMA) family of univariate models, and Vector Autoregressive (VAR) and Autoregressive Conditional Heteroscedastic (ARCH) family of multivariate models [135].

An **ARMA(p)** model has the following general form [136]:

$$x_t = a_1 x_{t-1} + a_2 x_{t-2} + \ldots + a_p x_{t-p} + \epsilon_t,$$

where the a and b are constant coefficients. This representation shows that the current value of the process x_t, is a linear combination of the past p terms of the series plus a white noise term, ϵ_t.

A **VAR(p)** model has the following general form [137]:

$$x_t = a_1 x_{t-1} + a_2 x_{t-2} + \ldots + a_p x_{t-p} + s_t + \epsilon_t,$$

where $x_t = (x_{1,t}, ..., x_{n,t})$ is a multivariate stochastic time series in vector notation, a_i; $i = 1, 2, ..., p$ are deterministic $n \times n$ matrices; $\epsilon_t = (\epsilon_{1,t}, ..., \epsilon_{n,t})$ is a multivariate white noise with variance-covariance matrix $\Omega = (\omega_1, ..., \omega_n)$; and $s_t = (s_{1,t}, ..., s_{n,t})$ is a vector of deterministic terms.

An **ARCH** model is the clustering of volatility (the variability of the time series) observed in many economic and financial series, defined as follows [137]:

$$x_t = \left[\sqrt{\beta + \lambda x_{t-1}^2}\right] z_t,$$

where x_t is the current value influenced by the past volatility x_{t-1}, z_t forms an independent and identically distributed (i.i.d.) sequence with zero mean and unitary variance, and the parameters and must be nonnegative.

A **GARCH** model generalises the ARCH model further by allowing past variances (volatility) to influence the current value, defined as follows [137]:

$$x_t = \sigma_t z_t',$$
$$\sigma_t^2 = \beta + \sum_{i=1}^{p} \lambda_i x_{t-i}^2 + \sum_{j=1}^{q} \delta_j \sigma_{t-j}^2,$$

where σ_t^2 is the current volatility, x_{t-i}^2 is the squared past returns, σ_{t-j}^2 is the past conditional volatility, and β, λ_i, δ_j must be nonnegative and estimated from the data.

Machine Learning Models

Besides traditional statistical approaches, **ML methods** are popular and successfully applied in the finance domain. This section will provide an overview of four frequently used ML models [71]: Artificial Neural Network (ANN), Support Vector Machines (SVM) and Support Vector Regression (SVR), Random Forest (RF), and Deep Learning (DL).

An **ANN** is a flexible model that can approximate nonlinear time series without a formal equation [138]. ANNs are structured in three main layers: input, hidden and output (Figure A.1). ANNs learn from historical data using backpropagation that achieves an acceptable accuracy by adjusting weights iteratively based on the difference between the predicted and actual output [71]. These features enable ANN to deal with noisy or incomplete data with robustness in real-world applications, making it a very useful tool in financial forecasting [138, 139]. Nevertheless, ANNs often perform inconsistently and unpredictably on financial data with high dimensionality [140] and inevitably suffer from overfitting [141].

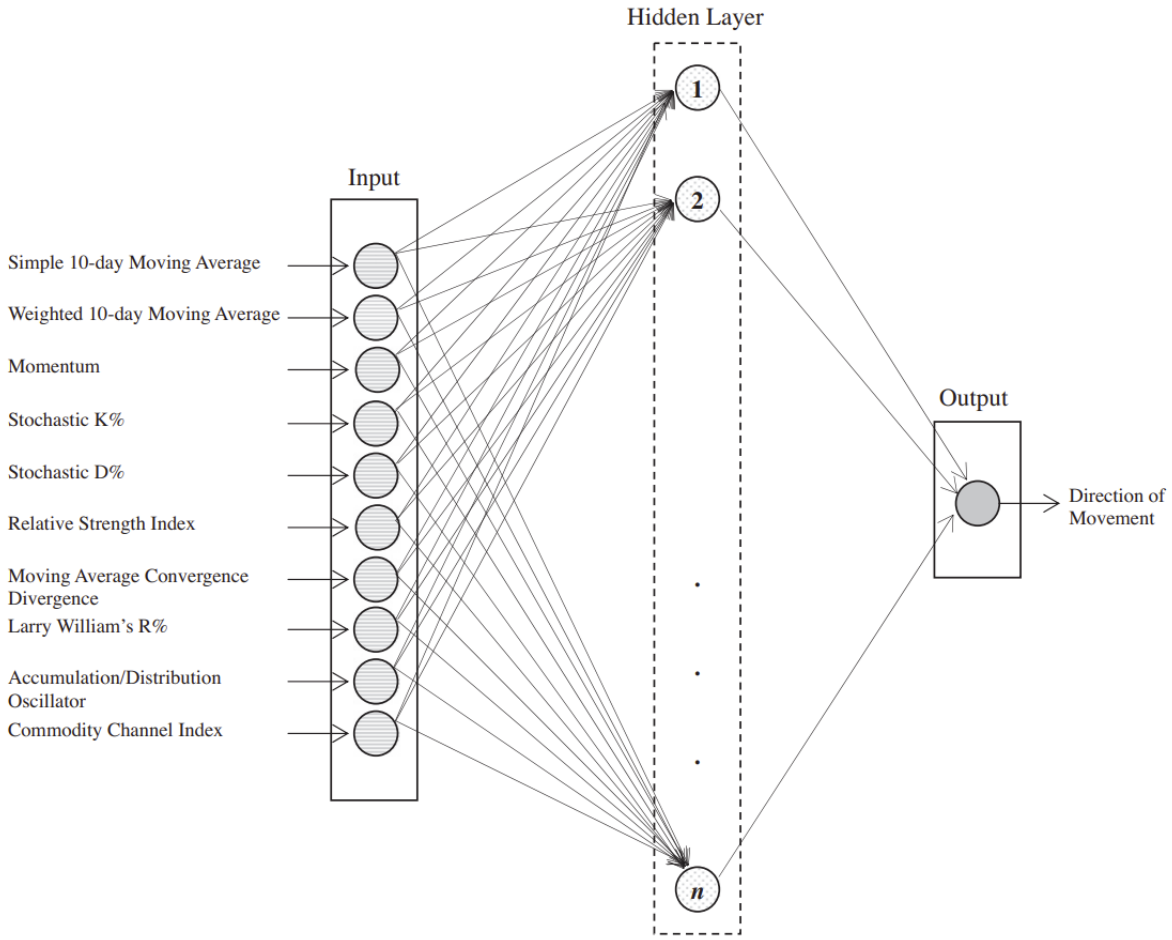
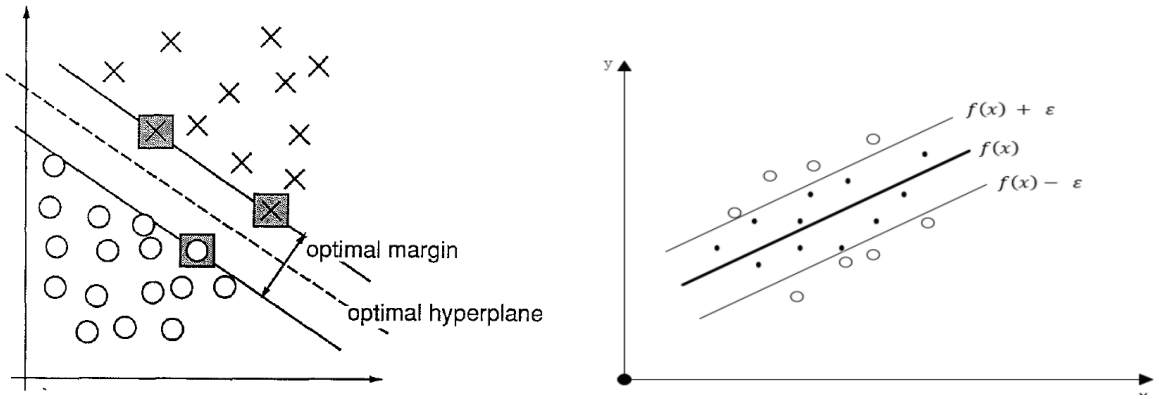


FIGURE A.1. The architecture of three layered feedforward ANN.

Reprinted from Figure 1 of Kara et al.'s study [142].

Different to ANN, a **SVM** scales the input data moderately well to high dimensions and easily avoids the overfitting problem due to its structural risk minimisation [71] (Figure A.2(a)). SVMs are mainly used for classification by realising a kernel function considering empirical error and a regularisation term. As a result of its ability in nonlinear approximation, SVMs are widely used to predict the direction of nonstationary time series, such as price increase or decrease [142, 143]. In contrast, a **SVR** is the regression version of the SVM but more useful for regression of time series, such as forecasting stock prices [71,142,143] (Figure A.2(b)). Although they outperform ANN, they require higher computational resources and they are sensitive to parameter selection and tuning [71], where extensive trial-and-error experimentations or cross-validations are required.



(a) Support vectors (grey squares) define the margin of largest separation between two classes.
Reprinted from Figure 2 of Cortes and Vapnik's study [144].

(b) Support vectors (solid black dots) are critical to define the regression function, f(x).
Reprinted from Figure 1 of Ye's study [143].

FIGURE A.2. Support Vector Machines (SVM) and Support Vector Regression (SVR).

Alternatively, a **RF** model is nonparametric and is based on a decision tree model, which constitutes classification and regression trees [71]. Classification trees are decision trees that forecast classified variables to categorise samples, whereas those that predict continuous variables are regression trees [145]. Each tree is trained independently on a random subset of the data computed by bootstrapping (bagging predictors) [146, 147]. RF takes the most frequent prediction across all individual trees as the final classification, and it averages the predictions of all individual trees for regression [147]. Furthermore, RF can generate a strong learner with several weak learners using weights (Figure A.3), which prevents recency bias and overfitting [148]. Nonetheless, RF consumes large computational resources with small changes in training data causing major changes in the model established by RF [71].

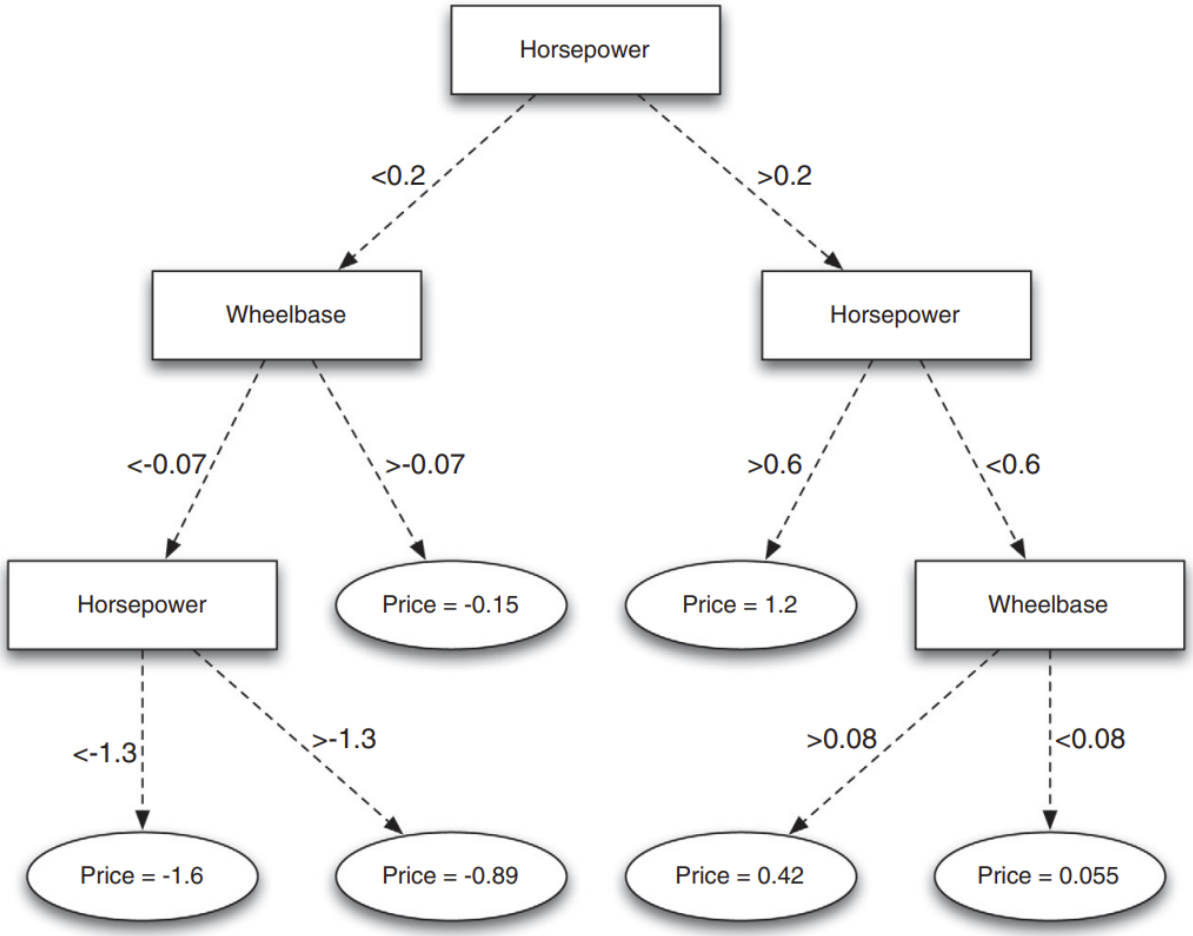


FIGURE A.3. Example of a regression tree for predicting price of cars.

Reprinted from Figure 4 of Booth et al.'s study [148].

A **DL** model is a learning method with a multi-level representation that is obtained by superimposing multiple simple but nonlinear modules [71]. Commonly applied DL models include Feedforward Neural Networks (FFNN), Convolutional Neural Networks (CNN), and Recurrent Neural Networks (RNN) [149–153]. CNNs are typically employed for image data, while **Long Short Term Memory (LSTM)** models, a variant of RNNs, are for capturing temporal dependencies in time-series data [152].

Compared with ANNs, SVMs and other algorithms, LSTM can extract robust features from long-term dependencies and noise resistance to describe more complex real data [153]. However, LSTM models are highly sensitive to hyperparameters and meanwhile difficult to tune effectively due to the complex architecture (Figure A.4). Additionally, LSTM requires large input datasets to prevent overfitting, while it is computationally expensive to train [71, 148].

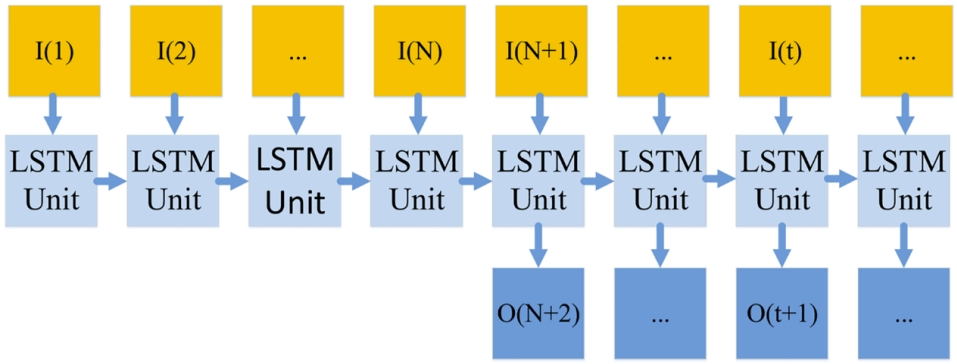


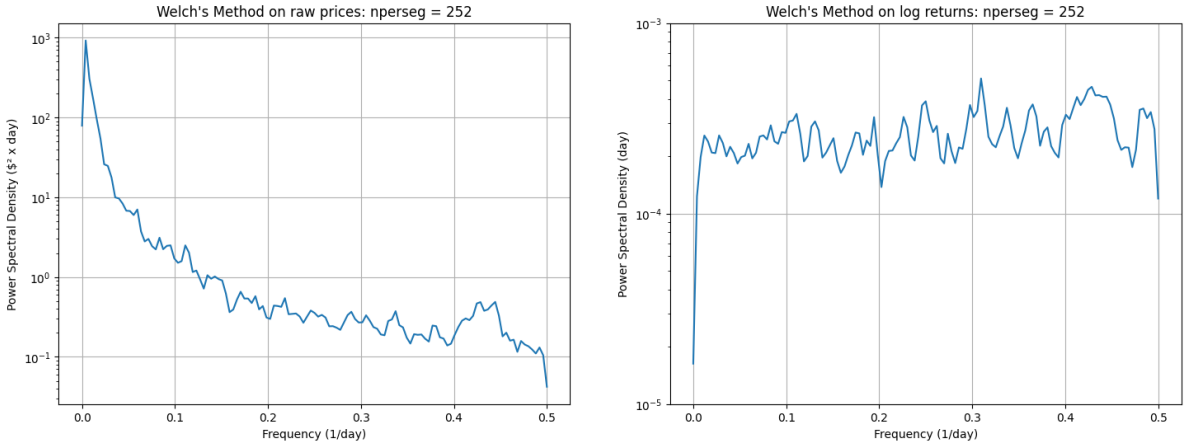
FIGURE A.4. The architecture of a LSTM model.

Excerpted from Figure 1 of Bao et al.'s study [153].

APPENDIX B

PSD Graphs on the Average Stocks

We have observed that there exist variations among stocks in *Section 3.2 Stock price dynamics*. Hence, to present an overview of the entire dataset, we will average the values across the 50 stocks in Figure B.1. Similar to Figure B.1(a), the average version of it in Figure B.1(a) reaffirms that all the power is in the low-frequency part of the price, and connectivity will be dominated by its long-timescale trends. Figure B.1(b) shows a clearer view of consistent signal fluctuations within the range just above 10^{-4} days across time, suggesting the stationarity of the time series (constant mean and variance). The PSD analysis indicates that we should utilise the log return dataset instead of the raw price dataset in our FC analysis to obtain a more stable connectivity measure.



(a) Raw Price

(b) Log Return

FIGURE B.1. PSD of the average of stocks.

APPENDIX C

Company Names and Market Sectors of Stocks

TABLE C.1. Company names and market sectors from Yahoo!Finance [1].

No.	Stock Symbol	Company Name	Market Sector
1	AAPL	Apple Inc.	Technology
2	ABT	Abbott Laboratories	Healthcare
3	AXP	American Express	Financial Services
4	BA	Boeing	Industrials
5	BAC	Bank of America	Financial Services
6	BHP	BHP Group Limited	Basic Materials
7	BMY	Bristol-Myers Squibb	Healthcare
8	C	Citigroup	Financial Services
9	CAT	Caterpillar	Industrials
10	COP	ConocoPhillips	Energy
11	CSCO	Cisco Systems, Inc.	Technology
12	CVX	Chevron	Energy
13	DE	Deere & Company	Industrials
14	DIS	Walt Disney	Communication Services
15	DUK	Duke Energy	Utilities
16	EXC	Exelon	Utilities
17	F	Ford Motor	Consumer Cyclical
18	GE	GE Aerospace	Industrials
19	HD	Home Depot	Consumer Cyclical
20	HON	Honeywell International	Industrials
21	IBM	IBM Corporation	Technology
22	JNJ	Johnson & Johnson	Healthcare
23	JPM	JPMorgan Chase	Financial Services
24	KO	Coca-Cola	Consumer Defensive
25	LLY	Eli Lilly	Healthcare
26	LOW	Lowe's	Consumer Cyclical

continued on next page

Table C.2 – *continued from previous page*

No.	Stock Symbol	Company Name	Market Sector
27	MCD	McDonald's	Consumer Cyclical
28	MMM	3M	Industrials
29	MO	Altria Group	Consumer Defensive
30	MRK	Merck & Co.	Healthcare
31	MS	Morgan Stanley	Financial Services
32	MSFT	Microsoft Corporation	Technology
33	NEE	NextEra Energy	Utilities
34	ORCL	Oracle Corporation	Technology
35	PFE	Pfizer	Healthcare
36	PG	Procter & Gamble	Consumer Defensive
37	PSA	Public Storage	Real Estate
38	RIO	Rio Tinto Group	Basic Materials
39	SHEL	Shell plc	Energy
40	SHW	Sherwin-Williams Company	Basic Materials
41	SO	Southern Company	Utilities
42	T	AT&T	Communication Services
43	TGT	Target	Consumer Defensive
44	UNP	Union Pacific	Industrials
45	USB	U.S. Bancorp	Financial Services
46	VZ	Verizon Communications	Communication Services
47	WELL	Welltower Inc.	Real Estate
48	WFC	Wells Fargo & Co	Financial Services
49	WMT	Walmart	Consumer Defensive
50	XOM	Exxon Mobil	Energy

APPENDIX D

DFC Networks with Other Time Windows

D.1 1 Month Window

This figure represents the DFC network based on a 1-month window length (21-trading-day duration), providing insights into the relationships between each time window pair with the highest resolution in this study. These correlations illustrate the dynamics of relationships between stocks over the period 1993 to 2025.

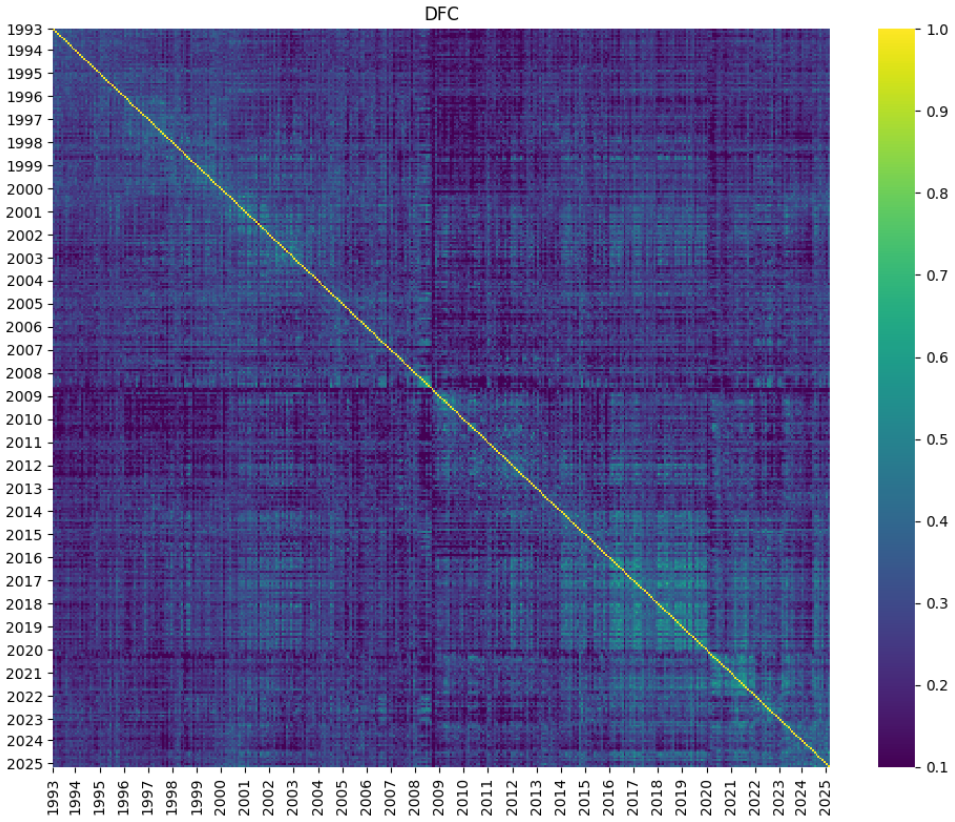


FIGURE D.1. DFC using the 1 Month (21 trading days) window.

D.2 DFC - 3 Months Window

This figure represents the DFC network based on a 3-month window length (63-trading-day duration), also providing insights into the relationships between each time window pair with higher resolution than the 6-month window length. Similarly, these correlations illustrate the dynamics of relationships between stocks over the period 1993 to 2025.

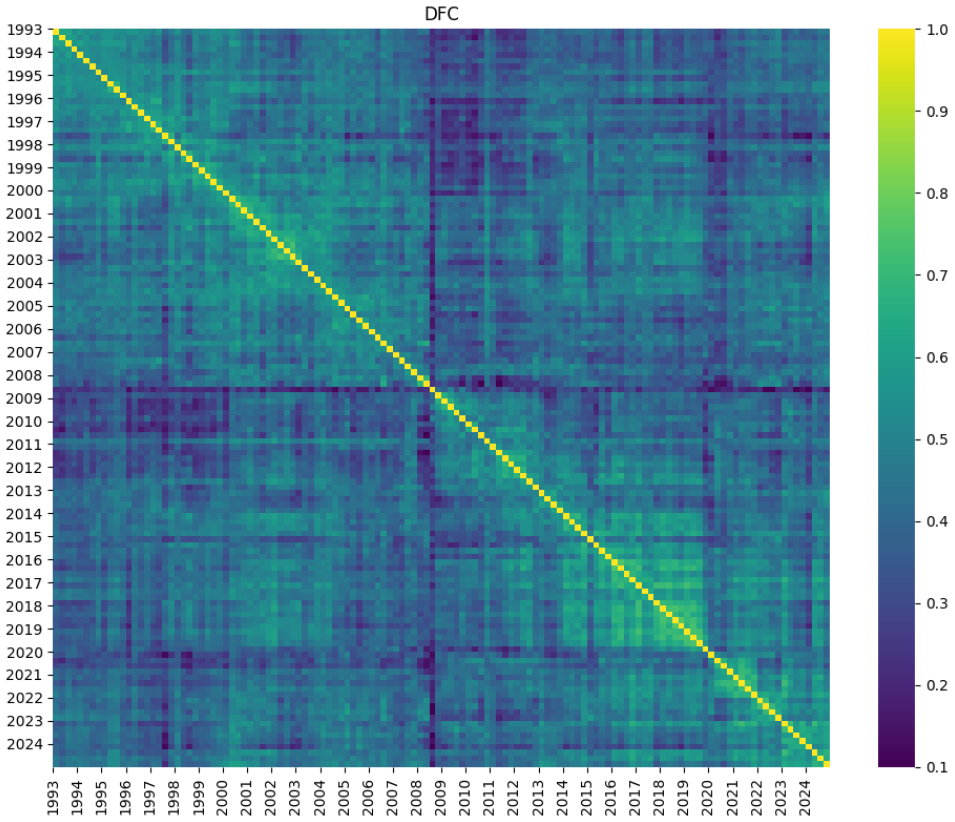


FIGURE D.2. DFC using the 3 Months (63 trading days) window.

D.3 1 Year Window

This figure represents the DFC network based on a 1-year window length (252-trading-day duration), providing insights into the relationships between each time window pair with the least resolution in this study. Similarly, these correlations illustrate the dynamics of relationships between stocks over the period 1993 to 2025.

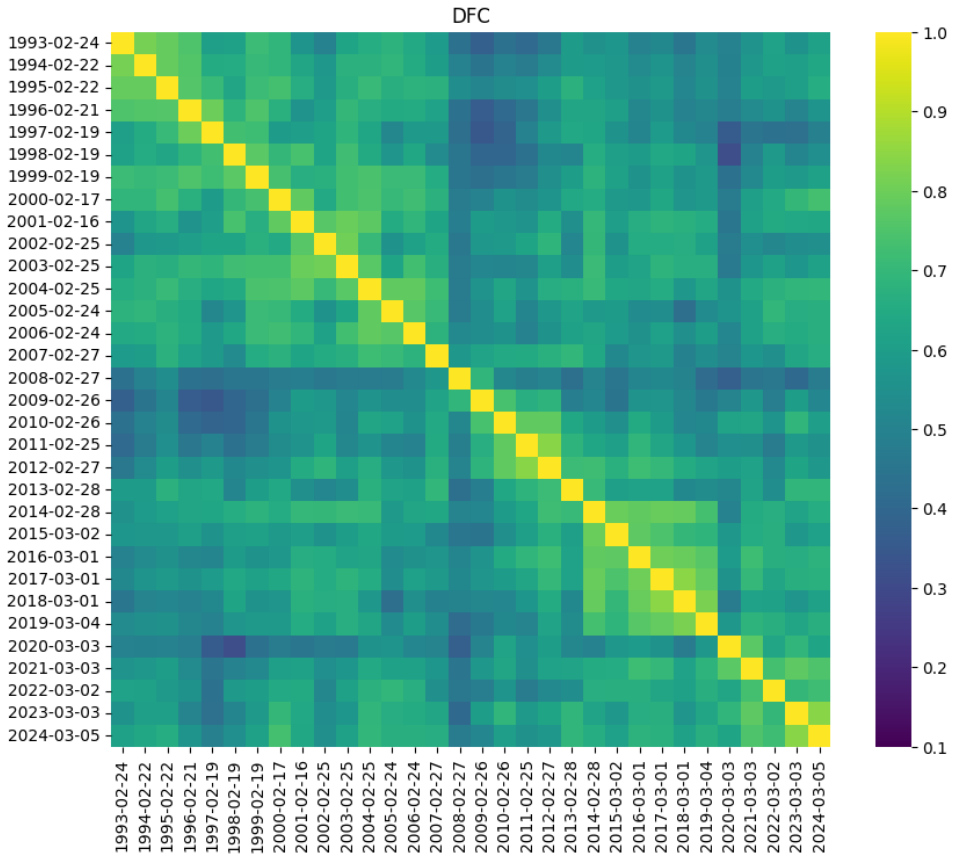


FIGURE D.3. DFC using the 1 Year (252 trading days) window.

APPENDIX E

Market States and Transitions with Other Time Windows

Figure E.1 is the market states identified based on the 1-month window length and Figure E.2 visualises the active times of each market state, using the methodology in *Section 5.2 Market states and transitions*.

E.1 1 Month Window: Market States

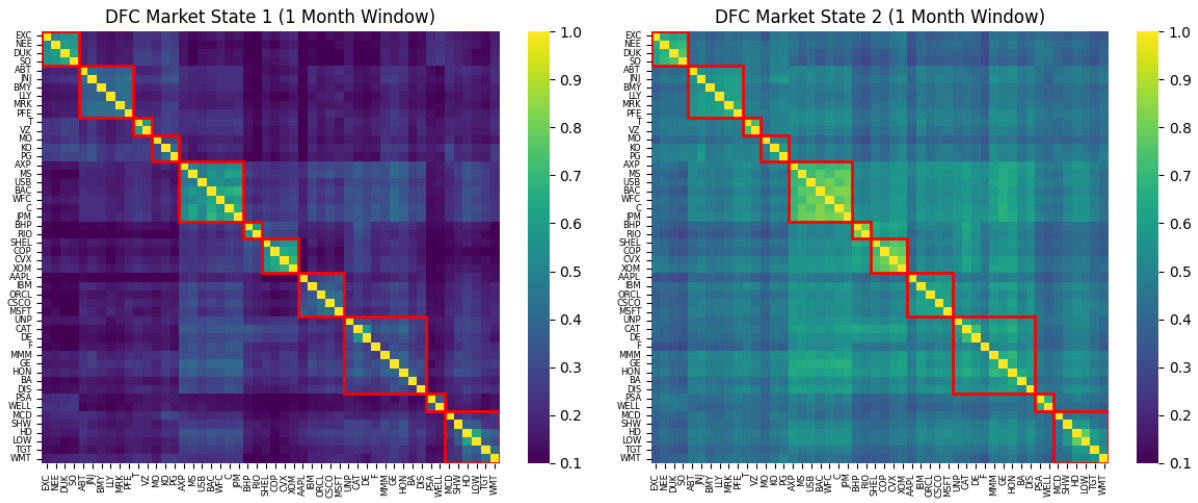


FIGURE E.1. Market States using the 1 Month (21 trading days) window.

E.2 1 Month Window: Time Vector

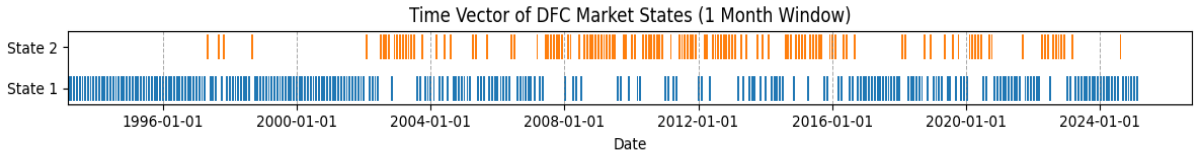


FIGURE E.2. Time Vector of the 1 Month (21 trading days) window.

Figure E.3 is the market states identified based on the 3-month window length and Figure E.4 visualises the active times of each market state, using the methodology in *Section 5.2 Market states and transitions*.

E.3 3 Months Window: Market States

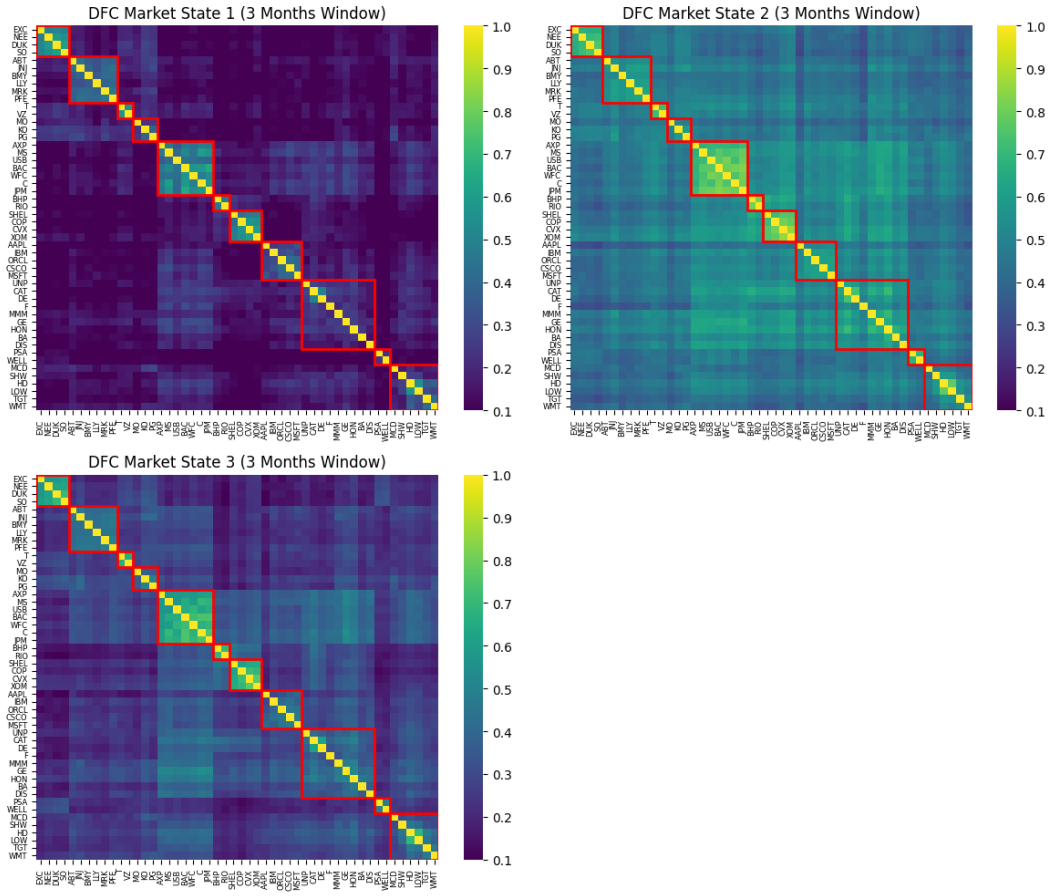


FIGURE E.3. Market States using the 3 Months (63 trading days) window.

E.4 3 Months Window: Time Vector

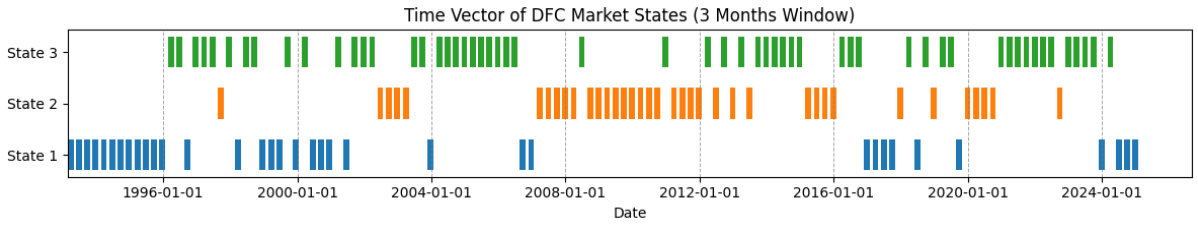


FIGURE E.4. Time Vector of the 3 Months (63 trading days) window.

Figure E.5 is the market states identified based on the 1-year window length and Figure E.6 visualises the active times of each market state, using the methodology in *Section 5.2 Market states and transitions*.

E.5 1 Year Window: Market States

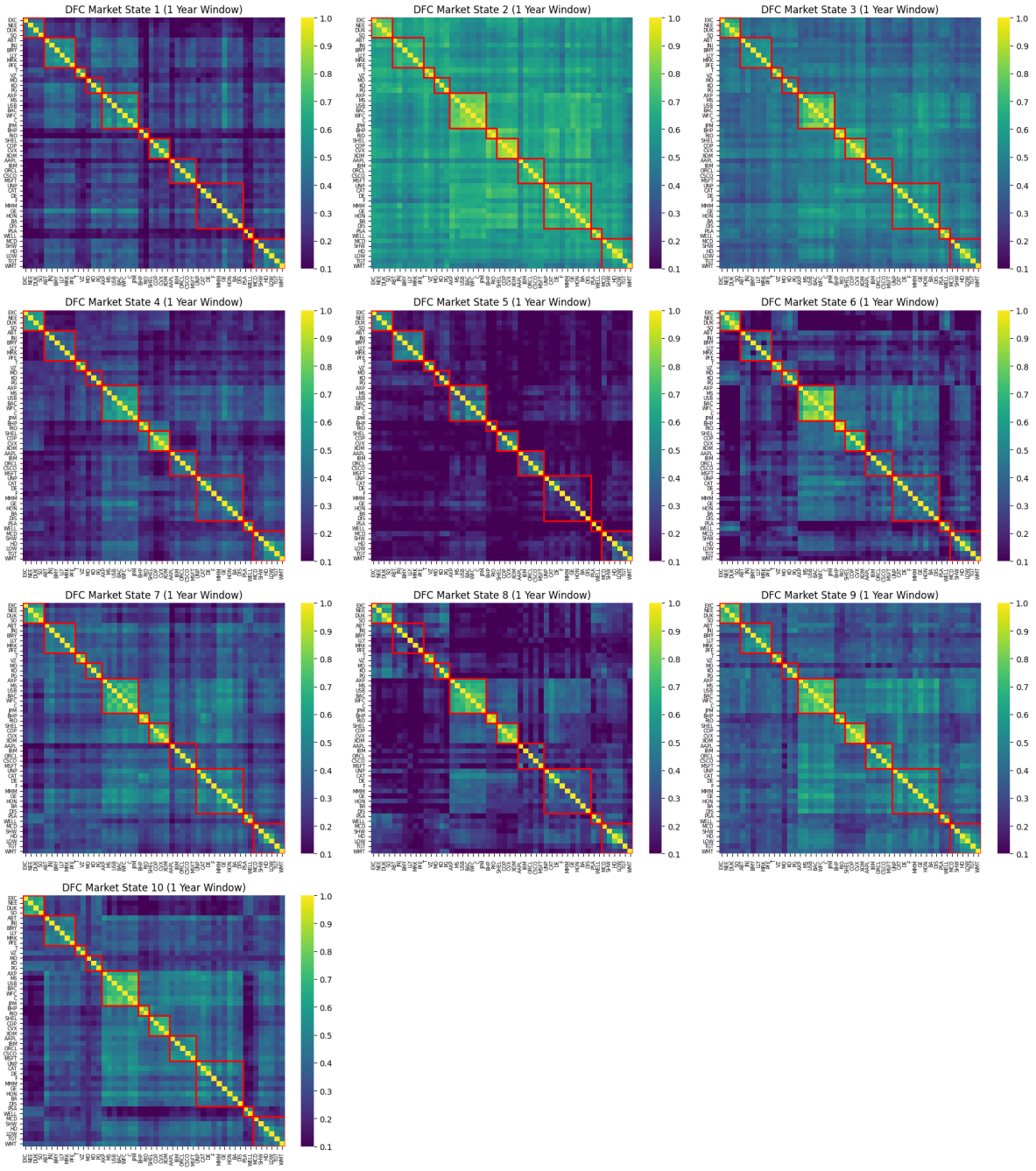


FIGURE E.5. Market States using the 1 Year (252 trading days) window.

E.6 1 Year Window: Time Vector

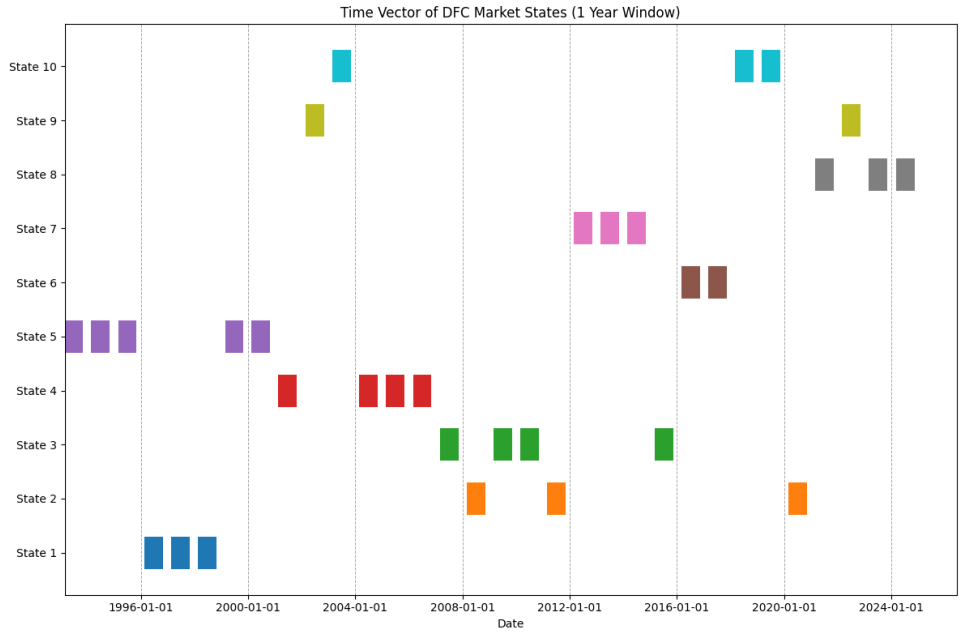


FIGURE E.6. Time Vector of the 1 Year (252 trading days) window.

APPENDIX F

HMM: matching states using time vector

Figure F.1 shows the 2 market states identified by the HMM, with their active times visualised in Figure F.2. The HMM states were aligned in the order of DFC states by correlating their time vectors. However, in Figure F.3, we can see that this reordering method did not work well as the correlations to VIX did not align.

F.1 Market States (n=2)

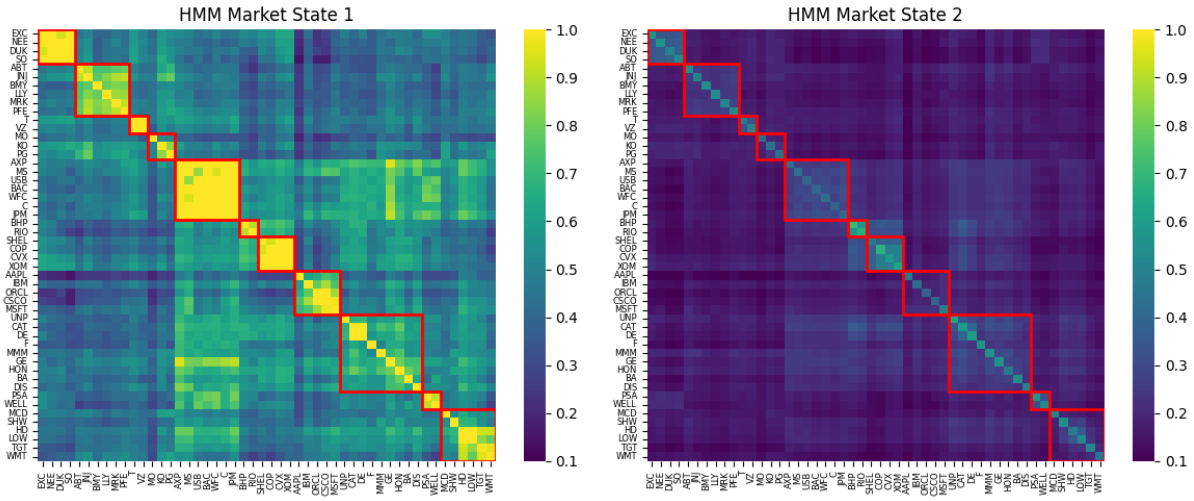


FIGURE F.1. HMM 2 Market States.

F.2 Time Vector - 2 States

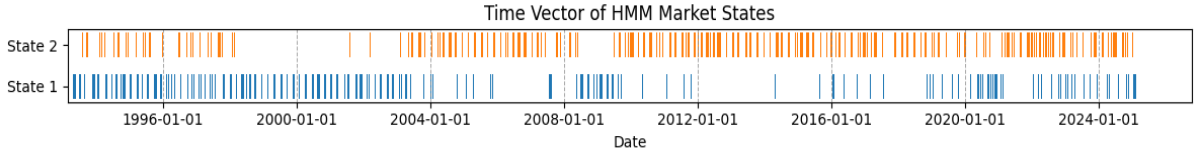


FIGURE F.2. HMM Time Vector - 2 states.

F.3 Correlation with VIX comparison - 2 States

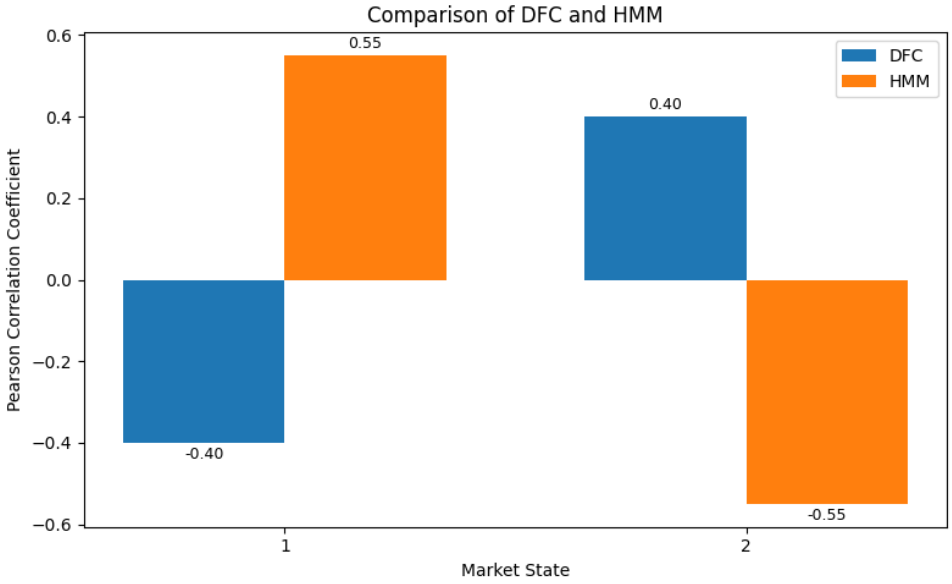


FIGURE F.3. Correlation with VIX - 1 Month Window Rolling Average.

Figure F.4 shows the 3 market states identified by the HMM, with their active times visualised in Figure F.5. The HMM states were aligned in the order of DFC states by correlating their time vectors. Similarly to 2 states, in Figure F.6, we can see that this reordering method did not work well as the correlations to VIX did not align.

F.4 Market States (n=3)

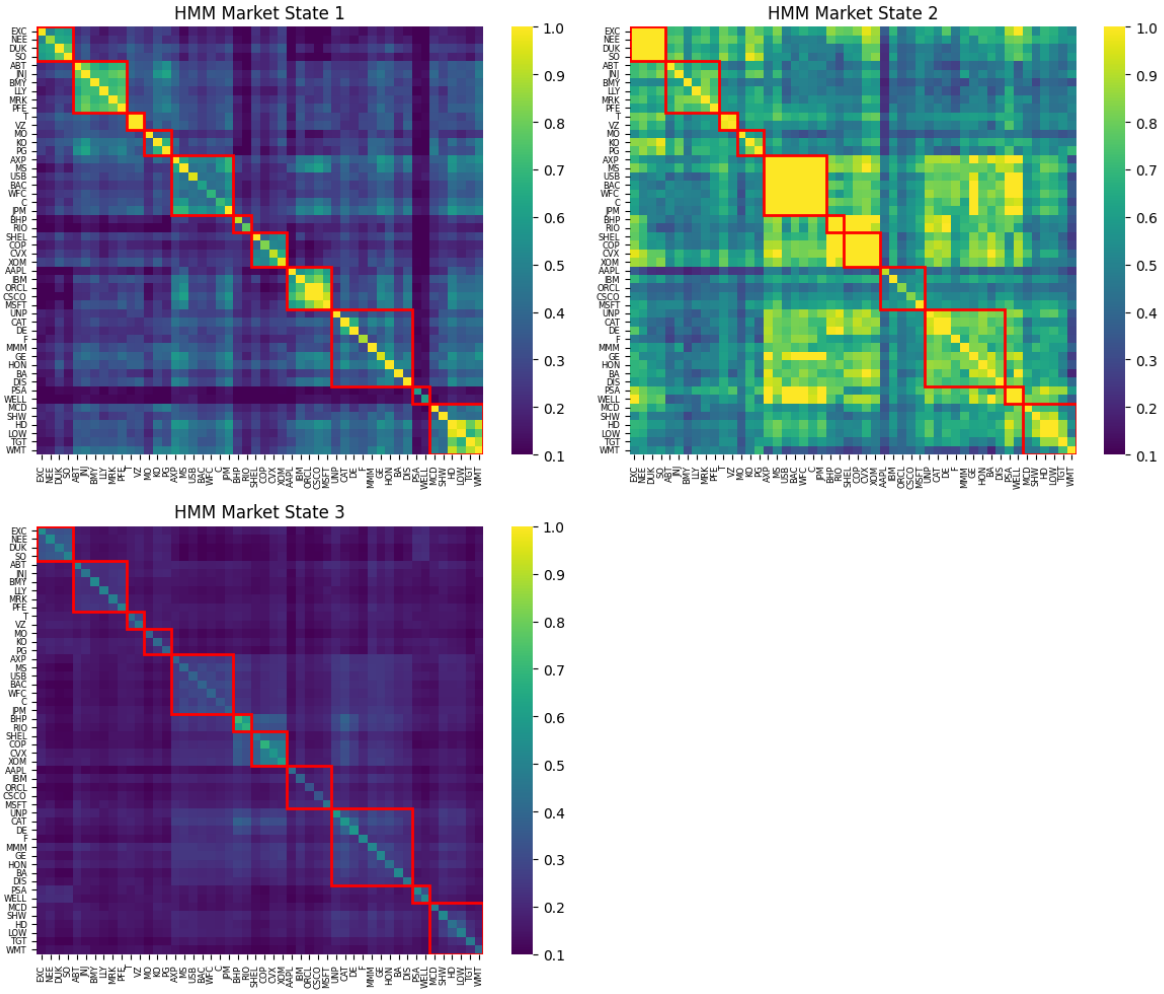


FIGURE F.4. HMM 3 Market States.

F.5 Time Vector - 3 States

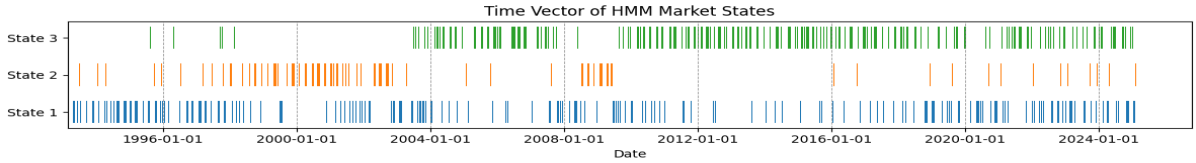


FIGURE F.5. HMM Time Vector - 3 states.

F.6 Correlation with VIX comparison - 3 States

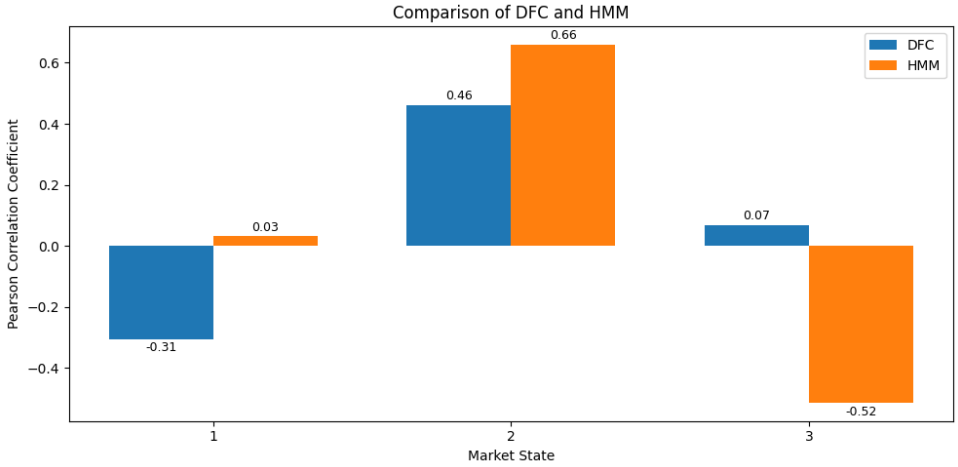


FIGURE F.6. Correlation with VIX - 6 Months Window Rolling Average.

Figure F.7 shows the 10 market states identified by the HMM, with their active times visualised in Figure F.8. The HMM states were aligned in the order of DFC states by correlating their time vectors. Similar to previously, in Figure F.9, this reordering method did not work well, as the correlations with VIX did not align.

F.7 Market States (n=10)

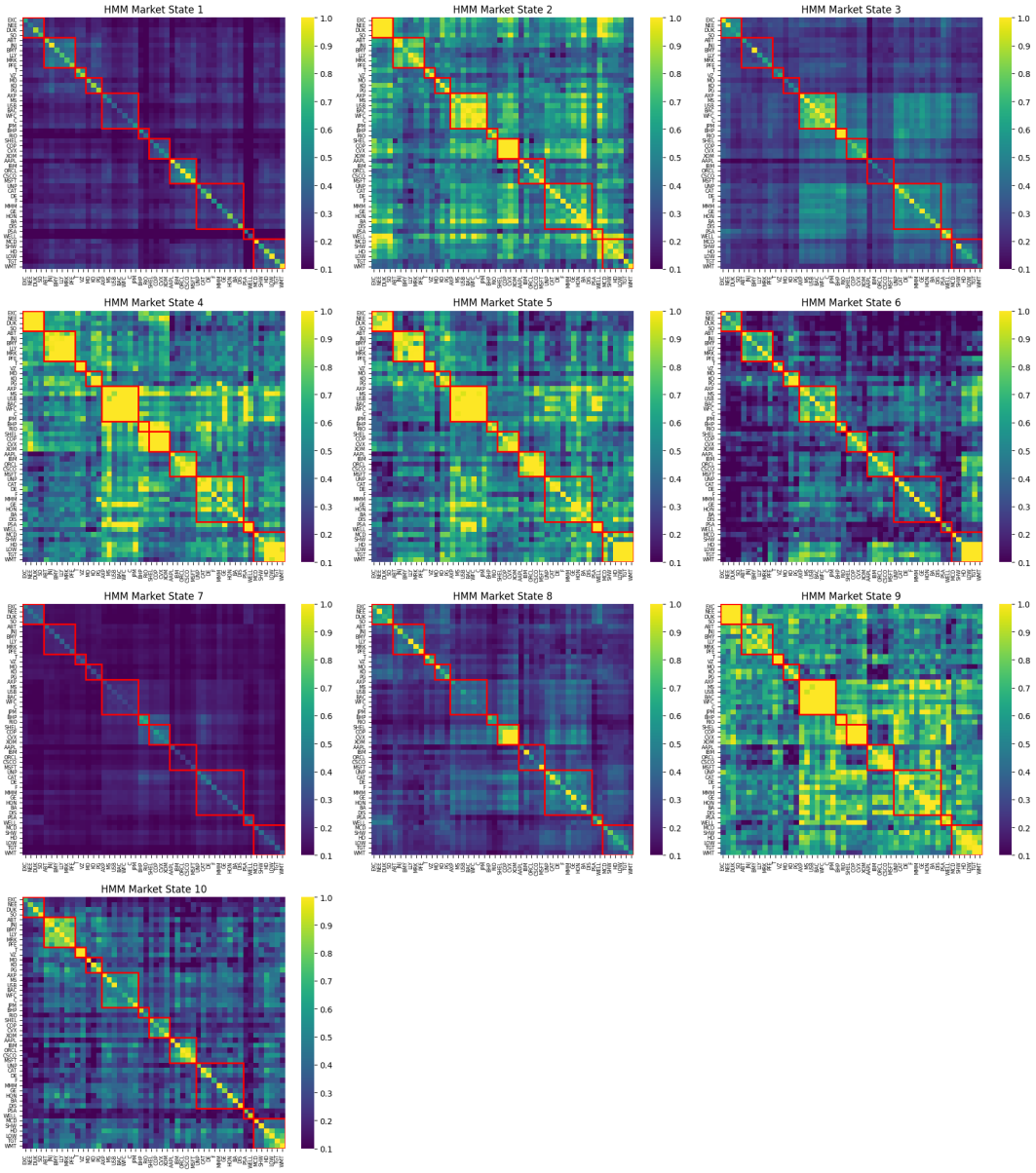


FIGURE F.7. HMM 10 Market States.

F.8 Time Vector - 10 states

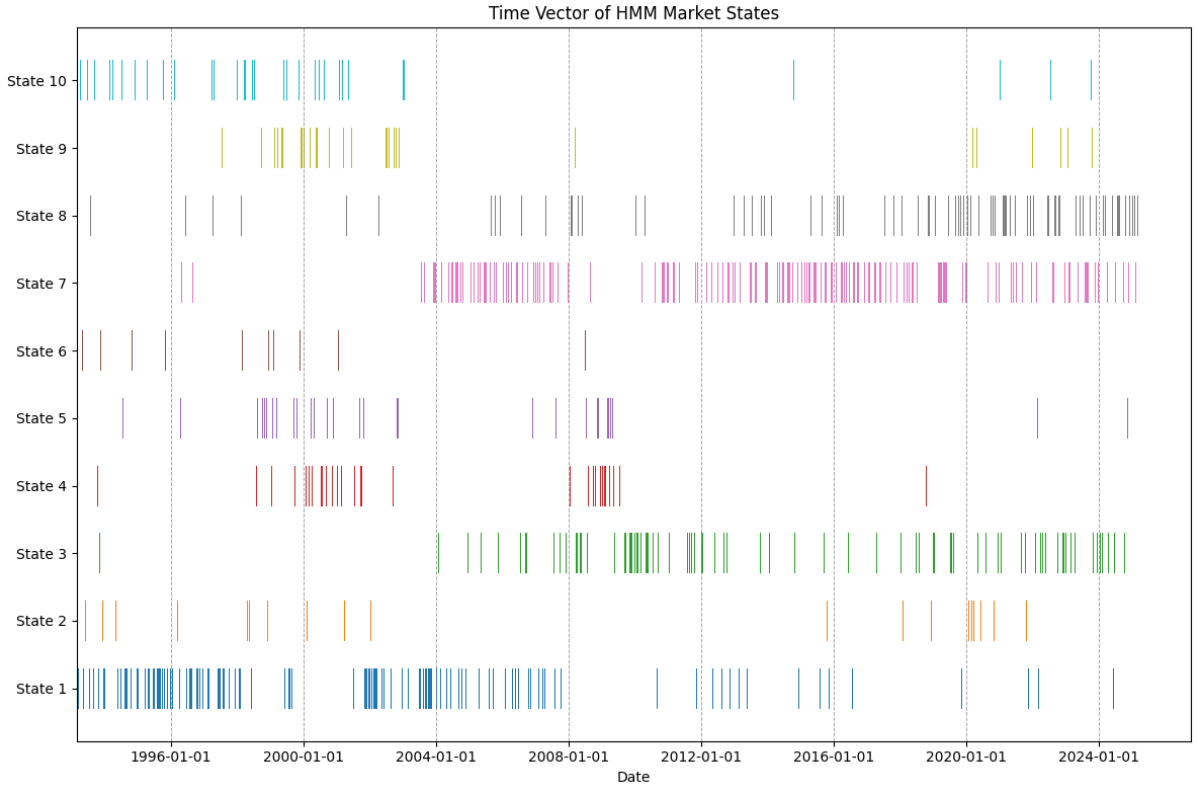


FIGURE F.8. HMM Time Vector - 10 states.

F.9 Correlation with VIX comparison - 10 states

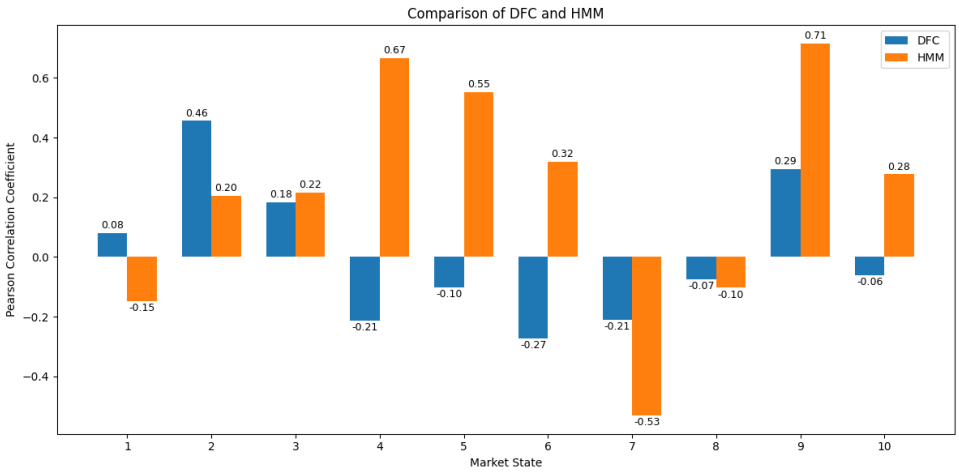


FIGURE F.9. Correlation with VIX - 1 Year Window Rolling Average.

APPENDIX G

HMM: matching states using state similarity

Figure G.1 shows the 2 market states identified by the HMM, with their active times visualised in Figure G.2. The HMM states were aligned in the order of DFC states by correlating their states. In Figure G.3, we can see that this reordering method worked better than *Appendix F* as the correlations to VIX were aligned.

G.1 Market States (n=2)

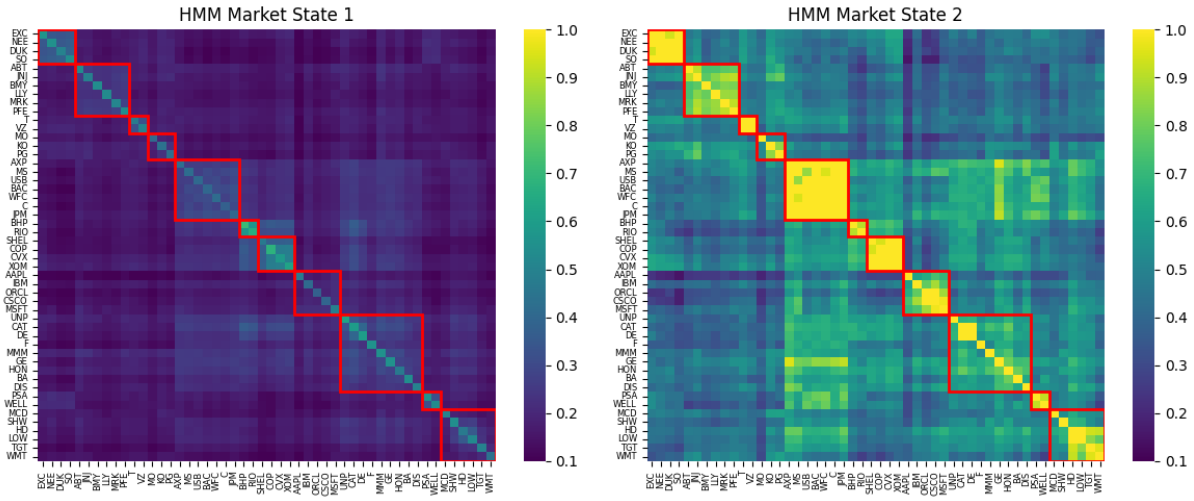


FIGURE G.1. HMM 2 Market States.

G.2 Time Vector - 2 states

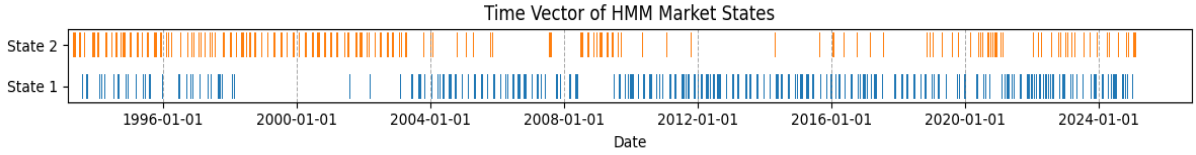


FIGURE G.2. HMM Time Vector - 2 states.

G.3 Correlation with VIX comparison - 2 states

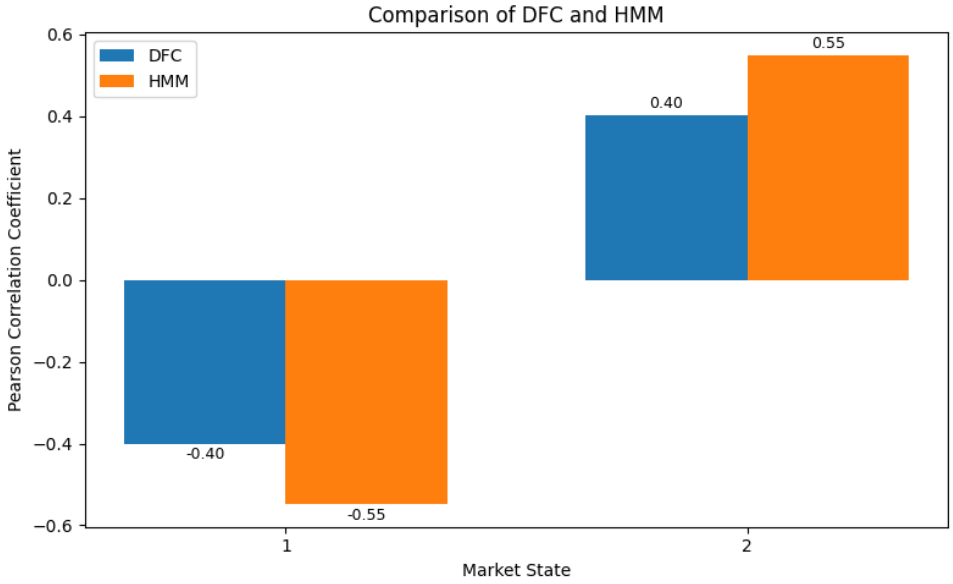


FIGURE G.3. Correlation with VIX - 1 Month Window Rolling Average.

Figure G.4 shows the 10 market states identified by the HMM, with their active times visualised in Figure G.5. The HMM states were aligned in the order of DFC states by correlating their states. In Figure G.6, we can see that this reordering method worked better than *Appendix F* as the correlations to VIX were mostly aligned.

G.4 Market States (n=10)

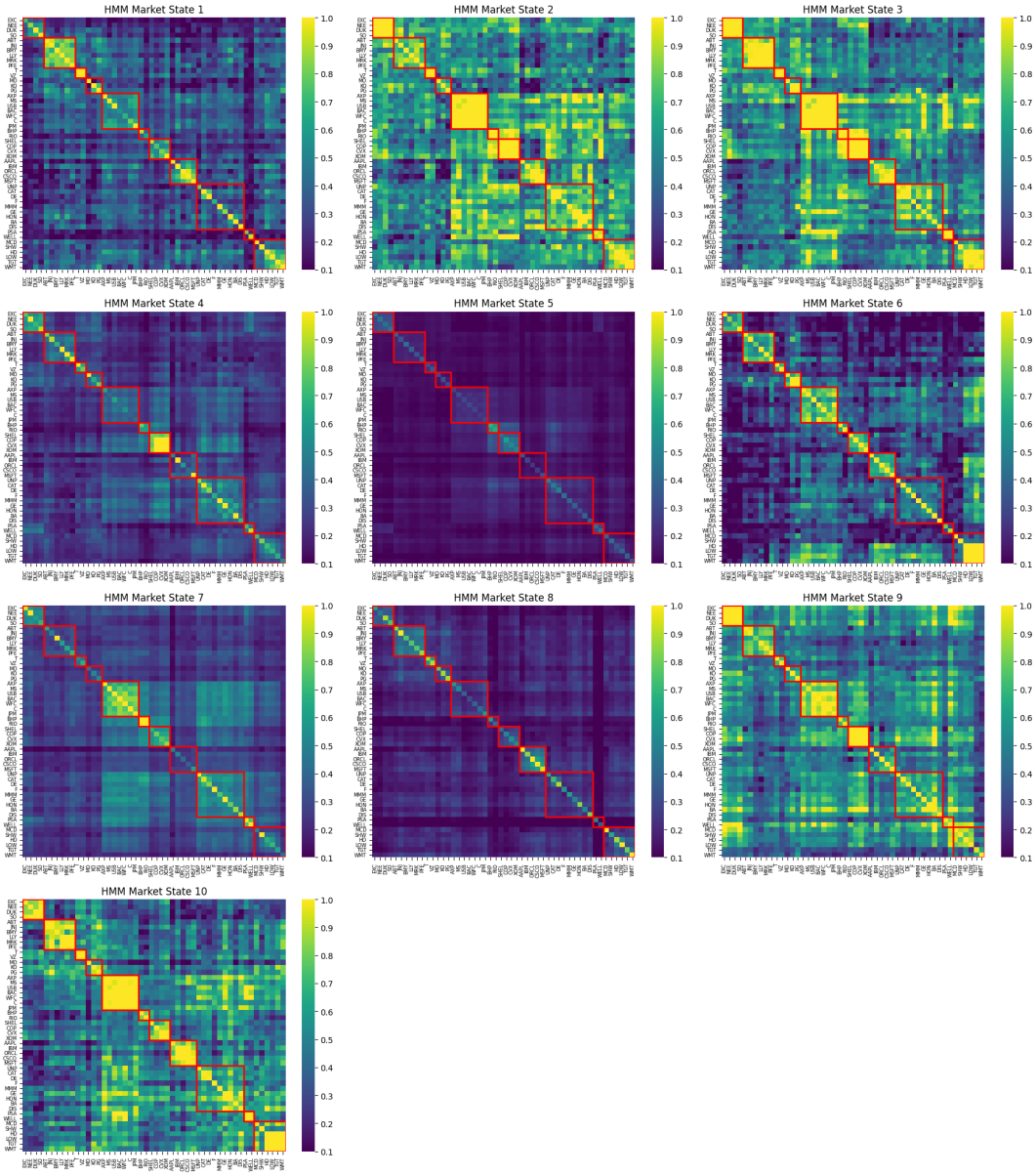


FIGURE G.4. HMM 10 Market States.

G.5 Time Vector - 10 states

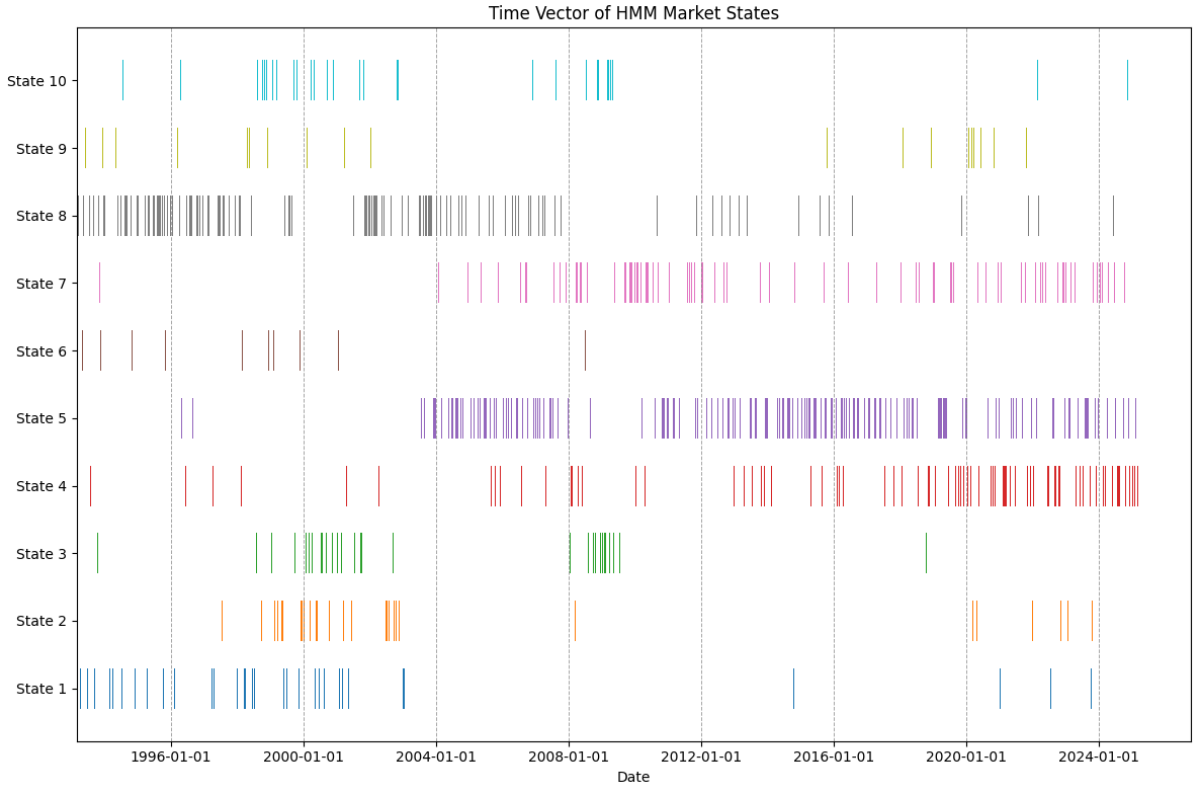


FIGURE G.5. HMM Time Vector - 10 states.

G.6 Correlation with VIX comparison - 10 states

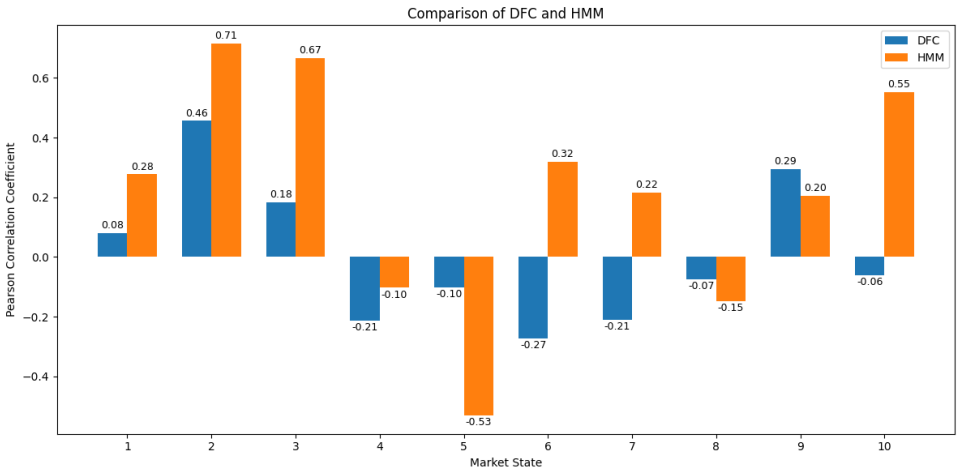


FIGURE G.6. Correlation with VIX - 1 Year Window Rolling Average.

APPENDIX H

DFC Networks with Lag-1 Data

H.1 Lag-1 static FC network of gaussian noise

This figure visualises the static FC network computed using a reconstructed dataset based on Gaussian noise to check whether our there were algorithm bias related to log differencing. Figure H.1 shows an evenly spread of positive and negative correlations.

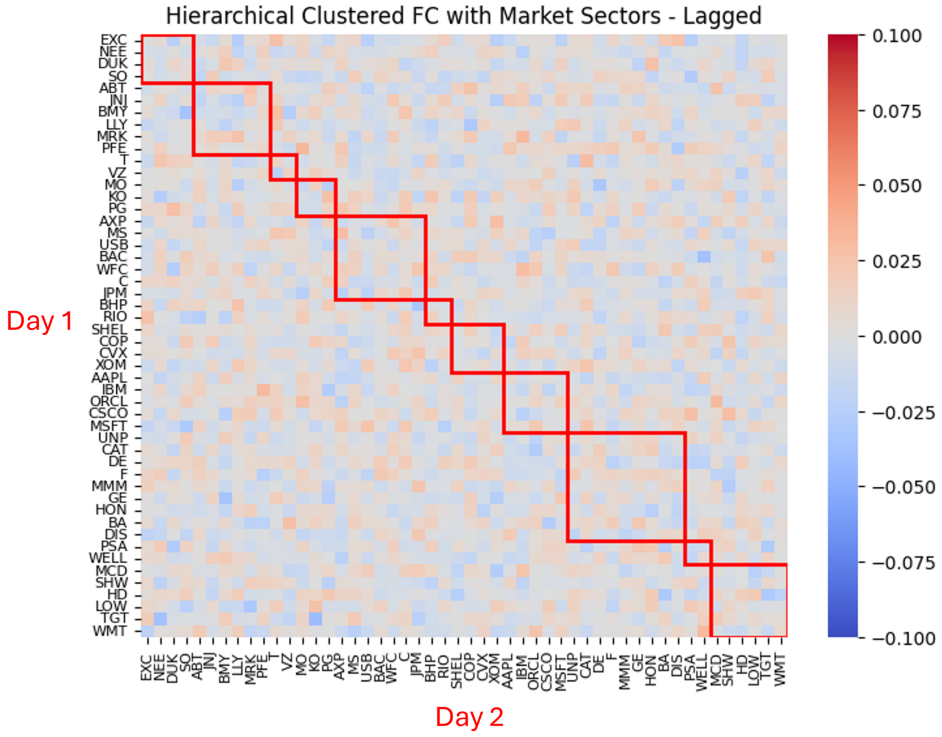


FIGURE H.1. Static FC of lag-1 gaussian noise.

H.2 1 Month Window

This figure represents the DFC network based on a directed measure (lag-1 correlation) of FC with a fixed 1-month window length (21-trading-day duration), providing insights into the relationships between each time window pair with the highest resolution in this study. These correlations illustrate the dynamics of relationships between stocks over the period 1993 to 2025.

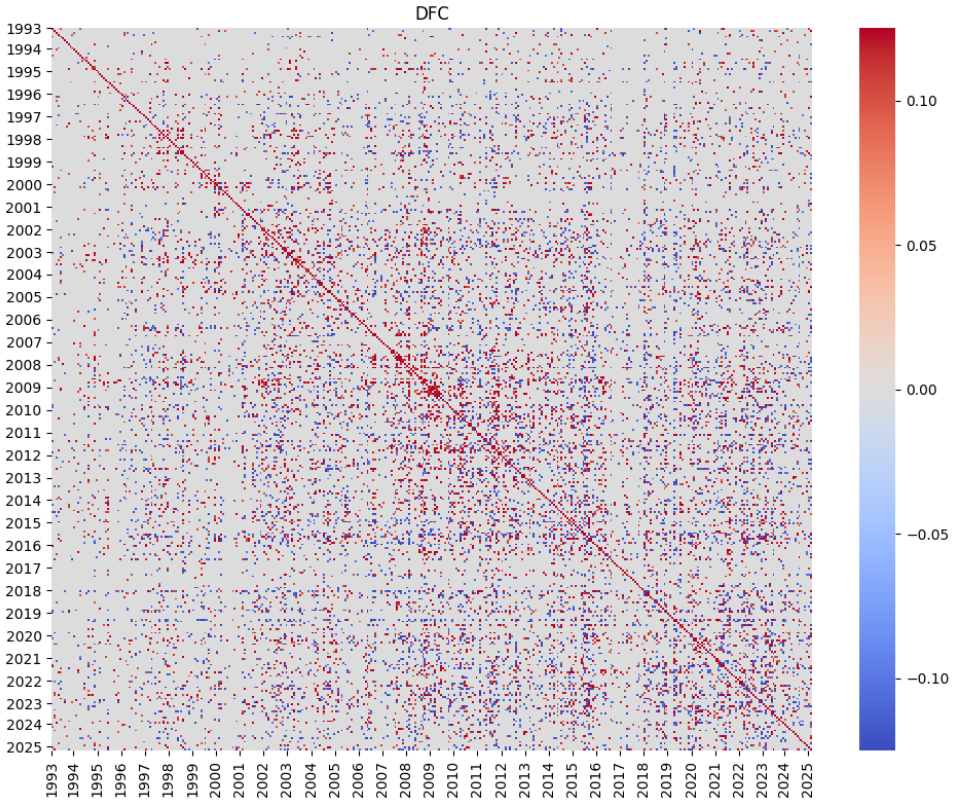


FIGURE H.2. DFC of lag-1 data - 1 Month Window.

H.3 3 Months Window

This figure represents the DFC network based on a directed measure (lag-1 correlation) of FC with a fixed 3-month window length (63-trading-day duration), providing insights into the relationships between each time window pair with a higher resolution than the 6-month window length. These correlations illustrate the dynamics of relationships between stocks over the period 1993 to 2025.

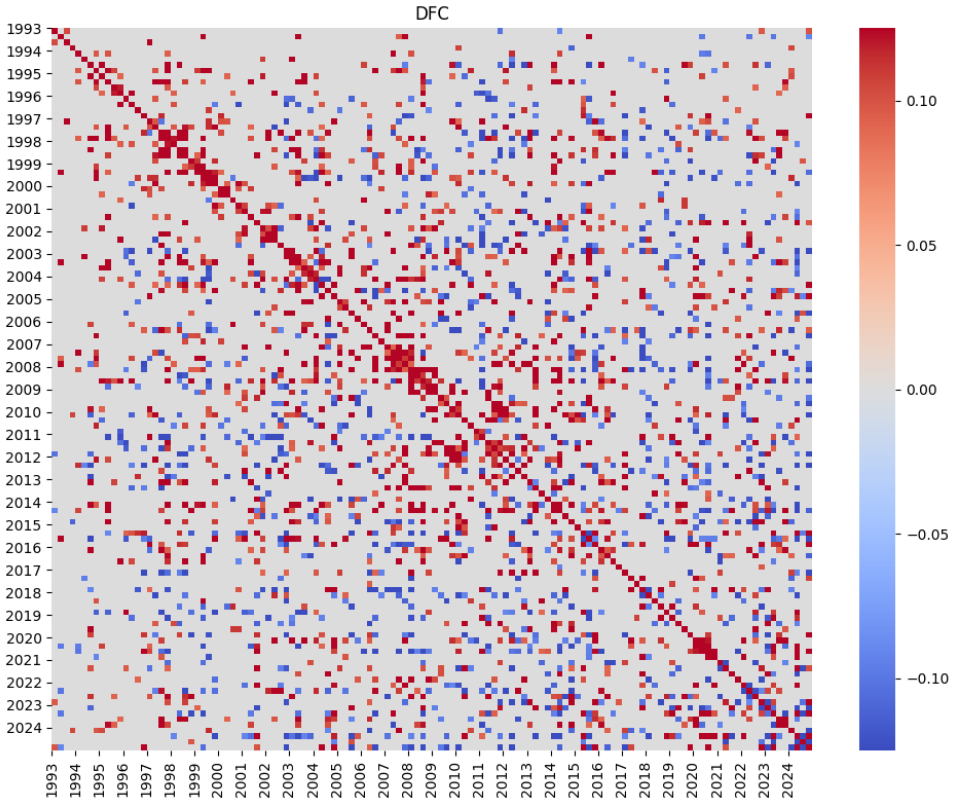


FIGURE H.3. DFC of lag-1 data - 3 Months Window.

H.4 1 Year Window

This figure represents the DFC network based on a directed measure (lag-1 correlation) of FC with a fixed 1-year window length (252-trading-day duration), providing insights into the relationships between each time window pair with the lowest resolution in this study. These correlations illustrate the dynamics of relationships between stocks over the period 1993 to 2025.

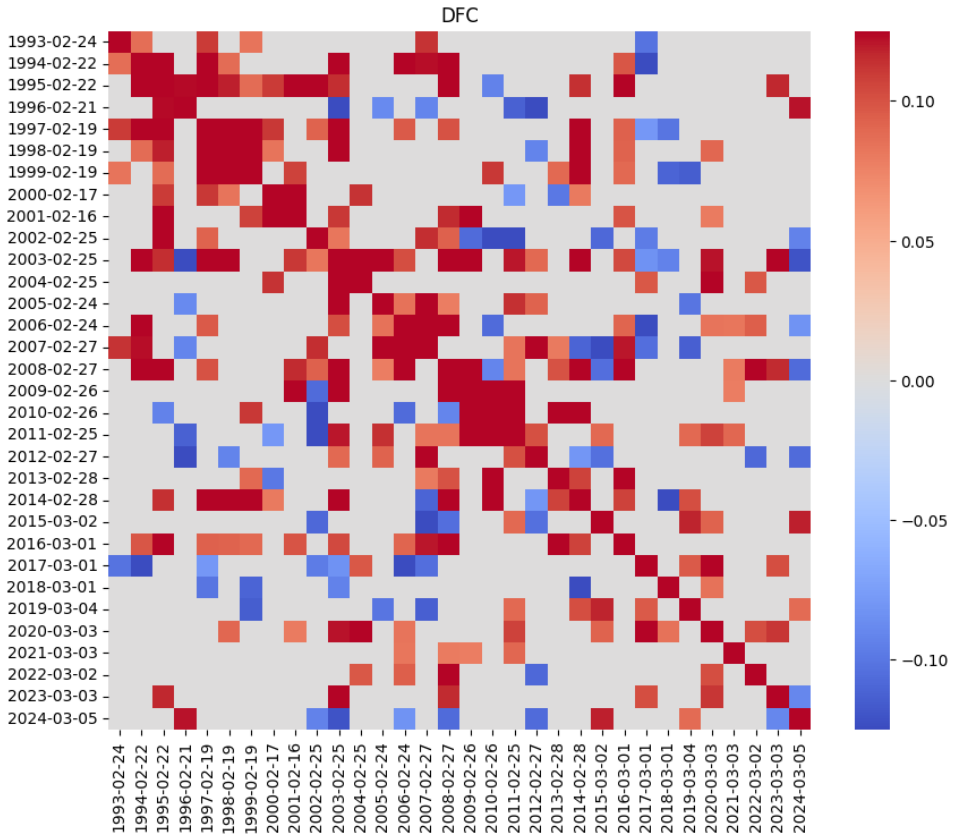


FIGURE H.4. DFC of lag-1 data - 1 Year Window.

APPENDIX I

Maket States and Transitions with Lag-1 Data

Figure I.1 is the "crisis" and "non-crisis" market states identified based on the directed FC measure (lag-1 correlation) with a fixed 1-year window length and Figure I.2 visualises the active times of each market state, using the methodology in *Section 5.2 Market states and transitions*.

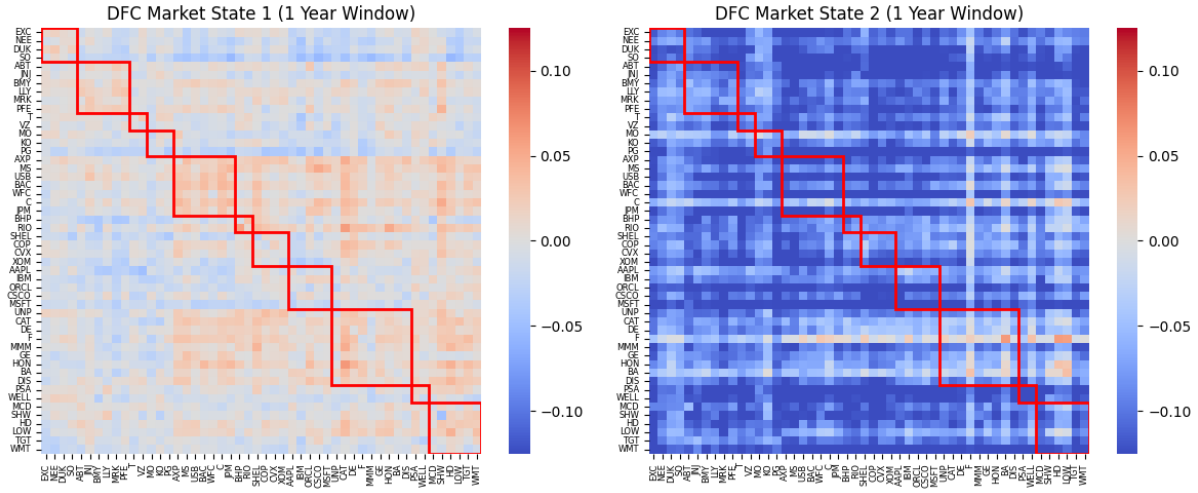


FIGURE I.1. States of lag-1 data - 1 Year Window.

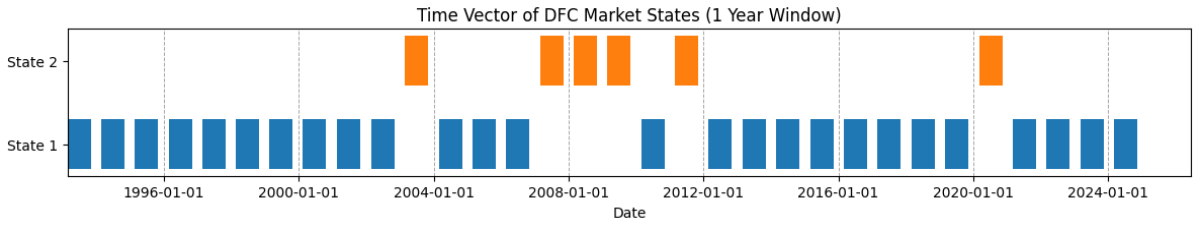


FIGURE I.2. Time Vector of lag-1 data - 1 Year Window.

APPENDIX J

Independent Component Analysis

To address the limitations of the sliding window approach, generative models, such as the Independent Component Analysis (ICA), can be implemented to minimise the modelling assumptions of the problem [17]. ICA assumes observations are linear mixtures of independent sources and can work with higher-order statistics to achieve independence [154], as illustrated in Figure I.1. Multi-subject data are aggregated and then processed by the ICA algorithm, which operates without an arbitrary window size. Subject-specific results are obtained through back-reconstruction, where individual components are estimated from the group data [154].

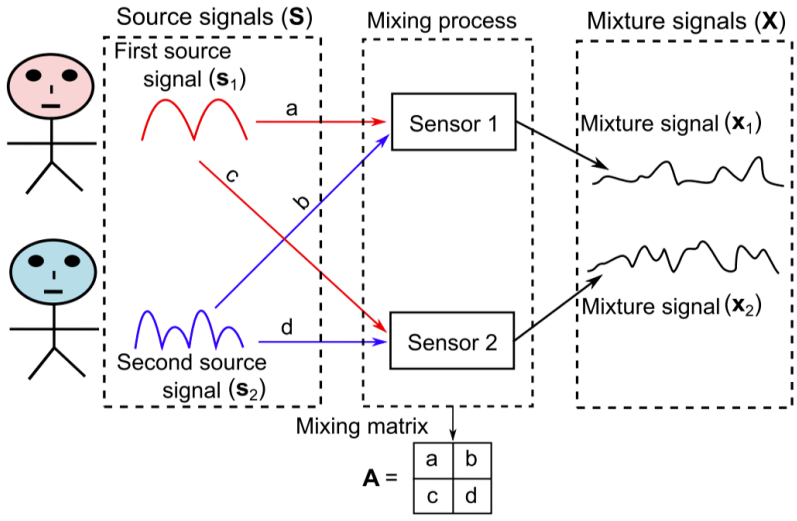


FIGURE J.1. An example of the ICA process.
Reprinted from Figure 2 of Tharwat's study [155].

Independent Component Analysis (ICA) was carried out to investigate the effect of removing three independent components each from 3 major market events: DotCom, the Global Financial Crisis and COVID-19. We standardised and transformed the raw price data into ICA signals. After the ICA transformation, we removed the three independent components that were most impacted by the three market events. Then, we reconstructed the filtered ICA signals into price data and logarithmic returns to reperform the entire pipeline and compare the results with the original ones. The results in this section are also based on the 6-month window length.

Nevertheless, in Figure J.2, we can see that multiple independent components were affected by the market events, and even removing one component from each event still leaves the impact of those events in the reconstructed time series. Hence, the DFC network in Figure J.4 still reflect the impact of GFC and COVID-19. Although the relationships between stocks and sectors changed in the static FC network (Figure J.3), we need to conduct further processes to analyse the results of ICA more meaningfully.

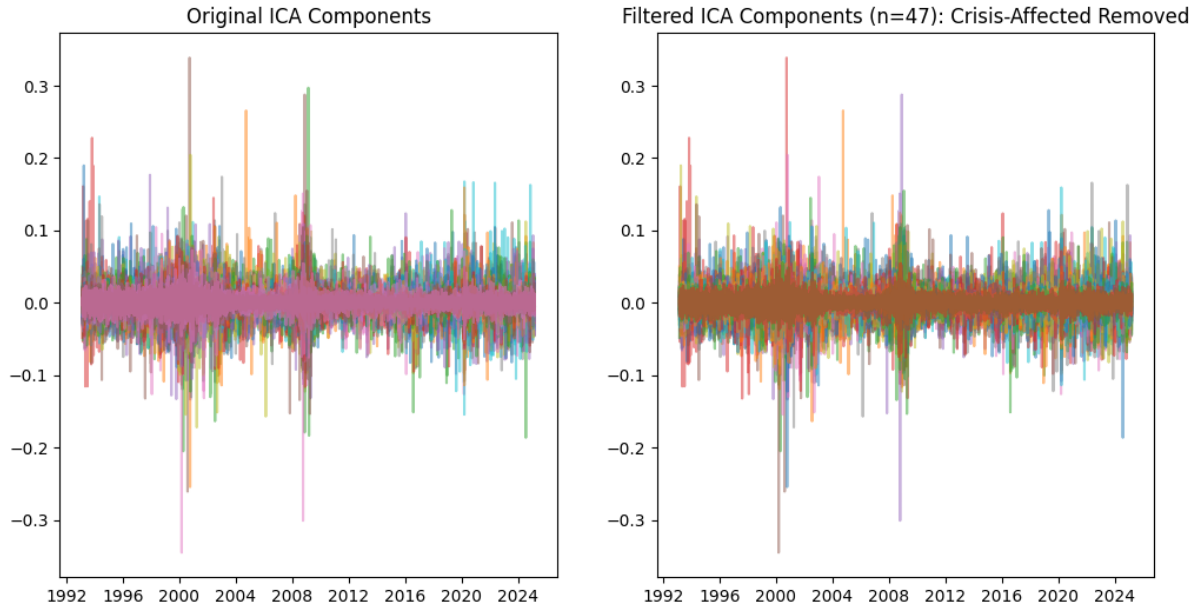
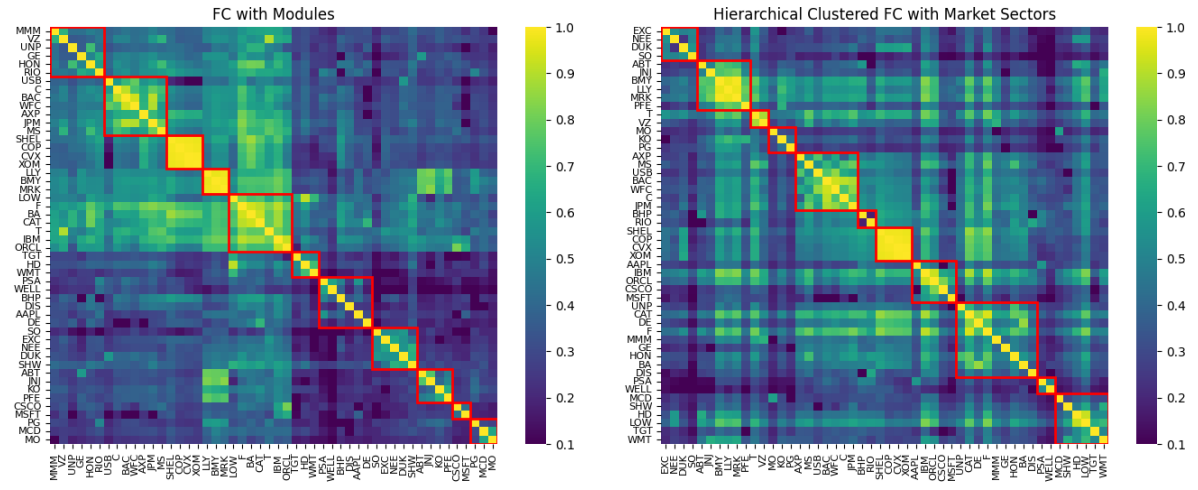


FIGURE J.2. Original vs Filtered ICA Signals.



(a) Re-clustered

(b) Same order as the original FC

FIGURE J.3. Static FC network after ICA.

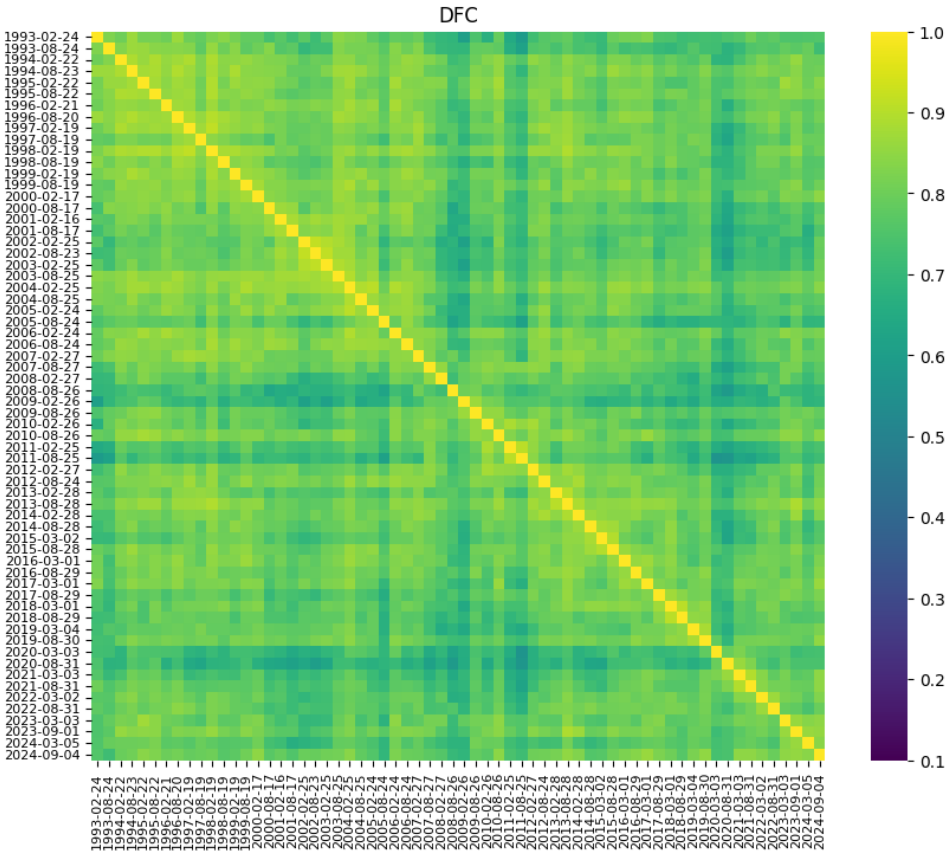


FIGURE J.4. DFC network after ICA.